

Glimm–Jaffe Parts I and II

Complete Lamport-notation reconstruction of the current proof chain

Fresh independent reconstruction and adversarial audit

16 August 2026

This volume expands the canonical Part I manuscript and the current Part II proof chain into numbered Lamport-style steps. It incorporates the explicit repairs identified by the accompanying critical audit; it is therefore both a reconstruction and a precise statement of what would have to replace the defective passages in the source files.

How to read this volume

The reconstruction is modular because the underlying Part II argument is modular. Every module is a separately compilable document with its own definitions and dependency interface; the remainder of this volume binds the resulting PDFs without altering their page images. The source manuscripts themselves were not silently rewritten.

The verdicts are deliberately asymmetric:

- Part I is **CONDITIONAL**. Its internal chain is valid after localizing the CE2 application cell by cell, but the claimed cutoff-uniform GJS scope remains an explicit external import rather than a reproduced cluster expansion.
- Part II is **FAIL** as written in the original V2 file: the assertion excluding four edge modes is false, and the shifted comparison drops an undefined remainder. The V2 module in this volume supplies the exact four-edge contribution and a complete fundamental-band/alias repair. With those replacements, V3–V4 close at the claimed rates.

The separate *Critical audit of Parts I and II* gives the finding-by-finding verdict, while the separate *Citation evidence dossier* reproduces every source page used to assess an imported statement. These three documents should be read together.

Module order and coverage

Part	Module	Coverage
I	Euclidean input to the Hamiltonian gap	Slab FKN representation, GJS clustering inputs, localized CE2 repair, dense polynomial vectors, and the gap inequality.
I	Gap to the main theorem	Path differentiability, inverse-energy filter, separation and clustering, one-path LPPL, full local norm convergence, local normality, spectral passage, invariances, and the conditional phase conclusion.

Part	Module	Coverage
II	Model sheet	Lattices, Weyl embeddings, Hamiltonians, closed coercive forms, compactness, ground-state simplicity, and Wick constants.
II	Quadratic obstruction	Exact finest-scale covariance obstruction and its consequences for naive continuum comparison.
II	Projective compactness	Cofinal compactness, consistency, and the distinction between subsequential and identified full-sequence limits.
II	Free fixed-coarse rate	D4 filter envelope/telescope, dispersion estimates, low/high-frequency split, and quantitative free convergence.
II	Weyl reduction	Pointwise Weyl criterion, extension to the coarse algebras, inductive-limit identification, and symmetry.
II	Gate scoping	Minimal gate logic, the sharp-time momentum-source no-go theorem, the mean-shift amendment, A1–A8 dependency map, and rejection of the slab fallback as load bearing.
II	V1	Weyl action, trace kernel, oscillator trace, Mehler bridge, exact jump shift, interacting marked-slice formula, continuum free formula, and thermal free rate.
II	V2	Same-noise coupling, edge covariance, Wick/convolution estimates, unshifted comparison with the four-edge correction, Wick dictionary, exact shifted fundamental-band/alias comparison, and Nelson bounds.
II	V3	Continuum interaction, uniform exponential bounds, finite-mode FKN, strong semigroup construction, generator/form identification, trace class and trace-norm approximation, periodic insertions, positivity improvement, and the gap.
II	V4	Coupled thermal comparison, normal-ordered partition convergence, full Laplace spectral transfer, derived lattice gap, thermal-to-ground interchange, gate, inductive-limit identification, and $\mathbb{Z}_2$ symmetry.

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Part I

Corrected structured proof I

GJS Euclidean estimates $\implies$ a cutoff-uniform local moment and physical Hamiltonian gap

Fresh reconstruction of `part_i/main.tex`

16 August 2026

Status. CONDITIONAL. The derivation below is valid from the explicitly listed cutoff-construction facts and the exact estimates in Glimm–Jaffe–Spencer (GJS), Theorems 1.1.7–1.1.8. It incorporates every local repair found in the audit: the dimensionless-coupling reduction, the correct FKN references, signed truncations, the two time-sign cases, the corrected unit-cell count, the missing exponential-moment prefactor, and the restriction $0 \leq \beta < m_1$.

This document proves no part of the GJS cluster expansion itself. That theorem is an import, displayed verbatim by page image in the separate citation dossier.

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1 Objects and exact imports

Let

$$\mathcal{H} = L^2(\mathcal{S}'(\mathbb{R}), d\phi_0), \quad K(g) = H_0 + V(g), \quad H(g) = K(g) - E(g),$$

where $E(g) = \inf \operatorname{spec} K(g)$ and

$$V(h) = \lambda \int_{\mathbb{R}} h(x) : \mathcal{P}(\varphi_0(x)) : dx, \quad \mathcal{P}(\xi) = \sum_{n=0}^{2p'} a_n \xi^n, \quad a_{2p'} > 0.$$

The scalar product is conjugate-linear in its first argument. The cutoffs satisfy $0 \leq g \leq 1$ and have compact support. Write $\Omega_0 = 1$ for the free vacuum.

Imported assumption 1.1 (cutoff Hamiltonian facts). For every such g , $K(g)$ is self-adjoint and bounded below, $E(g)$ is a simple eigenvalue, and its normalized eigenvector Ω_g can be chosen strictly positive $d\phi_0$ -almost everywhere. Consequently

$$c_g := \langle \Omega_g, \Omega_0 \rangle > 0.$$

For every compactly supported real h , the time-zero multiplication operator $V(h)$ is self-adjoint on its maximal domain.

Imported assumption 1.2 (slab FKN input). For a fixed g , the interacting Feynman–Kac–Nelson formula holds on every finite time slab. The correct source locations are Simon, Theorem III.6 for the free FKN formula and Corollary V.14/Theorem V.15 for its interacting form; GRS, Theorems II.14 and II.16, give the same transfer formula. Simon’s Theorem III.12, cited in the audited manuscript, is not an FKN theorem.

Let $d\Phi$ be the Euclidean Gaussian measure with covariance $(-\Delta + m_0^2)^{-1}$. For compactly supported measurable $0 \leq h \leq 1$, put

$$dq_h = Z(h)^{-1} \exp \left[-\lambda \int h(z) : \mathcal{P}(\Phi(z)) : dz \right] d\Phi.$$

GJS define localized sharp-time monomials Q and a norm $\|Q\|_{\text{GJS}}$ in (1.1.7)–(1.1.9). If $N(\Delta)$ is the total degree in a unit square Δ , a localized term with kernel w has norm

$$K_1 \left[\prod_{\Delta} K_2^{2N(\Delta)} N(\Delta)! \right] \|w\|_2,$$

and a nonlocalized polynomial is normed by summing its localized pieces.

Imported assumption 1.3 (CE1: GJS Theorem 1.1.8). There is $C_8 < \infty$, independent of the cutoff h , such that every admissible sharp-time polynomial obeys

$$\left| \int Q(s) dq_h \right| \leq C_8 \|Q(s)\|_{\text{GJS}}.$$

The theorem statement says “uniformly as $h \uparrow 1$ ”; GJS page 629 explicitly states that the estimates of Theorem 1.1.8 are uniform in the space cutoff h .

Imported assumption 1.4 (CE2: GJS Theorem 1.1.7). If Q_1, Q_2 have disjoint localization squares separated by a strip of maximal width d , then

$$\left| \int Q_1 Q_2 dq_h - \int Q_1 dq_h \int Q_2 dq_h \right| \leq C_7 e^{-m_1 d} \|Q_1\|_{\text{GJS}} \|Q_2\|_{\text{GJS}}, \quad (\text{CE2})$$

where $m_1 > 0$ and C_7 are independent of h, Q_1, Q_2 .

Remark 1.5 (no circular gap assumption). Assumptions 1.1–1.4 include simplicity of the cutoff ground state but do not include isolation of its eigenvalue. The gap proved below is therefore not used to obtain the slab limit.

2 The structured proof

Theorem 2.1 (Euclidean estimates imply (M) and (G)). *There is a dimensionless threshold $\varepsilon_{\mathcal{P}} > 0$ such that, if $0 \leq \lambda/m_0^2 < \varepsilon_{\mathcal{P}}$, then the following statements hold for every cutoff g .*

(i) *If h is real, $\|h\|_{\infty} \leq 1$, and $\text{supp } h$ is contained in an interval of length 2, then*

$$\|V(h)\Omega_g\| \leq M, \quad M^2 = 9C_8 K_1 K_2^{8p'} (4p')! \lambda^2 \left(\sum_{n=0}^{2p'} |a_n| \right)^2. \quad (\text{M})$$

(ii)

$$\text{spec } H(g) \subseteq \{0\} \cup [m_1, \infty), \quad (\text{G})$$

and 0 is a simple eigenvalue.

Proof in Lamport form

⟨1⟩ 1 Reduce the GJS smallness hypothesis to $\lambda/m_0^2 < \varepsilon_{\mathcal{P}}$.

⟨2⟩ 1 Fix source-admissible numbers $M_* > 0$ and $\lambda_* > 0$ such that the GJS expansion is valid at mass M_* whenever $0 \leq \lambda' < \lambda_*$.

Because: GJS formulate the construction for sufficiently large mass and sufficiently small coupling. Choosing one admissible pair merely names the constants in that hypothesis.

⟨2⟩ 2 For arbitrary (m_0, λ) set $s = m_0/M_*$ and apply the unitary space-time dilation stated on GJS page 590. It sends

$$(m_0, \lambda) \mapsto (s^{-1}m_0, s^{-2}\lambda) = \left(M_*, \frac{M_*^2}{m_0^2}\lambda\right). \quad (2.1)$$

⟨2⟩ 3 The dilation introduces no change in the coefficients of $\mathcal{P}$ or in Wick ordering.

Because: In two dimensions the scalar field has scaling dimension zero. More explicitly, if $\tilde{\Phi}(y) = \Phi(y/m_0)$, then

$$\mathbb{E}[\tilde{\Phi}(y)\tilde{\Phi}(y')] = C_{m_0}((y - y')/m_0) = C_1(y - y'),$$

by the substitution $p = m_0 q$ in the Fourier integral for C_{m_0} . Equality of all covariance contractions implies equality of the corresponding Wick polynomials. The Jacobian contributes m_0^{-2} to the interaction.

⟨2⟩ 4 Set $\varepsilon_{\mathcal{P}} = \lambda_*/M_*^2$. Then $\lambda/m_0^2 < \varepsilon_{\mathcal{P}}$ implies $s^{-2}\lambda < \lambda_*$, so the transformed model satisfies the source hypothesis.

⟨2⟩ 5 This proves ⟨1⟩1. □

⟨1⟩ 2 Establish the exact slab transfer identities without assuming a gap.

Fix g and define

$$h_{T,g}(t, x) = \mathbf{1}_{[-T, T]}(t)g(x), \quad \psi_a = e^{-aH(g)}\Omega_0, \quad \hat{\psi}_a = \psi_a/\|\psi_a\|.$$

⟨2⟩ 1 For bounded nonnegative time-zero functions F_1, F_2 , FKN gives, for $0 \leq t \leq 2T$ and $|s| \leq T$,

$$\int F_1(\Phi(-t/2))F_2(\Phi(t/2)) dq_{h_{T,g}} = \frac{\langle F_1\psi_{T-t/2}, e^{-tH(g)}F_2\psi_{T-t/2} \rangle}{\|\psi_T\|^2}, \quad (2.2)$$

$$\int F_1(\Phi(s)) dq_{h_{T,g}} = \frac{\langle \psi_{T-s}, F_1\psi_{T+s} \rangle}{\|\psi_T\|^2}. \quad (2.3)$$

Because: The unshifted formula contains $K(g)$ and the same factor $e^{-2TE(g)}$ in numerator and partition function. Multiplying every semigroup by the corresponding $e^{aE(g)}$ replaces $K(g)$ by $H(g)$ and cancels that factor exactly.

⟨2⟩ 2 Equations (2.2)–(2.3) hold for bounded signed functions.

Because: Write $F = F^+ - F^-$ and expand bilinearly. Each of the four terms is a finite nonnegative FKN integral, so linearity is legitimate.

⟨2⟩ 3 They extend to the unbounded functions used below.

Because: For a real multiplication function F , set $F^{[M]} = (-M) \vee (F \wedge M)$. Apply the bounded identity to $F^{[M]}$. Whenever $F\psi_a \in \mathcal{H}$, $F^{[M]}\psi_a \rightarrow F\psi_a$ by dominated convergence, because $|F - F^{[M]}|^2|\psi_a|^2 \leq |F|^2|\psi_a|^2$. On path space, use $|F_1 F_2| \leq (F_1^2 + F_2^2)/2$ after translating both factors to their respective time slices. The CE1 estimates below supply the required square integrability.

$\langle 2 \rangle 4$ This proves $\langle 1 \rangle 2$. □

$\langle 1 \rangle 3$ Prove the slab vectors converge to the cutoff ground state using simplicity only.

Let $P_g = |\Omega_g\rangle\langle\Omega_g|$. Positivity gives $P_g\Omega_0 = c_g\Omega_g$ with $c_g > 0$. Hence

$$\psi_a = c_g\Omega_g + e^{-aH(g)}(1 - P_g)\Omega_0. \quad (3.1)$$

$\langle 2 \rangle 1$ The second term of (3.1) converges to zero in norm.

Because: If μ is the spectral measure of $(1 - P_g)\Omega_0$, then

$$\|e^{-aH(g)}(1 - P_g)\Omega_0\|^2 = \int_{(0,\infty)} e^{-2aE} d\mu(E) \rightarrow 0$$

by dominated convergence. Simplicity says precisely that $\mu(\{0\}) = 0$; no positive distance from zero is used.

$\langle 2 \rangle 2$ Consequently

$$\hat{\psi}_a \rightarrow \Omega_g, \quad \|\psi_a\| \downarrow c_g, \quad 1 \leq \frac{\|\psi_b\|}{\|\psi_a\|} \leq c_g^{-1} \quad (0 \leq b \leq a), \quad (3.2)$$

and every ratio in (3.2) tends to 1 when both indices tend to infinity with bounded difference.

Because: $e^{-aH(g)}$ is a contraction and $\|\Omega_0\| = 1$.

$\langle 2 \rangle 3$ This proves $\langle 1 \rangle 3$. □

$\langle 1 \rangle 4$ Derive the local interaction moment (M).

Let h satisfy the hypotheses of (M). In Euclidean notation define

$$Q_V = \lambda^2 \sum_{n,n'=0}^{2p'} a_n a_{n'} \iint h(x)h(y) : \Phi(0,x)^n :: \Phi(0,y)^{n'} : dx dy. \quad (4.1)$$

$\langle 2 \rangle 1$ Q_V is exactly the multiplication function $V(h)^2$.

Because: Expanding the square of $\lambda \sum_n a_n \int h(x) : \Phi(0,x)^n : dx$ gives (4.1). No Wick reordering of the product is performed or required; GJS (1.1.5) permits products of individually Wick-ordered powers.

$\langle 2 \rangle 2$ The support of h meets at most three half-open unit spatial cells in positive measure.

Because: An interval of length 2 meets at most three members of the partition $[j, j+1)$ except possibly in endpoints, which have zero Lebesgue measure.

$\langle 2 \rangle 3$ For one ordered pair of cells and degrees n, n' , the GJS local weight is bounded by $K_1 K_2^{8p'} (4p')!$.

Because: The total local degree is at most $n + n' \leq 4p'$. If both variables belong to one square, the weight is $K_1 K_2^{2(n+n')}(n+n')!$. If they belong to two squares, it is $K_1 K_2^{2n} n! K_2^{2n'} n'!$. In both cases

$$K_2^{2(n+n')} \leq K_2^{8p'}, \quad n!n'! \leq (n+n')! \leq (4p')!,$$

after increasing K_2 to at least 1, as GJS do.

⟨2⟩ 4 Summing the at most $3^2 = 9$ ordered cell pairs and all coefficients gives

$$\|Q_V\|_{\text{GJS}} \leq 9K_1K_2^{8p'}(4p')!\lambda^2 \left(\sum_{n=0}^{2p'} |a_n| \right)^2 = M^2/C_8. \quad (4.2)$$

⟨2⟩ 5 For every $T > 0$,

$$\|V(h)\hat{\psi}_T\|^2 = \int V(h)^2 dq_{hT,g} \leq C_8 \|Q_V\|_{\text{GJS}} \leq M^2. \quad (4.3)$$

Because: Apply (2.2) at $t = 0$ with the two individually nonnegative functions $F_1 = F_2 = |V(h)|$. Their product is $V(h)^2 = Q_V$. Then apply CE1 and (4.2). This is the required correction to the manuscript's invalid phrase " $F_1 F_2 = V(h)^2 \geq 0$ ", which did not specify nonnegative F_1, F_2 .

⟨2⟩ 6 Pass (4.3) to Ω_g .

Because: Choose $T_k \rightarrow \infty$ such that $\|V(h)\hat{\psi}_{T_k}\|$ approaches its liminf. By (4.3), a subsequence converges weakly to some y . The graph of the closed linear operator $V(h)$ is a closed linear subspace of $\mathcal{H} \oplus \mathcal{H}$, hence is weakly closed. Since $\hat{\psi}_{T_k} \rightarrow \Omega_g$ strongly, (Ω_g, y) lies in the graph. Thus $y = V(h)\Omega_g$ and weak lower semicontinuity gives

$$\|V(h)\Omega_g\| \leq \liminf_k \|V(h)\hat{\psi}_{T_k}\| \leq M.$$

⟨2⟩ 7 This proves (M). □

⟨1⟩ 5 Obtain correct GJS norm bounds for polynomial time-zero vectors.

Let $Q = \prod_{i=1}^r \varphi_0(f_i)$ with real $f_i \in C_c^\infty(\mathbb{R})$, all supported in an interval I of length L . Let

$$N_I := \#\{j : |I \cap [j, j+1)| > 0\}.$$

⟨2⟩ 1 $N_I \leq \lfloor L \rfloor + 2 \leq L + 2$.

Because: If $j_{\min}$ and $j_{\max}$ are the extreme intersecting cells, then $j_{\max} - j_{\min} - 1 < L$, hence the number $j_{\max} - j_{\min} + 1$ is at most $\lfloor L \rfloor + 2$. The manuscript's bound $L + 1$ is false for nonintegral L ; an interval of length 0.2 can meet two cells.

⟨2⟩ 2 Decomposing each f_i by the unit cells yields

$$\|Q\|_{\text{GJS}} \leq K_1 K_2^{2r} r! N_I^{r/2} \prod_{i=1}^r \|f_i\|_2, \quad (5.1)$$

$$\|Q^2\|_{\text{GJS}} \leq K_1 K_2^{4r} (2r)! N_I^r \prod_{i=1}^r \|f_i\|_2^2 =: \mathbf{n}(Q)^2. \quad (5.2)$$

Because: Expand the kernel into cell assignments. For any assignment, the product of local factorials is at most the factorial of total degree, and the K_2 exponents sum to twice that degree. For each i ,

$$\sum_{j: I \cap [j, j+1) \neq \emptyset} \|f_i \mathbf{1}_{[j, j+1)}\|_2 \leq N_I^{1/2} \|f_i\|_2$$

by Cauchy–Schwarz. Multiplying these r estimates proves (5.1); applying it to the $2r$ factors in Q^2 proves (5.2).

⟨2⟩ 3 The norm is independent of the common sharp time s .

Because: At every s all factors occupy one temporal row. Changing s changes only that common row label; the spatial cell assignments, local degrees, kernel norms, and product of weights are unchanged. This is the precise statement; arbitrary time translation does not literally preserve the fixed lattice squares.

⟨2⟩ 4 This proves ⟨1⟩5. □

⟨1⟩ 6 Transfer CE1 and CE2 to connected cutoff-ground-state semigroup bounds.

⟨2⟩ 1 CE1 and (2.2) at $t = 0$ imply

$$\sup_T \|Q\hat{\psi}_T\|^2 \leq C_8 \mathfrak{n}(Q)^2. \quad (6.1)$$

Because: Apply the signed-truncation construction of ⟨2⟩3 to Q ; equivalently use $F_1 = F_2 = |Q|$. Then the Euclidean integrand is Q^2 , to which CE1 and (5.2) apply. The weak-graph argument of ⟨1⟩4 also gives $\Omega_g \in D(Q)$.

⟨2⟩ 2 For fixed $s \in \mathbb{R}$,

$$\int Q(s) dq_{h_{T,g}} \longrightarrow \langle \Omega_g, Q\Omega_g \rangle. \quad (6.2)$$

Because: For $T > |s|$, (2.3) gives

$$\int Q(s) dq_{h_{T,g}} = \langle \hat{\psi}_{T-s}, Q\psi_{T+s}/\|\psi_T\| \rangle \frac{\|\psi_{T-s}\|}{\|\psi_T\|}. \quad (6.3)$$

The first vector tends strongly to Ω_g , and the last factor tends to 1. Write the middle vector as

$$\frac{Q\psi_{T+s}}{\|\psi_T\|} = \frac{\|\psi_{T+s}\|}{\|\psi_T\|} Q\hat{\psi}_{T+s}. \quad (6.4)$$

If $s \geq 0$, the scalar ratio in (6.4) is at most 1. If $s < 0$, it is at most c_g^{-1} by (3.2). Thus (6.4) is bounded in either case. Every weak subsequential limit equals $Q\Omega_g$ by weak closedness of the graph and because the scalar ratio tends to 1. Hence the whole middle vector converges weakly to $Q\Omega_g$, proving (6.2). This separate treatment of $s < 0$ repairs the false inequality $\|\psi_{T+s}\| \leq \|\psi_T\|$ used for both signs in the manuscript.

⟨2⟩ 3 For $t \geq 0$,

$$\langle Q\Omega_g, e^{-tH(g)} Q\Omega_g \rangle \leq \liminf_{T \rightarrow \infty} \int Q(-t/2) Q(t/2) dq_{h_{T,g}}. \quad (6.5)$$

Because: With $u_T = Q\psi_{T-t/2}/\|\psi_T\|$, the signed-truncation extension of (2.2) gives the exact equality

$$\int Q(-t/2) Q(t/2) dq_{h_{T,g}} = \|e^{-tH(g)/2} u_T\|^2. \quad (6.6)$$

The ratio bound (3.2) and (6.1) make u_T bounded. Write $u_T = (\|\psi_{T-t/2}\|/\|\psi_T\|) Q\hat{\psi}_{T-t/2}$; the scalar tends to 1 and is uniformly bounded. From every subsequence choose a further one on which u_T converges weakly. Since $\hat{\psi}_{T-t/2} \rightarrow \Omega_g$ strongly, weak closedness of the graph of the closed operator Q identifies that weak limit as $Q\Omega_g$. Thus the full net converges weakly to $Q\Omega_g$. Boundedness of $e^{-tH/2}$ and weak lower semicontinuity of the norm prove (6.5).

⟨2⟩ 4 For $t > 2$, CE2 implies

$$\left| \int Q(-t/2) Q(t/2) dq_h - \int Q(-t/2) dq_h \int Q(t/2) dq_h \right| \leq C_7 e^{2m_1} e^{-m_1 t} \|Q\|_{\text{GJS}}^2. \quad (6.7)$$

Because: Write the source-prescribed finite localization decomposition as $Q = \sum_{a \in A} Q_a$, so that $\|Q\|_{\text{GJS}} = \sum_a \|Q_a\|_{\text{GJS}}$. The two temporal rows of every pair $Q_a(-t/2), Q_b(t/2)$ are separated by a horizontal strip of width at least $t - 2$. Therefore CE2 and the triangle inequality give

$$\begin{aligned} |\text{Cov}_{dq_h}(Q(-t/2), Q(t/2))| &\leq \sum_{a,b} |\text{Cov}_{dq_h}(Q_a(-t/2), Q_b(t/2))| \\ &\leq C_7 e^{-m_1(t-2)} \sum_{a,b} \|Q_a\|_{\text{GJS}} \|Q_b\|_{\text{GJS}} \\ &= C_7 e^{2m_1} e^{-m_1 t} \|Q\|_{\text{GJS}}^2. \end{aligned}$$

This decomposition is required because GJS Theorem 1.1.7 is stated for localized monomials; applying it directly to a general Q would be invalid.

$\langle 2 \rangle 5$ Let $\psi_Q = Q\Omega_g - \langle \Omega_g, Q\Omega_g \rangle \Omega_g$. Then

$$\langle \psi_Q, e^{-tH(g)} \psi_Q \rangle \leq \hat{C}_Q e^{-m_1 t} \quad (t \geq 0), \quad (6.8)$$

where

$$\hat{C}_Q = \max\{C_7 e^{2m_1} \|Q\|_{\text{GJS}}^2, e^{3m_1} \|\psi_Q\|^2\}.$$

Because: For $t > 2$, expand the connected expression, use (6.5), then (6.7), and use (6.2) for both one-point terms. Explicitly, if a_T is the connected Euclidean difference and b_T the product of one-point functions, then

$$\liminf_T (a_T + b_T) \leq \sup_T |a_T| + \lim_T b_T.$$

For $0 \leq t \leq 3$, positivity of $H(g)$ gives $\langle \psi_Q, e^{-tH(g)} \psi_Q \rangle \leq \|\psi_Q\|^2 \leq e^{3m_1} \|\psi_Q\|^2 e^{-m_1 t}$.

$\langle 2 \rangle 6$ This proves $\langle 1 \rangle 6$. □

$\langle 1 \rangle 7$ Prove polynomial vectors are dense.

Define $d\mu_g = |\Omega_g|^2 d\phi_0$. Since $\Omega_g > 0$ almost everywhere, multiplication by Ω_g is a unitary map from $L^2(d\mu_g)$ to $\mathcal{H}$. It therefore suffices to prove that cylinder polynomials are dense in $L^2(d\mu_g)$.

$\langle 2 \rangle 1$ Every real cylinder variable $X_f = \varphi_0(f)$ has a nonzero exponential-moment radius.

Because: Let f have support in an interval of length L , set $K' = K_2^2 N_I^{1/2} \|f\|_2$, and apply (6.1) to $Q = X_f^n$. Then

$$\mathbb{E}_{\mu_g}[X_f^{2n}] \leq C_8 K_1 (K')^{2n} (2n)!. \quad (7.1)$$

Since $e^{s|x|} \leq 2 \cosh(sx)$,

$$\mathbb{E}_{\mu_g} e^{s|X_f|} \leq 2 \max\{1, C_8 K_1\} \sum_{n=0}^{\infty} (sK')^{2n} < \infty \quad (0 \leq s < K'^{-1}). \quad (7.2)$$

The factor K_1 in (7.2), omitted in the manuscript, affects only the prefactor, not the radius.

$\langle 2 \rangle 2$ If $\zeta \in L^2(d\mu_g)$ is orthogonal to all cylinder polynomials, then its conditional expectation on every finite cylinder sigma-algebra is zero.

Because: Fix real $e_1, \dots, e_k$ and define

$$F(z_1, \dots, z_k) = \mathbb{E}_{\mu_g} \left[\bar{\zeta} \exp \left(\sum_{j=1}^k z_j X_{e_j} \right) \right].$$

By Cauchy–Schwarz, Hölder, and (7.2), F is analytic in a nonempty polystrip about $i\mathbb{R}^k$. Differentiation under the integral is justified by the same exponential dominator. Every derivative at zero

is an inner product with a polynomial, hence is zero. Repeated application of the one-variable identity theorem gives $F = 0$ throughout the connected polystrip, in particular on $i\mathbb{R}^k$. Thus the finite complex measure

$$B \mapsto \mathbb{E}_{\mu_g} [\bar{\zeta} \mathbf{1}_{\{(X_{e_1}, \dots, X_{e_k}) \in B\}}]$$

has zero Fourier transform. Uniqueness of Fourier transforms of finite measures gives $\mathbb{E}_{\mu_g} [\zeta \mid \sigma(X_{e_1}, \dots, X_{e_k})] = 0$.

⟨2⟩ 3 A countable increasing family of these finite sigma-algebras exhausts the cylinder sigma-algebra.

Because: Choose (e_j) dense in real $L^2(\mathbb{R})$. The free covariance $C = (2\sqrt{-\partial_x^2 + m_0^2})^{-1}$ is bounded, so

$$\mathbb{E}_{\phi_0} |X_f - X_e|^2 = \langle f - e, C(f - e) \rangle \leq \|C\| \|f - e\|_2^2.$$

For an approximating sequence, a subsequence converges ϕ_0 -almost surely and hence μ_g -almost surely because $\mu_g \ll \phi_0$. Therefore every X_f is measurable with respect to $\bigvee_k \sigma(X_{e_1}, \dots, X_{e_k})$ modulo null sets.

⟨2⟩ 4 The martingale convergence theorem now gives $\zeta = 0$.

Because: The conditional expectations in ⟨2⟩2 are all zero and converge in L^2 to the conditional expectation on the exhausted cylinder sigma-algebra, which is ζ itself. Hence cylinder polynomials, and therefore $\text{span}\{Q\Omega_g\}$, are dense.

⟨2⟩ 5 This proves ⟨1⟩7. □

⟨1⟩ 8 Deduce the physical Hamiltonian gap (G).

Fix $0 \leq \beta < m_1$ and let $P_\beta = \mathbf{1}_{[0, \beta]}(H(g))$.

⟨2⟩ 1 For every centered polynomial vector ψ_Q , $P_\beta \psi_Q = 0$.

Because: The spectral theorem and (6.8) give, for every $t \geq 0$,

$$e^{-\beta t} \|P_\beta \psi_Q\|^2 \leq \langle \psi_Q, e^{-tH(g)} \psi_Q \rangle \leq \hat{C}_Q e^{-m_1 t}.$$

Multiply by $e^{\beta t}$ and let $t \rightarrow \infty$. Since $m_1 - \beta > 0$, $\|P_\beta \psi_Q\| = 0$.

⟨2⟩ 2 $P_\beta \Omega_g = \Omega_g$.

Because: $H(g)\Omega_g = 0$ and $0 \in [0, \beta]$.

⟨2⟩ 3 $P_\beta = |\Omega_g\rangle\langle\Omega_g|$.

Because: By ⟨1⟩7, the complex span of Ω_g and the centered polynomial vectors is dense. Steps ⟨2⟩1–⟨2⟩2 show that P_β maps this dense subspace into $\mathbb{C}\Omega_g$. Continuity of the bounded projection extends this containment to all of $\mathcal{H}$, while $P_\beta \Omega_g = \Omega_g$ shows its range is exactly that line.

⟨2⟩ 4 There is no spectrum in $(0, m_1)$.

Because: If an open interval $J \subseteq (0, m_1)$ had a nonzero spectral projection, choose $\beta < m_1$ above its right endpoint. Then $\mathbf{1}_J(H(g)) \leq P_\beta - |\Omega_g\rangle\langle\Omega_g| = 0$, a contradiction. Thus $\text{spec } H(g) \subseteq \{0\} \cup [m_1, \infty)$; simplicity was imported in Assumption 1.1.

⟨2⟩ 5 This proves (G), hence Theorem 2.1. □

3 Audit map back to the manuscript

Location	Defect	Why the written step fails	Exact correction above
lines 305, 536	wrong source	Simon III.12 is a projection-product estimate, not FKN.	Assumption 1.2; use Simon III.6 and V.14–V.15 or GRS II.14, II.16.
line 352	positivity hypothesis unstated	The slab lemma requires each $F_i \geq 0$, not merely $F_1 F_2 \geq 0$.	$\langle 1 \rangle 4 \langle 2 \rangle 5$: $F_1 = F_2 = V(h) $.
lines 376–377	false cell count	A length- L interval can meet more than $L + 1$ unit cells.	$\langle 1 \rangle 5$: use $N_I \leq \lfloor L \rfloor + 2 \leq L + 2$.
lines 404–420	incomplete unbounded passage	Signed observables are outside the stated nonnegative FKN lemma.	$\langle 1 \rangle 2 \langle 2 \rangle 2 - \langle 2 \rangle 3$ and $\langle 1 \rangle 6$: signed truncations plus an explicit L^2 dominator.
line 412	false for $s < 0$	$\ \psi_{T+s}\ \leq \ \psi_T\ $ holds only for $s \geq 0$.	$\langle 1 \rangle 6 \langle 2 \rangle 2$: split the two signs and use the bound c_g^{-1} .
line 455	missing factor	$C_8 K_1$ cannot disappear from the moment series.	$\langle 1 \rangle 7 \langle 2 \rangle 1$: keep it outside the geometric series.
line 484	missing range restriction	$P_{[0,\beta]} \Omega_g = \Omega_g$ is false for $\beta < 0$.	$\langle 1 \rangle 8$: state $0 \leq \beta < m_1$.
standing assumption	scaling suppressed	The source smallness conditions are dimensional until scaling is written.	$\langle 1 \rangle 1$: exact dilation and definition of $\varepsilon_{\mathcal{P}}$.

4 Logical conclusion

The argument establishes (M) and (G) for every cutoff g , with constants independent of g , conditional on the enumerated construction facts and the GJS estimates. None of the identified defects forces a change in the claimed gap m_1 or in the qualitative local-moment conclusion; each is repaired explicitly above. The proof does not establish the GJS estimates, cutoff-Hamiltonian self-adjointness, ground-state simplicity, or FKN; those remain imported theorems and must not be hidden by the phrase “conditional only on GJS Theorems 1.1.7–1.1.8.”

Corrected structured proof II

Uniform gap and local moment $\implies$ local-norm convergence and the main theorem

Fresh reconstruction of `part_i/main.tex`

16 August 2026

Status. CONDITIONAL. Given the explicitly listed inputs in Section 1, the full cutoff net is Cauchy in the norm of every local dual, with a super-polynomial rate. The resulting state is locally normal, translation invariant, parity invariant for even interaction, and a spectral ground state.

The derivation below is independent of Euclidean uniqueness. The current Part I document correctly states its two-phase exclusion only conditionally on the load-bearing phase-decomposition hypothesis B4; the final section records that scope.

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1 Exact inputs

For a spatial cutoff $g \in \mathcal{G}$, let

$$K(g) = H_0 + V(g), \quad E(g) = \inf \text{spec } K(g), \quad H(g) = K(g) - E(g), \quad P_g = |\Omega_g\rangle\langle\Omega_g|,$$

and let $\omega_g(A) = \langle\Omega_g, A\Omega_g\rangle$ and $\beta_t^g = \text{Ad } e^{itH(g)}$. Let $\mathfrak{A}(O)$ be the time-zero local von Neumann algebra of a bounded interval O and let O_r be its open r -neighbourhood.

Imported assumption 1.1 (FKN, stability, and free hypercontractivity (F)). For every affine cutoff path $g_s = (1-s)g + sg'$ and every $T > 0$, the interacting FKN formula (3.1) below holds. Its finite-slab interaction random variables satisfy $\mathcal{U}_v \in L^p$ for every finite p and $e^{-a\mathcal{U}_g} \in L^1$ for every finite $a > 0$. The free time-zero Markov semigroup satisfies Nelson hypercontractivity with $(q-1)/(p-1) = e^{2m_0T}$. These are the precise standard inputs used to prove path regularity; they are not consequences of (G) or (M).

Imported assumption 1.2 (uniform physical gap (G)). There is $\gamma > 0$, independent of g , such that

$$\text{spec } H(g) \subseteq \{0\} \cup [\gamma, \infty), \quad \dim \ker H(g) = 1. \quad (\text{G})$$

Imported assumption 1.3 (uniform local interaction moment (M)). There is $M < \infty$ such that

$$\|V(h)\Omega_g\| \leq M \quad (\text{M})$$

for every g and every real $h \in C_c^\infty(\mathbb{R})$ satisfying $\|h\|_\infty \leq 1$ and whose support is contained in an interval of length 2.

Imported assumption 1.4 (local domains and affiliation). For every compactly supported real h , $\Omega_g \in D(V(h))$, and $V(h)$ is self-adjoint and affiliated with $\mathfrak{A}(U)$ whenever U contains $\text{supp } h$. Along every affine path there is a common core $\mathcal{C}$ for the K_s such that $K_s \xi = K_{s_0} \xi + (s - s_0)V(g' - g)\xi$ for $\xi \in \mathcal{C}$. The correct model-specific smoothing source is Glimm–Jaffe Expositions, Theorem 7.4. Theorem 7.3 alone assumes that the multiplication potential is bounded below and therefore does not, by itself, justify this application.

Imported assumption 1.5 (finite propagation and patching). For every cutoff,

$$\beta_t^g(\mathfrak{A}(O)) \subseteq \mathfrak{A}(O_{|t|}). \quad (\text{FP})$$

There is a cutoff-independent quasi-local dynamics α such that

$$g = 1 \text{ on } O_{|t|} \implies \alpha_t(A) = \beta_t^g(A) \quad (A \in \mathfrak{A}(O)). \quad (\text{P})$$

Imported assumption 1.6 (covariance). If $(\tau_a g)(x) = g(x - a)$, then

$$\omega_{\tau_a g} = \omega_g \circ \tau_{-a}. \quad (\text{T})$$

For even $\mathcal{P}$, each ω_g is invariant under $\theta = \text{Ad } \Gamma(-1)$.

The first corrected proof document derives Assumptions 1.2 and 1.3 from GJS in the weak-coupling region. Assumptions 1.1 and 1.4–1.6 are standard but logically independent cutoff-construction inputs.

2 Statement to be proved

For a cutoff g and interval O , write

$$d(g, O) = \text{dist}(O, \mathbb{R} \setminus \{g = 1\}^\circ).$$

Theorem 2.1 (local-norm convergence and spectral ground state). *Under Assumptions 1.1–1.6, there is a state ω_∞ on the quasi-local algebra $\mathfrak{A}$ such that, for every bounded interval O ,*

$$\lim_{g \in \mathcal{G}} \|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| = 0. \quad (2.1)$$

If $g = 1$ on O_d , then

$$\|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| \leq \Psi_\gamma(d), \quad (2.2)$$

where $\Psi_\gamma(d) = O_N((1 + d)^{-N})$ for every N . The limit is locally normal, is a spectral ground state for $(\mathfrak{A}, \alpha)$, is translation invariant, and, for even $\mathcal{P}$, is θ -invariant.

3 Structured proof

⟨1⟩ 1 Prove differentiability of the ground-state projection along an affine cutoff path.

Let g, g' equal 1 on a common nonempty interval, put

$$g_s = (1-s)g + sg', \quad v = g' - g, \quad K_s = K(g_s), \quad H_s = K_s - E_s, P_s = |\Omega_s\rangle\langle\Omega_s|, \quad Q_s = 1 - P_s.$$

⟨2⟩ 1 The family $s \mapsto e^{-TK_s}$ is norm-continuous on $[0, 1]$ and norm- C^1 on $(0, 1)$ for each $T > 0$.

Because: On free Euclidean path space define

$$\mathcal{U}_s = \lambda \int_0^T \int g_s(x) : \mathcal{P}(\Phi(t, x)) : dx dt, \quad w_s = e^{-\mathcal{U}_s}, \quad \mathcal{U}_v = \mathcal{U}_1 - \mathcal{U}_0.$$

For bounded time-zero f, h , the interacting FKN formula is

$$\langle f, e^{-TK_s} h \rangle = \int \overline{J_T f} J_0 h w_s d\mu_0. \quad (3.1)$$

The correct references are Simon Corollary V.14/Theorem V.15 or GRS Theorem II.16, not Simon Theorem III.12.

Set $a = e^{-m_0 T}$ and

$$q_T = \frac{2}{1-a}, \quad r_T = \frac{2}{1+a}, \quad p_T = 1+a, \quad p'_T = 1+a^{-1}.$$

Then $1/r_T + 1/q_T = 1$, $r_T p_T = 2$, and $(p'_T - 1)/(p_T - 1) = e^{2m_0 T}$. Nelson hypercontractivity—Simon Theorem I.17, or its time-region version Theorem III.17—therefore gives

$$\|P_T F\|_{p'_T} \leq \|F\|_{p_T}.$$

The Markov property and Hölder yield

$$\begin{aligned} \|(J_T f)(J_0 h)\|_{r_T}^{r_T} &= \int |h|^{r_T} P_T(|f|^{r_T}) d\nu_0 \\ &\leq \| |h|^{r_T} \|_{p_T} \|P_T(|f|^{r_T})\|_{p'_T} \\ &\leq \|h\|_2^{r_T} \|f\|_2^{r_T}. \end{aligned}$$

Thus

$$\|(J_T f)(J_0 h)\|_{r_T} \leq \|f\|_2 \|h\|_2. \quad (3.2)$$

Finite-volume stability gives $\mathcal{U}_v \in L^p(\mu_0)$ for every finite p and $e^{-a\mathcal{U}_i} \in L^1(\mu_0)$ for every $a < \infty$, $i = 0, 1$ (Simon Theorem V.7). Since $\mathcal{U}_s = (1-s)\mathcal{U}_0 + s\mathcal{U}_1$, Hölder gives

$$\|w_s\|_a^a \leq \left(\int e^{-a\mathcal{U}_0} d\mu_0 \right)^{1-s} \left(\int e^{-a\mathcal{U}_1} d\mu_0 \right)^s. \quad (3.3)$$

Convergence in probability together with (3.3) at a larger exponent gives $w_{s_n} \rightarrow w_s$ in L^a by Vitali. Moreover

$$\frac{w_{s+h} - w_s}{h} = -\mathcal{U}_v \int_0^1 w_{s+\vartheta h} d\vartheta. \quad (3.4)$$

Choose finite p, a with $q_T^{-1} = p^{-1} + a^{-1}$. Hölder and the just-proved L^a continuity show that (3.4) converges in L^{q_T} to $-\mathcal{U}_v w_s$ for $0 < s < 1$. Finally (3.1), (3.2), and duality give

$$\|e^{-TK_s} - e^{-TK_r}\| \leq \|w_s - w_r\|_{q_T},$$

and the same estimate for difference quotients. This proves the asserted norm regularity.

$\langle 2 \rangle 2$ $s \mapsto P_s$ is norm-continuous on $[0, 1]$ and C^1 on $(0, 1)$.

Because: $L_s = e^{-TK_s}$ has the simple largest eigenvalue $\lambda_s = e^{-TE_s} = \|L_s\|$. By (G),

$$\text{spec } L_s \setminus \{\lambda_s\} \subset [0, e^{-T\gamma}\lambda_s].$$

The relative spectral separation $1 - e^{-T\gamma}$ is uniform. A Riesz contour around λ_s , together with the norm regularity from $\langle 2 \rangle 1$, gives exactly the stated regularity of P_s .

$\langle 2 \rangle 3$ Locally on $(0, 1)$ choose normalized C^1 vectors in the parallel gauge $\langle \Omega_s, \dot{\Omega}_s \rangle = 0$. Then

$$\dot{E}_s = \langle \Omega_s, V(v)\Omega_s \rangle, \quad \dot{\Omega}_s = -H_s^{-1}Q_s V(v)\Omega_s. \quad (3.5)$$

Because: Let $\mathcal{C}$ be the common core supplied by cutoff construction. For $\xi \in \mathcal{C}$, $K_s \xi = K_{s_0} \xi + (s - s_0)V(v)\xi$. Differentiate the weak eigenvalue identity $\langle K_s \xi, \Omega_s \rangle = E_s \langle \xi, \Omega_s \rangle$ to obtain

$$\langle H_s \xi, \dot{\Omega}_s \rangle = \langle \xi, (\dot{E}_s - V(v))\Omega_s \rangle. \quad (3.6)$$

Assumption 1.4 makes the vector on the right well defined. Since $\mathcal{C}$ is a core for the self-adjoint H_s , (3.6) says $\dot{\Omega}_s \in D(H_s)$ and $H_s \dot{\Omega}_s = (\dot{E}_s - V(v))\Omega_s$. Pairing with Ω_s proves the first formula in (3.5). The parallel gauge puts $\dot{\Omega}_s$ in $Q_s \mathcal{H}$; (G) makes $H_s^{-1}Q_s$ bounded by γ^{-1} , proving the second formula.

$\langle 2 \rangle 4$ This proves $\langle 1 \rangle 1$. □

$\langle 1 \rangle 2$ Construct an exact inverse-energy filter with super-polynomial time tail.

$\langle 2 \rangle 1$ Choose even $\chi \in C^\infty(\mathbb{R}; [0, 1])$ with $\chi = 0$ on $[-\gamma/2, \gamma/2]$ and $\chi = 1$ outside $(-\gamma, \gamma)$, and set

$$\widehat{W}_\gamma(E) = \begin{cases} \chi(E)/E, & E \neq 0, \\ 0, & E = 0. \end{cases} \quad (3.7)$$

$\langle 2 \rangle 2$ Its inverse transform $W_\gamma(t) = (2\pi)^{-1} \int e^{-itE} \widehat{W}_\gamma(E) dE$ is an L^1 function satisfying

$$\int (1 + |t|)^q |W_\gamma(t)| dt < \infty \quad (q \geq 0). \quad (3.8)$$

Because: Oddness gives

$$W_\gamma(t) = -\frac{i}{\pi} \int_0^\infty \frac{\sin(tE)\chi(E)}{E} dE.$$

For $0 < |t| \leq 1$, the compact transition part is absolutely bounded and the tail is uniformly bounded after $u = |t|E$ by Dirichlet's test. For $|t| \geq 1$, every derivative $\widehat{W}_\gamma^{(k)}$ with $k \geq 1$ is integrable, so k integrations by parts give

$$|W_\gamma(t)| \leq \frac{\|\widehat{W}_\gamma^{(k)}\|_1}{2\pi|t|^k}.$$

Choose $k > q + 1$. This proves (3.8), including integrability near zero.

$\langle 2 \rangle 3$ If H satisfies (G), then

$$H^{-1}(1 - P_{\ker H}) = \int_{\mathbb{R}} W_\gamma(t) e^{itH} (1 - P_{\ker H}) dt. \quad (3.9)$$

Because: Fourier inversion gives $\widehat{W}_\gamma(E) = E^{-1}$ on $[\gamma, \infty)$ and zero at $E = 0$. The spectral theorem gives (3.9); it is a norm-convergent Bochner integral because $W_\gamma \in L^1$.

Define

$$T_\gamma(r) = \int_{|t|>r} |W_\gamma(t)| dt.$$

From (3.8), Markov's inequality gives $T_\gamma(r) \leq C_{q,\gamma}(1+r)^{-q}$ for every q .

⟨2⟩ 4 This proves ⟨1⟩2. □

⟨1⟩ 3 Prove the spectral-separation estimate used for clustering.

Suppose

$$F(t) = \int_{[\gamma, \infty)} e^{itE} d\rho(E), \quad G(t) = \int_{(-\infty, -\gamma]} e^{itE} d\sigma(E),$$

with $|\rho|(\mathbb{R}), |\sigma|(\mathbb{R}) \leq M_0$, and $F = G$ on $(-R, R)$, where $\gamma R \geq 1$.

⟨2⟩ 1 For $a, \varepsilon > 0$ define

$$f_{a,\varepsilon}(t) = \frac{1}{2\pi i} \frac{e^{-t^2/(2a)}}{t - i\varepsilon}.$$

Its transform is

$$\check{f}_{a,\varepsilon}(E) = \int_0^\infty \sqrt{\frac{a}{2\pi}} e^{-a(E-E')^2/2} e^{-\varepsilon E'} dE', \quad (3.10)$$

which lies in $[0, 1]$ and tends to the standard normal distribution function $\Phi(\sqrt{a}E)$ as $\varepsilon \downarrow 0$.

Because: The transform of $(2\pi i)^{-1}(t - i\varepsilon)^{-1}$ is $\mathbf{1}_{(0,\infty)}(E)e^{-\varepsilon E}$ by a one-pole contour integral. Multiplication by the Gaussian in t convolves this function with the normalized Gaussian displayed in (3.10).

⟨2⟩ 2 The part of $f_{a,\varepsilon}$ outside $(-R, R)$ satisfies

$$\int_{|t|\geq R} |f_{a,\varepsilon}(t)| dt \leq \frac{a}{\pi R^2} e^{-R^2/(2a)}. \quad (3.11)$$

Because: $|t - i\varepsilon| \geq |t|$ and, for $t \geq R$, $t^{-1} \leq t/R^2$. Integrating $te^{-t^2/(2a)}$ exactly proves (3.11).

⟨2⟩ 3 Using $F = G$ on $(-R, R)$ and Fubini gives

$$\left| \int \Phi(\sqrt{a}E) d\rho(E) - \int \Phi(\sqrt{a}E) d\sigma(E) \right| \leq \frac{2M_0a}{\pi R^2} e^{-R^2/(2a)}. \quad (3.12)$$

Because: Before $\varepsilon \downarrow 0$, the difference of the two time integrals is supported in $|t| \geq R$ and is at most $(|\rho|(\mathbb{R}) + |\sigma|(\mathbb{R}))$ times (3.11). Dominated convergence in energy proves (3.12).

⟨2⟩ 4 Therefore

$$|F(0)| \leq 2M_0 e^{-\gamma R/2}. \quad (3.13)$$

Because: Insert and subtract the two integrals in (3.12). On the supports, $1 - \Phi(\sqrt{a}E) \leq \frac{1}{2}e^{-a\gamma^2/2}$ and $\Phi(\sqrt{a}E) \leq \frac{1}{2}e^{-a\gamma^2/2}$, respectively. Hence

$$|F(0)| \leq M_0 e^{-a\gamma^2/2} + \frac{2M_0a}{\pi R^2} e^{-R^2/(2a)}.$$

Set $a = R/\gamma$. Both exponentials become $e^{-\gamma R/2}$ and $2a/(\pi R^2) = 2/(\pi\gamma R) \leq 2/\pi < 1$, which proves (3.13).

⟨2⟩ 5 This proves ⟨1⟩3. □

⟨1⟩ 4 Derive clustering with one unbounded local observable.

Let H, Ω satisfy (G), let $X = X^*$ be affiliated with every $\mathfrak{A}(U)$ containing a compact set B , assume $\Omega \in D(X)$, and let $A \in \mathfrak{A}(O)$. Put $R = \text{dist}(B, O) > 0$ and suppose $\gamma R \geq 1$.

⟨2⟩ 1 Define

$$\begin{aligned} F(t) &= \langle X\Omega, (e^{itH} - P_0)A\Omega \rangle, \\ G(t) &= \langle A^*\Omega, (e^{-itH} - P_0)X\Omega \rangle, \quad P_0 = |\Omega\rangle\langle\Omega|. \end{aligned}$$

Then F and G have spectral measures supported in $[\gamma, \infty)$ and $(-\infty, -\gamma]$, with variations at most $M_0 = \|X\Omega\|\|A\|$.

Because: Insert the spectral resolution of H on $(1 - P_0)\mathcal{H}$ and use Cauchy–Schwarz for the variation of the resulting complex measures.

⟨2⟩ 2 $F(t) = G(t)$ for $|t| < R$.

Because: By finite propagation, $\beta_t(A) \in \mathfrak{A}(O_{|t|})$. Choose a bounded open $U \supset B$ disjoint from $O_{|t|}$. Affiliation implies that $\beta_t(A) \in \mathfrak{A}(U)'$ preserves $D(X)$ and commutes with X there. Since $e^{-itH}\Omega = \Omega$,

$$\begin{aligned} \langle X\Omega, e^{itH}A\Omega \rangle &= \langle \Omega, X\beta_t(A)\Omega \rangle = \langle \Omega, \beta_t(A)X\Omega \rangle \\ &= \langle A^*\Omega, e^{-itH}X\Omega \rangle. \end{aligned}$$

The P_0 terms agree because $\langle \Omega, X\Omega \rangle$ is real.

⟨2⟩ 3 Apply (3.13) to obtain

$$|\langle X\Omega, A\Omega \rangle - \langle X\Omega, \Omega \rangle \langle \Omega, A\Omega \rangle| \leq 2\|X\Omega\|\|A\|e^{-\gamma R/2}. \quad (3.14)$$

⟨2⟩ 4 This proves ⟨1⟩4. □

⟨1⟩ 5 Differentiate the state and localize the perturbation.

⟨2⟩ 1 For self-adjoint $A \in B(\mathcal{H})$ and $0 < s < 1$,

$$\frac{d}{ds}\omega_{g_s}(A) = -2 \operatorname{Re} \int_{\mathbb{R}} W_\gamma(t) [\langle V(v)\Omega_s, \beta_t^{g_s}(A)\Omega_s \rangle - \langle V(v)\Omega_s, \Omega_s \rangle \omega_{g_s}(A)] dt. \quad (3.15)$$

Because: Differentiate the vector state and insert (3.5):

$$\frac{d}{ds}\omega_{g_s}(A) = -2 \operatorname{Re} \langle V(v)\Omega_s, H_s^{-1}Q_s A\Omega_s \rangle.$$

Insert the exact filter (3.9). Since $e^{itH_s}Q_s A\Omega_s = \beta_t^{g_s}(A)\Omega_s - \omega_{g_s}(A)\Omega_s$, this is (3.15). The integral is absolutely convergent because $W_\gamma \in L^1$ and $V(v)\Omega_s$ exists; v has compact support.

⟨2⟩ 2 Choose a smooth partition of unity $\sum_j \eta_j = 1$ with $0 \leq \eta_j \leq 1$ and $\operatorname{supp} \eta_j \subset (j-1, j+1)$. Put $h_j = \eta_j v$, $V_j = V(h_j)$. Then

$$V(v)\xi = \sum_j V_j \xi, \quad \|V_j \Omega_s\| \leq M. \quad (3.16)$$

Because: Only finitely many h_j are nonzero. Also $|v| \leq 1$, so $\|h_j\|_\infty \leq 1$, and each support has length at most 2. Apply (M) to the cutoff g_s .

⟨2⟩ 3 This proves ⟨1⟩5. □

⟨1⟩ 6 Prove the one-path local-perturbations estimate.

Assume $g = g' = 1$ on O_d , where

$$d \geq d_0 := 4 + 2/\gamma.$$

For a nonzero h_j set $B_j = \text{supp } h_j$ and $r_j = \text{dist}(B_j, O)$.

⟨2⟩ 1 $r_j \geq d$ for every nonzero h_j .

Because: On O_d , $v = g' - g = 0$. Therefore the support of $h_j = \eta_j v$ is contained in $\mathbb{R} \setminus O_d$.

⟨2⟩ 2 For self-adjoint $A \in \mathfrak{A}(O)$, $\|A\| \leq 1$, define

$$C_{j,s}(t) = \langle V_j \Omega_s, \beta_t^{g_s}(A) \Omega_s \rangle - \langle V_j \Omega_s, \Omega_s \rangle \omega_{g_s}(A).$$

Then

$$|C_{j,s}(t)| \leq \begin{cases} 2M e^{-\gamma r_j/4}, & |t| \leq r_j/2, \\ 2M, & t \in \mathbb{R}. \end{cases} \quad (3.17)$$

Because: In the first case, finite propagation places $\beta_t^{g_s}(A)$ in $\mathfrak{A}(O_{|t|})$ and $\text{dist}(B_j, O_{|t|}) \geq r_j/2$. Moreover $\gamma r_j/2 \geq 1$. Apply (3.14) with $X = V_j$, then (3.16). For arbitrary t , bound each of the two terms defining $C_{j,s}(t)$ by M .

⟨2⟩ 3 Equations (3.15)–(3.17) imply

$$\left| \frac{d}{ds} \omega_{g_s}(A) \right| \leq 4M \sum_j \left(\|W_\gamma\|_1 e^{-\gamma r_j/4} + T_\gamma(r_j/2) \right). \quad (3.18)$$

Because: Split the t -integral for each j at $|t| = r_j/2$. The outside factor 2 in (3.15), the factor $2M$ in either line of (3.17), and the two integral regions give exactly the coefficient $4M$ in (3.18).

⟨2⟩ 4 At most eight indices j have $r_j \in [n, n+1)$ for any integer $n \geq \lfloor d \rfloor$.

Because: Write $O = (a, b)$. Since $d \geq 4$ and $B_j \subset [j-1, j+1]$, a nonempty B_j lies wholly on one side of O_d . On the right, compactness gives $x_j \in B_j$ with $x_j - b = r_j$. Since $|x_j - j| \leq 1$ and $r_j \in [n, n+1)$,

$$j \in [b + n - 1, b + n + 2],$$

which contains at most four integers. The left side gives four more.

⟨2⟩ 5 Integrating (3.18) along the path gives

$$\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\| \leq 32M \sum_{n \geq \lfloor d \rfloor} \left(\|W_\gamma\|_1 e^{-\gamma n/4} + T_\gamma(n/2) \right). \quad (3.19)$$

Because: Both functions of r_j in (3.18) decrease with r_j . Apply the shell count from ⟨2⟩4, producing $4M \cdot 8 = 32M$. Integrate first on $[\varepsilon, 1 - \varepsilon]$, where differentiability holds, and then let $\varepsilon \downarrow 0$ using the endpoint norm continuity of P_s proved in ⟨1⟩1. Taking the supremum over self-adjoint contractions gives the norm of the hermitian functional $\omega_g - \omega_{g'}$.

⟨2⟩ 6 This proves ⟨1⟩6. □

⟨1⟩ 7 Define the rate and prove that it is super-polynomial.

For $d \geq d_0$ set

$$\Psi_\gamma(d) = 32M \sum_{n \geq \lfloor d \rfloor} \left(\|W_\gamma\|_1 e^{-\gamma n/4} + T_\gamma(n/2) \right), \quad (3.20)$$

and for $d < d_0$ enlarge it to at least 2.

⟨2⟩ 1 For every $N \geq 1$, there is $C_{N,\gamma,M}$ such that

$$\Psi_\gamma(d) \leq C_{N,\gamma,M}(1+d)^{-N}. \quad (3.21)$$

Because: The first series has an exponential tail. For the second choose $q = N + 2$ in $T_\gamma(r) \leq C_{q,\gamma}(1+r)^{-q}$; then $\sum_{n \geq [d]} (1+n)^{-N-2} = O((1+d)^{-N-1})$.

⟨2⟩ 2 This proves ⟨1⟩7. □

⟨1⟩ 8 Prove the full cutoff net is locally norm Cauchy and construct ω_∞ .

⟨2⟩ 1 Fix O and $\varepsilon > 0$. Choose $d \geq d_0$ with $\Psi_\gamma(d) < \varepsilon$, and choose $g_0 \in \mathcal{G}$ with $g_0 = 1$ on O_d .

⟨2⟩ 2 If $g, g' \succcurlyeq g_0$, then

$$\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\| < \varepsilon. \quad (3.22)$$

Because: The directed order is inclusion of plateau interiors, so both cutoffs equal 1 on O_d . Apply (3.19).

⟨2⟩ 3 The limit functionals on local algebras are compatible and extend uniquely to a state ω_∞ on the norm-dense quasi-local union.

Because: Each $\mathfrak{A}(O)^*$ is complete, so (3.22) gives a limit on $\mathfrak{A}(O)$. If $O_1 \subset O_2$, uniqueness of the norm limit shows that the O_2 limit restricts to the O_1 limit. Positivity and normalization pass to the limit. A compatible bounded functional of norm one on the algebraic union extends uniquely by norm continuity to $\mathfrak{A}$.

⟨2⟩ 4 If one fixed g equals 1 on O_d , then (2.2) holds.

Because: The subset of cutoffs g' equal to 1 on O_d is cofinal. Apply (3.19) to (g, g') and let g' tend through this subset to the local norm limit.

⟨2⟩ 5 This proves local-norm convergence and its rate. □

⟨1⟩ 9 Prove local normality.

Fix O and let $0 \leq A_i \uparrow A$ in $\mathfrak{A}(O)$. For any cutoff g ,

$$0 \leq \omega_\infty(A) - \omega_\infty(A_i) \leq 2\delta\|A\| + \omega_g(A - A_i) \quad (3.23)$$

whenever $\|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| < \delta$.

Because: Add and subtract $\omega_g(A)$ and $\omega_g(A_i)$, use $\|A_i\| \leq \|A\|$, and bound each state difference by the local dual norm. The cutoff vector state is normal on $\mathfrak{A}(O)$, so its last term in (3.23) tends to zero with i . First let i increase, then let $\delta \downarrow 0$. Thus ω_∞ is normal on $\mathfrak{A}(O)$.

⟨1⟩ 10 Prove the spectral ground-state condition.

Choose cutoffs g_n with $g_n = 1$ on $(-n, n)$. For local $A, B \in \mathfrak{A}(O)$ define the finite complex measure

$$\rho_n(S) = \langle A^* \Omega_{g_n}, \mathbf{1}_S(H(g_n)) B \Omega_{g_n} \rangle, \quad G_n(t) = \int_{[0,\infty)} e^{itE} d\rho_n(E). \quad (3.24)$$

⟨2⟩ 1 For $|t| < d(g_n, O)$,

$$\omega_{g_n}(A\alpha_t(B)) = G_n(t). \quad (3.25)$$

Because: Patching gives $\alpha_t(B) = \beta_t^{g_n}(B)$, and $e^{-itH(g_n)}\Omega_{g_n} = \Omega_{g_n}$. Moving the unitary to the middle gives (3.25).

$\langle 2 \rangle 2$ If $f \in L^1(\mathbb{R})$ and $\check{f}(E) = \int f(t)e^{itE}dt$ vanishes on $[0, \infty)$, then

$$\int_{\mathbb{R}} f(t)G_n(t)dt = 0. \quad (3.26)$$

Because: Fubini is justified by $|\rho_n|(\mathbb{R}) \leq \|A\|\|B\|$, and the spectral support in (3.24) is nonnegative.

$\langle 2 \rangle 3$ The limit function $G(t) = \omega_\infty(A\alpha_t(B))$ is continuous and satisfies $\int fG = 0$.

Because: For each T , eventually $d(g_n, O) > T$, and $A\alpha_t(B) \in \mathfrak{A}(O_T)$ for $|t| \leq T$. Hence

$$\sup_{|t| \leq T} |G(t) - G_n(t)| \leq \|A\|\|B\| \|(\omega_\infty - \omega_{g_n})|_{\mathfrak{A}(O_T)}\| \longrightarrow 0. \quad (3.27)$$

Thus G is a locally uniform limit of continuous functions. Let $M_0 = \|A\|\|B\|$ and $\epsilon_T = \int_{|t| > T} |f(t)|dt$. Using (3.26),

$$\left| \int fG \right| \leq \|f\|_1 M_0 \|(\omega_\infty - \omega_{g_n})|_{\mathfrak{A}(O_T)}\| + 2M_0 \epsilon_T.$$

First let $n \rightarrow \infty$, then $T \rightarrow \infty$.

$\langle 2 \rangle 4$ $\omega_\infty \circ \alpha_t = \omega_\infty$.

Because: For local A and fixed t , eventually patching gives $\omega_{g_n}(\alpha_t(A)) = \omega_{g_n}(A)$. Both observables lie in a fixed local algebra, so local norm convergence passes the identity to the limit. Density extends it to $\mathfrak{A}$.

$\langle 2 \rangle 5$ Steps $\langle 2 \rangle 3$ – $\langle 2 \rangle 4$ are exactly the spectral ground-state condition. □

$\langle 1 \rangle 11$ Prove spatial and parity invariance.

$\langle 2 \rangle 1$ $\omega_\infty \circ \tau_a = \omega_\infty$ for every $a \in \mathbb{R}$.

Because: $g \mapsto \tau_a g$ is an order isomorphism of the cutoff net and hence is cofinal. For local A , covariance (T) and local norm convergence give

$$\omega_\infty(A) = \lim_g \omega_{\tau_a g}(A) = \lim_g \omega_g(\tau_{-a}A) = \omega_\infty(\tau_{-a}A).$$

Extend by norm density.

$\langle 2 \rangle 2$ For even $\mathscr{P}$, $\omega_\infty \circ \theta = \omega_\infty$.

Because: Every cutoff state has this invariance by Assumption 1.6; pass to the local norm limit, then use density.

$\langle 2 \rangle 3$ Together with $\langle 1 \rangle 8$ – $\langle 1 \rangle 10$, this proves Theorem 2.1. □

4 The averaged Glimm–Jaffe vacuum

The manuscript’s canonical-vacuum corollary does follow. Let $g_n(x) = g(x/n)$ with $g = 1$ on $[-3, 3]$, and let $h \geq 0$, $\text{supp } h \subset [-1, 1]$, $\int h = 1$. The averaged state of Glimm–Jaffe III has the form

$$\omega_n(A) = \int_{-1}^1 h(u)\omega_{\tau_{-enu}g_n}(A)du, \quad \epsilon \in \{\pm 1\}.$$

For $O \subset [-R, R]$, every shifted cutoff in the integral equals 1 on an interval containing $[-2n, 2n]$, so

$$d(\tau_{-\epsilon nu} g_n, O) \geq 2n - R$$

uniformly in u . Therefore

$$\|(\omega_n - \omega_\infty)|_{\mathfrak{A}(O)}\| \leq \int_{-1}^1 h(u) \Psi_\gamma(2n - R) du = \Psi_\gamma(2n - R) \longrightarrow 0.$$

No subnet and no pre-established translation invariance of the limit is used.

5 The conditional two-phase exclusion

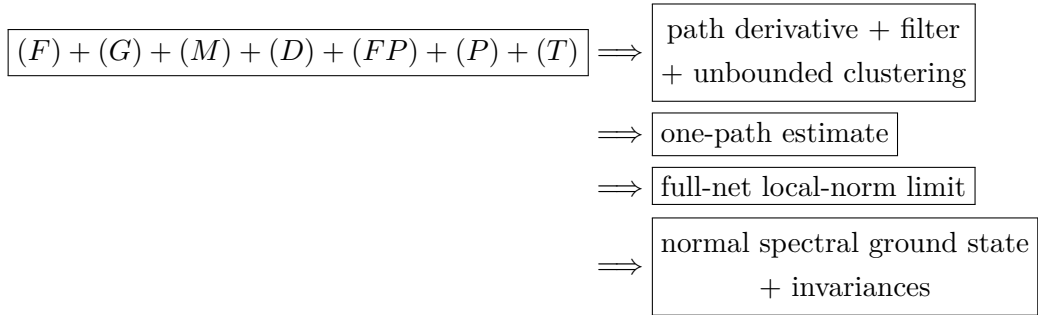
The current Part I theorem correctly makes the following four assumptions explicit:

- (B1) there exist two translation-invariant clustering ground states ω_+, ω_- ;
- (B2) $\theta\omega_+ = \omega_-$;
- (B3) a local order parameter $A = A^*$ has $\omega_\pm(A) = \pm m$, $m \neq 0$;
- (B4) every θ -invariant ground state relevant there is the midpoint $\frac{1}{2}(\omega_+ + \omega_-)$.

The contradiction uses (B4) to identify the locally obtained invariant limit with that midpoint. Without (B4), invariance alone does not imply the identification. GJS 1975–1976 prove phase-transition and pure-phase results in specified strong-coupling models, but the cited pages do not state all four assumptions as a classification of the Hamiltonian ground-state space. Hence the exclusion is rigorous exactly as a conditional theorem and is not an unconditional consequence of Theorem 2.1.

6 Dependency conclusion

The exact dependency chain is



Every arrow has been derived above. The theorem remains conditional on all seven input groups; it is not logically “conditional only” on the two GJS estimates.

Part II

Part II finite-scale model sheet

MMST nearest-neighbour kinematics, the Daubechies $D4$ scaling map, and a fixed Wick-quartic convention

Lamport-form definition and consistency proof

16 August 2026

Scope fixed. The resolution limit is first formulated on one fixed spatial torus $\mathbb{T}_L = \mathbb{R}/(2L\mathbb{Z})$. The lattice kinetic energy is the MMST nearest-neighbour Hamiltonian, the observable embeddings are induced by the orthonormal Daubechies $D4$ mask, and the interaction convention is

$$\lambda \varepsilon_N \sum_{x \in \Lambda_N} : \Phi_N(x)^4 :_{c_{N,L}}.$$

No Daubechies–Galerkin kinetic energy, infinite-line Hamiltonian, or continuum Hamiltonian is identified with this operator. Spatial-cutoff and thermodynamic limits are separate limits and require separately stated hypotheses.

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2	The $D4$ observable embeddings	2
3	Nearest-neighbour Hamiltonian and Wick convention	4
4	Scope ledger	6

1 Geometric and canonical data

⟨1⟩ 1 Fix the torus and the dyadic lattices.

Fix $L > 0$, choose $s_0 \in \mathbb{N}_0$, put $r_0 = 2^{s_0}$ and

$$\varepsilon_0 := \frac{L}{r_0}, \quad \varepsilon_N := 2^{-N} \varepsilon_0, \quad r_N := 2^N r_0. \quad (1.1)$$

The position and momentum lattices are

$$\Lambda_N := \{\varepsilon_N n : -r_N \leq n \leq r_N - 1\} \subset \mathbb{T}_L, \quad (1)$$

$$\Gamma_N := \left\{ \frac{\pi j}{L} : -r_N \leq j \leq r_N - 1 \right\}. \quad (1.2)$$

All position indices are taken modulo $2r_N$. There are exactly $2r_N$ sites and $2r_N \varepsilon_N = 2L$.

⟨1⟩ 2 Fix the Fourier normalization.

For $f : \Lambda_N \rightarrow \mathbb{C}$, define

$$(F_N f)(k) := \sum_{x \in \Lambda_N} f(x) e^{-ikx}, \quad k \in \Gamma_N. \quad (1.3)$$

Then

$$f(x) = \frac{1}{2r_N} \sum_{k \in \Gamma_N} (F_N f)(k) e^{ikx}, \quad \sum_x |f(x)|^2 = \frac{1}{2r_N} \sum_k |(F_N f)(k)|^2. \quad (1.4)$$

For the real canonical coordinates below, set

$$\hat{q}_N := \varepsilon_N^{1/2} F_N q, \quad \hat{p}_N := \varepsilon_N^{1/2} F_N p. \quad (1.5)$$

$\langle 1 \rangle 3$ Fix the symplectic and Weyl conventions.

Let $h_{N,L} = \ell^2(\Lambda_N)$. Write every one-particle vector uniquely as

$$\xi = \varepsilon_N q + ip, \quad q, p : \Lambda_N \rightarrow \mathbb{R}. \quad (1.6)$$

The symplectic form is

$$\sigma_{N,L}((q, p), (q', p')) = \varepsilon_N \sum_{x \in \Lambda_N} (q(x)p'(x) - p(x)q'(x)). \quad (1.7)$$

The Weyl algebra $\mathcal{W}_{N,L}$ is generated by unitaries $W_N(\xi)$ satisfying

$$W_N(\xi)W_N(\eta) = e^{-i\sigma_{N,L}(\xi,\eta)/2} W_N(\xi + \eta). \quad (1.8)$$

In the regular Schrödinger representation,

$$W_N(\varepsilon_N q + ip) = e^{i(\Phi_N(q) + \Pi_N(p))}, \quad (2)$$

$$\Phi_N(q) = \varepsilon_N \sum_x q(x) \Phi_N(x), \quad \Pi_N(p) = \varepsilon_N \sum_x p(x) \Pi_N(x), \quad (1.9)$$

and therefore

$$[\Phi_N(x), \Pi_N(y)] = i\varepsilon_N^{-1} \delta_{xy} I. \quad (1.10)$$

2 The D4 observable embeddings

$\langle 2 \rangle 1$ Fix the low-pass mask, including its normalization.

Put

$$h_0 = \frac{1 + \sqrt{3}}{4\sqrt{2}}, \quad h_1 = \frac{3 + \sqrt{3}}{4\sqrt{2}}, \quad (3)$$

$$h_2 = \frac{3 - \sqrt{3}}{4\sqrt{2}}, \quad h_3 = \frac{1 - \sqrt{3}}{4\sqrt{2}}, \quad (2.1)$$

and $h_n = 0$ otherwise. Define

$$m_0(t) := 2^{-1/2} \sum_{n=0}^3 h_n e^{-int}. \quad (2.2)$$

With $z = e^{-it}$, direct substitution gives

$$m_0(t) = \frac{1}{8} ((1 + \sqrt{3}) + (3 + \sqrt{3})z + (3 - \sqrt{3})z^2 + (1 - \sqrt{3})z^3) \quad (4)$$

$$= \left(\frac{1+z}{2} \right)^2 \underbrace{\frac{(1 + \sqrt{3}) + (1 - \sqrt{3})z}{2}}_{=: D(t)}. \quad (2.3)$$

In particular $m_0(0) = 1$. Moreover

$$|D(t)|^2 = 2 - \cos t, \quad |m_0(t)|^2 = \cos^4(t/2)(2 - \cos t). \quad (2.4)$$

Writing $c = \cos t$, expand

$$|m_0(t)|^2 + |m_0(t + \pi)|^2 = \frac{(1 + c)^2(2 - c) + (1 - c)^2(2 + c)}{4} = 1. \quad (2.5)$$

Indeed, the two cubic polynomials are $2 + 3c - c^3$ and $2 - 3c + c^3$, whose sum is 4. Thus $0 \leq |m_0(t)| \leq 1$, and (2.5) is the quadrature-mirror identity used below.

$\langle 2 \rangle$ 2 Define the one-step refinement map.

For $q, p \in \ell^2(\Lambda_N; \mathbb{R})$, define

$$(R_N^{N+1}(q, p))(y) := \sqrt{2} \sum_{x \in \Lambda_N} \sum_{n=0}^3 h_n(q(x), p(x)) \delta_{x+n\varepsilon_{N+1}}(y). \quad (2.6)$$

For $M < N$, set

$$R_M^N := R_{N-1}^N \circ \dots \circ R_M^{M+1}. \quad (2.7)$$

The QMF identity (2.5), equivalently $\sum_n h_n h_{n+2j} = \delta_{j0}$, gives by direct expansion

$$\sigma_{N+1,L}(R_N^{N+1}\xi, R_N^{N+1}\eta) = \sigma_{N,L}(\xi, \eta). \quad (2.8)$$

Hence the rule

$$\alpha_M^N(W_M(\xi)) := W_N(R_M^N \xi) \quad (2.9)$$

defines an injective unital $*$ -homomorphism and

$$\alpha_N^K \circ \alpha_M^N = \alpha_M^K. \quad (2.10)$$

$\langle 2 \rangle$ 3 Record the exact support enlargement.

If q, p are supported in $X \subset \Lambda_M$, every site occurring in $R_M^N(q, p)$ has the form

$$x + \sum_{j=1}^{N-M} n_j \varepsilon_{M+j}, \quad x \in X, \quad n_j \in \{0, 1, 2, 3\}. \quad (2.11)$$

Consequently, before periodization,

$$\text{supp } R_M^N(q, p) \subset X + [0, 3\varepsilon_M(1 - 2^{-(N-M)})] \subset X + [0, 3\varepsilon_M]. \quad (2.12)$$

This is the support enlargement to be used in every local estimate.

$\langle 2 \rangle$ 4 Fix the continuum observable embedding.

Let ϕ be the compactly supported orthonormal $D4$ scaling function, normalized by $\int \phi = 1$ and supported in $[0, 3]$. Let

$$\phi_{M,x}(y) := \varepsilon_M^{-1/2} \phi((y - x)/\varepsilon_M), \quad (2.13)$$

periodized on $\mathbb{T}_L$. Define

$$R_M^\infty(q, p) := \varepsilon_M^{1/2} \sum_{x \in \Lambda_M} (q(x) + ip(x)) \phi_{M,x}. \quad (2.14)$$

The scaling equation gives

$$R_N^\infty R_M^N = R_M^\infty. \quad (2.15)$$

Thus

$$\beta_M(W_M(\xi)) := W_{\text{ct}}(R_M^\infty \xi) \quad (2.16)$$

is an injective $*$ -homomorphism and

$$\beta_N \circ \alpha_M^N = \beta_M. \quad (2.17)$$

3 Nearest-neighbour Hamiltonian and Wick convention

⟨3⟩ 1 Fix the free operator.

Let $m > 0$. On $\mathcal{H}_{N,L} = L^2(\mathbb{R}^{2r_N}, d\varphi)$ take the Schrödinger realization of (1.10) and define on Schwartz space

$$H_{N,L}^{(0)} := \frac{\varepsilon_N}{2} \sum_{x \in \Lambda_N} \left[\Pi_N(x)^2 + m^2 \Phi_N(x)^2 + \varepsilon_N^{-2} (\Phi_N(x + \varepsilon_N) - \Phi_N(x))^2 \right]. \quad (3.1)$$

Its one-particle dispersion is exactly

$$\gamma_N(k) = \sqrt{m^2 + 4\varepsilon_N^{-2} \sin^2(\varepsilon_N k/2)}, \quad k \in \Gamma_N. \quad (3.2)$$

The free vacuum state is

$$\omega_{N,L}^{(0)}(W_N(\varepsilon_N q + ip)) = \exp \left[-\frac{1}{4(2r_N)} \sum_{k \in \Gamma_N} \left(\gamma_N(k)^{-1} |\hat{q}_N(k)|^2 + \gamma_N(k) |\hat{p}_N(k)|^2 \right) \right]. \quad (3.3)$$

⟨3⟩ 2 Fix the Wick covariance with no hidden additive constant.

Translation invariance gives

$$c_{N,L} := \omega_{N,L}^{(0)}(\Phi_N(x)^2) \quad (5)$$

$$= \frac{1}{2r_N} \sum_{k \in \Gamma_N} \frac{1}{2\varepsilon_N \gamma_N(k)}. \quad (3.4)$$

The finite-scale Wick polynomial is, by definition,

$$:\Phi_N(x)^4:_{c_{N,L}} := \Phi_N(x)^4 - 6c_{N,L} \Phi_N(x)^2 + 3c_{N,L}^2 I. \quad (3.5)$$

For comparison, after the thermodynamic limit at fixed N ,

$$c_N = \frac{1}{4\pi} \int_{-\pi}^{\pi} \frac{d\theta}{\sqrt{(m\varepsilon_N)^2 + 4\sin^2(\theta/2)}} = \frac{1}{2\pi} \log \frac{8}{m\varepsilon_N} + o(1). \quad (3.6)$$

Equation (3.4), not (3.6), is the convention in the fixed-torus Hamiltonian.

⟨3⟩ 3 Fix the interacting Hamiltonian.

For $\lambda \geq 0$ and a function $g_N : \Lambda_N \rightarrow [0, 1]$, define

$$V_{N,L}(g_N) := \lambda \varepsilon_N \sum_{x \in \Lambda_N} g_N(x) :\Phi_N(x)^4:_{c_{N,L}}, \quad (6)$$

$$K_{N,L}(g_N) := H_{N,L}^{(0)} + V_{N,L}(g_N). \quad (3.7)$$

The resolution problem treated in the companion documents uses $g_N \equiv 1$. The factor λ , rather than $\lambda/4!$, is part of the definition. The second line of (3.7) denotes the self-adjoint operator obtained from the closed quadratic form constructed below; it is not an assertion about an unspecified operator domain.

Proposition 3.1 (finite-volume operator and ground state). *For every $N, L, m > 0$, $\lambda \geq 0$, and $0 \leq g_N \leq 1$, the polynomial Schrödinger expression (3.7) has a lower-bounded closed quadratic-form realization $K_{N,L}(g_N)$. This realization has compact resolvent, and its lowest eigenvalue is simple.*

Proof in Lamport form

⟨1⟩ 1 Bound the negative Wick term.

For every $u \in \mathbb{R}$ and $a \geq 0$, completing the square gives

$$u^4 - 6au^2 + 3a^2 = (u^2 - 3a)^2 - 6a^2 \geq -6a^2. \quad (3.8)$$

Thus $V_{N,L}(g_N)$ is bounded below by a finite constant.

⟨1⟩ 2 Establish coercivity of the total potential.

Write $d = 2r_N$, $c = c_{N,L}$, and let $U : \mathbb{R}^d \rightarrow \mathbb{R}$ be the complete multiplication potential in (3.7). Its gradient contribution is nonnegative, and its remaining, coordinatewise part is

$$\varepsilon_N \sum_x f_x(\varphi_x) + 3\lambda \varepsilon_N c^2 \sum_x g_N(x), \quad f_x(u) := \lambda g_N(x) u^4 + \left(\frac{m^2}{2} - 6\lambda c g_N(x) \right) u^2. \quad (3.9)$$

If $\lambda = 0$, then $f_x(u) = m^2 u^2 / 2 \rightarrow \infty$ at every site. Suppose $\lambda > 0$. For each site with $g_N(x) > 0$, the polynomial f_x has positive leading coefficient and hence tends to $+\infty$ as $|u| \rightarrow \infty$; for each site with $g_N(x) = 0$, it is again $m^2 u^2 / 2$. Because the set of sites is finite, for every $A > 0$ there is $R_A < \infty$ such that

$$\max_x |\varphi_x| \geq R_A \implies \sum_x f_x(\varphi_x) \geq A. \quad (3.10)$$

Indeed, choose R_A so that $f_x(u) \geq A - \sum_{y \neq x} \inf_{v \in \mathbb{R}} f_y(v)$ whenever $|u| \geq R_A$, simultaneously for all x . Since $\|\varphi\| \rightarrow \infty$ implies $\max_x |\varphi_x| \rightarrow \infty$, (3.10) proves

$$U(\varphi) \longrightarrow +\infty \quad \text{as} \quad \|\varphi\| \longrightarrow \infty. \quad (3.11)$$

In the special case $g_N \equiv 1$, one also has the explicit inequality

$$\sum_x \varphi_x^4 \geq \frac{1}{2r_N} \left(\sum_x \varphi_x^2 \right)^2. \quad (3.12)$$

⟨1⟩ 3 Construct the self-adjoint operator by a closed form.

Choose $C > 0$ so that $W := U + C \geq 1$, which is possible by (3.8). On $L^2(\mathbb{R}^d)$, define

$$\mathcal{D}(\mathfrak{q}_C) := H^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d, W d\varphi), \quad (7)$$

$$\mathfrak{q}_C[\psi] := \frac{1}{2\varepsilon_N} \int_{\mathbb{R}^d} |\nabla \psi|^2 d\varphi + \int_{\mathbb{R}^d} W |\psi|^2 d\varphi. \quad (3.13)$$

This domain is dense because it contains $C_c^\infty(\mathbb{R}^d)$. It is complete for the norm $\|\psi\|_{\mathfrak{q}_C}^2 := \mathfrak{q}_C[\psi]$: if (ψ_j) is Cauchy, it converges in H^1 to some ψ , while $W^{1/2}\psi_j$ converges in L^2 to some v . A subsequence of ψ_j converges pointwise almost everywhere to ψ , and the same subsequence has a further subsequence for which $W^{1/2}\psi_j \rightarrow v$ pointwise almost everywhere. Hence $v = W^{1/2}\psi$, proving $\psi \in \mathcal{D}(\mathfrak{q}_C)$ and form-norm convergence. Thus $\mathfrak{q}_C$ is closed. The representation theorem for closed positive forms yields a unique positive self-adjoint operator $K_{N,L}(g_N) + C$; this defines $K_{N,L}(g_N)$. If $\psi, \zeta \in C_c^\infty(\mathbb{R}^d)$, integration by parts gives

$$\mathfrak{q}_C(\zeta, \psi) = \left\langle \zeta, \left(-\frac{1}{2\varepsilon_N} \Delta + U + C \right) \psi \right\rangle. \quad (3.14)$$

Consequently the form operator extends exactly the polynomial differential expression (3.7) on $C_c^\infty(\mathbb{R}^d)$.

⟨1⟩ 4 Prove compactness of the resolvent.

Let (ψ_j) be bounded in the form norm. Given $B > 0$, coercivity (3.11) supplies R_B such that $W(\varphi) \geq B$ for $\|\varphi\| \geq R_B$. Therefore

$$\int_{\|\varphi\| \geq R_B} |\psi_j|^2 d\varphi \leq B^{-1} \int W |\psi_j|^2 d\varphi \leq B^{-1} \sup_j \mathfrak{q}_C[\psi_j]. \quad (3.15)$$

On each ball, (ψ_j) is bounded in H^1 , so the Rellich compactness theorem gives an L^2 -convergent subsequence there. A diagonal subsequence, together with the uniform tail bound (3.15) and then $B \rightarrow \infty$, is Cauchy in $L^2(\mathbb{R}^d)$. Hence the form-domain inclusion into L^2 is compact. The resolvent $(K_{N,L}(g_N) + C)^{-1}$ maps L^2 boundedly into the form domain by the spectral theorem; composing it with the compact inclusion proves that this resolvent is compact.

⟨1⟩ 5 Prove simplicity of the lowest eigenvalue.

Compact resolvent and lower boundedness give a lowest eigenvalue. For $t > 0$, the Feynman–Kac formula gives the integral kernel

$$e^{-tK_{N,L}(g_N)}(\mathbf{x}, \mathbf{y}) = \left(\frac{\varepsilon_N}{2\pi t}\right)^{d/2} e^{-\varepsilon_N \|\mathbf{x} - \mathbf{y}\|^2 / (2t)} \mathbb{E}_{\mathbf{x}, \mathbf{y}}^t \left[\exp\left(-\int_0^t U(B_s) ds\right) \right]. \quad (3.16)$$

Here $\mathbb{E}_{\mathbf{x}, \mathbf{y}}^t$ is expectation for the Brownian bridge with diffusion coefficient ε_N^{-1} . Formula (3.16) follows first for bounded truncations of the continuous potential from the Trotter product formula and then for U by monotone form convergence. A bridge path is continuous almost surely, so its image is compact and the integral of the polynomial U along it is finite. Every factor on the right of (3.16) is therefore strictly positive. Thus $e^{-tK_{N,L}(g_N)}$ is positivity improving.

Choose a ground eigenfunction f . Since $|\nabla|f|| \leq |\nabla f|$ almost everywhere, the form satisfies $\mathfrak{q}_C[|f|] \leq \mathfrak{q}_C[f]$. The variational definition of the lowest eigenvalue forces equality, so $|f|$ is also a ground eigenfunction. Applying the positivity-improving semigroup to $|f| \neq 0$ shows that the normalized ground eigenfunction $\psi_0 := |f|/\|f\|$ is strictly positive almost everywhere.

Suppose that the ground eigenspace had dimension at least two. Since the operator commutes with complex conjugation, that eigenspace contains a real ground eigenfunction $h \neq 0$ orthogonal to ψ_0 . The strict positivity of ψ_0 and $\langle \psi_0, h \rangle = 0$ imply that both h_+ and h_- are nonzero. The strictly positive kernel in (3.16) then gives, almost everywhere,

$$|e^{-tK} h| < e^{-tK} |h|. \quad (3.17)$$

Pairing (3.17) with ψ_0 gives

$$e^{-tE_0} \langle \psi_0, |h| \rangle < \langle \psi_0, e^{-tK} |h| \rangle = e^{-tE_0} \langle \psi_0, |h| \rangle,$$

a contradiction. Hence the lowest eigenvalue is simple. □

Denote the normalized ground vector by $\Omega_{N,L,\lambda,g}$, subtract the ground energy, and set

$$\omega_{N,L,\lambda,g}(A) := \langle \Omega_{N,L,\lambda,g}, A \Omega_{N,L,\lambda,g} \rangle. \quad (3.18)$$

4 Scope ledger

Statement	Status	Exact reason
MMST nearest-neighbour free kinematics	PASS	Equations (1.1)–(1.10) and (3.1)–(3.3) fix the one-dimensional specialization.

Statement	Status	Exact reason
$D4$ wavelet observable embeddings	PASS	The mask, QMF identity, maps, and support enlargement are explicit in (2.1)–(2.17).
Finite-torus Wick quartic convention	PASS	Equations (3.4)–(3.7) fix the covariance, Wick polynomial, lattice measure, and coupling normalization.
Interacting resolution limit	FAIL	Defining the Hamiltonians supplies states but does not prove convergence of their pullbacks.
Infinite-volume/spatial-cutoff limit	FAIL	It is a distinct limit and is not a consequence of a fixed-torus definition.
Daubechies–Galerkin kinetic energy	FAIL	It is not the nearest-neighbour operator (3.1) and is outside this track.

References

- [1] V. Morinelli, G. Morsella, A. Stottmeister, Y. Tanimoto, *Scaling limits of lattice quantum fields by wavelets*, Commun. Math. Phys. **387** (2021), 1299–1367.

Part II, corrected at the first falsifiable checkpoint

The quadratic OAR model obstructs diagonal local state-norm convergence

Lamport-form proof and source audit

16 August 2026

Decisive status. The Part II target in `strategy.md` is not a well-defined statement as written, because an injective wavelet map embeds *observables*; it does not push a lattice state canonically to the continuum algebra. Under the natural MMST comparison, the intended diagonal local-norm estimate is false already for the free massive scalar field.

More precisely, for every bounded interval O containing a smaller interval, there are finest-scale Weyl unitaries U_N whose lattice support and continuum wavelet support lie in O and for which

$$\liminf_{N \rightarrow \infty} |\omega_N(U_N) - \omega(\beta_N(U_N))| \geq e^{-\sqrt{2+\sqrt{2}}/4} - e^{-3\pi/16} = 0.0752053795 \dots \quad (\text{R})$$

Consequently no estimate $C\varepsilon_N^\theta$, $\theta > 0$, can hold in local state norm on the diagonal scale. This does not contradict the MMST theorem: MMST take the fine-scale limit while the tested observable is held at a *fixed coarse scale*. Section 8 gives the corrected Part II target.

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1 Exact model and comparison map

This section fixes every normalization. The construction is the one-dimensional specialization of Morinelli–Morsella–Stottmeister–Tanimoto (MMST), equations (3.2)–(3.10), (4.1)–(4.5), and Proposition 4.1.

Fix $L > 0$. For $N \geq 0$, choose integers $r_N = 2^N r_0$ and put

$$\varepsilon_N = L/r_N, \quad \Lambda_N = \{\varepsilon_N n : -r_N \leq n \leq r_N - 1\} \subset \mathbb{T}_L, \quad \mathbb{T}_L = \mathbb{R}/(2L\mathbb{Z}).$$

The dual momenta and their dimensionless representatives are

$$k_j = \frac{\pi j}{L}, \quad \theta_j = \varepsilon_N k_j = \frac{\pi j}{r_N}, \quad -r_N \leq j \leq r_N - 1. \quad (1.1)$$

For a real sequence $p = (p_n)$, define

$$\tilde{p}_j = \sum_{n=-r_N}^{r_N-1} p_n e^{-i\theta_j n}.$$

Parseval’s identity, with no suppressed constant, is

$$\sum_n |p_n|^2 = \frac{1}{2r_N} \sum_j |\tilde{p}_j|^2. \quad (1.2)$$

Let $\mathcal{W}_N$ be the Weyl algebra over $\mathbb{C}^{2r_N}$ with its standard symplectic form. In one space dimension the MMST decomposition (3.7) is $\xi = \varepsilon_N q + ip$, q, p real. Let $m > 0$ be the physical mass and put $\delta_N = m\varepsilon_N$. The nearest-neighbour lattice dispersion is

$$\gamma_N(k_j) = \left(m^2 + 2\varepsilon_N^{-2}(1 - \cos \theta_j) \right)^{1/2} \quad (1)$$

$$= \varepsilon_N^{-1} a_{\delta_N}(\theta_j), \quad a_{\delta}(\theta) = \left(\delta^2 + 4 \sin^2(\theta/2) \right)^{1/2}. \quad (1.3)$$

The centered lattice vacuum is the quasifree state

$$\omega_N(W_N(\varepsilon_N q + ip)) = \exp \left[-\frac{1}{4} \left(\|\gamma_N^{-1/2} \hat{q}\|_N^2 + \|\gamma_N^{1/2} \hat{p}\|_N^2 \right) \right], \quad (1.4)$$

where $\hat{p}_j = \varepsilon_N^{1/2} \tilde{p}_j$ and $\|z\|_N^2 = (2r_N)^{-1} \sum_j |z_j|^2$.

Let $\phi \in L^2(\mathbb{R})$ be a real compactly supported scaling function such that its integer translates are orthonormal. We assume the exact decay used in MMST,

$$|\hat{\phi}(\zeta)| \leq C(1 + |\zeta|)^{-\rho}, \quad \rho > 1, \quad (1.5)$$

which holds for the Daubechies choices admitted in MMST in one dimension. Write

$$\phi_{N,n}(x) = \varepsilon_N^{-1/2} \phi(x/\varepsilon_N - n).$$

Periodize these functions with period $2L$. The MMST symplectic map and the induced injective morphism of Weyl C^* -algebras are

$$R_N(q, p) = \varepsilon_N^{1/2} \sum_n (q_n + ip_n) \phi_{N,n}, \quad (1.6)$$

$$\beta_N(W_N(\varepsilon_N q + ip)) = W_{\text{ct}}(R_N(q, p)). \quad (1.7)$$

Let ω be the continuum mass- m vacuum on $\mathcal{W}(\mathbb{T}_L)$.

Definition 1.1 (the only canonical finite-scale comparison). The lattice state ω_N and continuum state ω are compared on $\mathcal{W}_N$ as

$$\omega_N \quad \text{versus} \quad \omega \circ \beta_N. \quad (1.8)$$

An injective $*$ -homomorphism is contravariant on states: it sends ω to $\omega \circ \beta_N$. It supplies no canonical state on $\mathcal{W}(\mathbb{T}_L)$ from ω_N .

2 Exact covariance symbols

Lemma 2.1 (lattice momentum symbol). *For every real $p \in \mathbb{R}^{2r_N}$,*

$$\omega_N(W_N(ip)) = \exp \left[-\frac{1}{4} Q_N^{\text{lat}}(p) \right], \quad Q_N^{\text{lat}}(p) = \frac{1}{2r_N} \sum_j a_{\delta_N}(\theta_j) |\tilde{p}_j|^2. \quad (2.1)$$

Proof in Lamport form

$\langle 1 \rangle 1$ Insert $q = 0$ in (1.4).

$\langle 1 \rangle 2$ Use $\hat{p}_j = \varepsilon_N^{1/2} \tilde{p}_j$ and $\gamma_N(k_j) = \varepsilon_N^{-1} a_{\delta_N}(\theta_j)$ to obtain

$$\|\gamma_N^{1/2} \hat{p}\|_N^2 = \frac{1}{2r_N} \sum_j \varepsilon_N^{-1} a_{\delta_N}(\theta_j) \varepsilon_N |\tilde{p}_j|^2,$$

which is (2.1). □

Lemma 2.2 (continuum pullback symbol). *For every real $p \in \mathbb{R}^{2r_N}$,*

$$\omega(\beta_N(W_N(ip))) = \exp \left[-\frac{1}{4} Q_N^{\text{ct}}(p) \right], \quad (2.2)$$

where

$$Q_N^{\text{ct}}(p) = \frac{1}{2r_N} \sum_j b_{\phi, \delta_N}(\theta_j) |\tilde{p}_j|^2, \quad (2.3)$$

$$b_{\phi, \delta}(\theta) = \sum_{\ell \in \mathbb{Z}} \left(\delta^2 + (\theta + 2\pi\ell)^2 \right)^{1/2} |\hat{\phi}(\theta + 2\pi\ell)|^2. \quad (2.4)$$

The series in (2.4) converges uniformly in $\theta \in [-\pi, \pi]$.

Proof in Lamport form

$\langle 1 \rangle 1$ Compute the Fourier coefficient of the embedded momentum smearing.

For continuum momentum $k \in (\pi/L)\mathbb{Z}$, (1.6) and $\widehat{\phi_{N,n}}(k) = \varepsilon_N^{1/2} e^{-ik\varepsilon_N n} \hat{\phi}(\varepsilon_N k)$ give

$$\Im \widehat{R_N(0, p)}(k) = \varepsilon_N \hat{\phi}(\varepsilon_N k) \sum_n p_n e^{-ik\varepsilon_N n}. \quad (2.5)$$

$\langle 1 \rangle 2$ Split every continuum momentum uniquely as

$$k = k_j + \frac{2\pi\ell}{\varepsilon_N}, \quad -r_N \leq j \leq r_N - 1, \quad \ell \in \mathbb{Z}. \quad (2.6)$$

Then $e^{-ik\varepsilon_N n} = e^{-i\theta_j n}$ and

$$\varepsilon_N(m^2 + k^2)^{1/2} = \left(\delta_N^2 + (\theta_j + 2\pi\ell)^2 \right)^{1/2}. \quad (2.7)$$

$\langle 1 \rangle 3$ Insert (2.5)–(2.7) into the continuum vacuum quadratic form.

The torus Fourier normalization is $(2L)^{-1} \sum_k$. Since $\varepsilon_N/(2L) = 1/(2r_N)$, grouping the sum by (2.6) gives exactly (2.3)–(2.4), with no limiting argument.

⟨1⟩ 4 Prove convergence of (2.4).

By (1.5), uniformly for $|\theta| \leq \pi$ and $|\ell| \geq 1$,

$$\left(\delta^2 + (\theta + 2\pi\ell)^2\right)^{1/2} |\widehat{\phi}(\theta + 2\pi\ell)|^2 \leq C_\delta(1 + |\ell|)^{1-2\rho}.$$

The majorant is summable because $\rho > 1$. The Weierstrass test proves uniform convergence. $\square$

Lemma 2.3 (alias partition of unity). *For every $\theta \in \mathbb{R}$,*

$$\sum_{\ell \in \mathbb{Z}} |\widehat{\phi}(\theta + 2\pi\ell)|^2 = 1. \quad (2.8)$$

Consequently, with $d(\theta) = \text{dist}(\theta, 2\pi\mathbb{Z})$,

$$b_{\phi,\delta}(\theta) \geq h_\delta(\theta) := \left(\delta^2 + d(\theta)^2\right)^{1/2}. \quad (2.9)$$

Proof in Lampion form

⟨1⟩ 1 Use orthonormality of the integer translates.

For every $n \in \mathbb{Z}$, Plancherel gives

$$\begin{aligned} \delta_{n0} &= \int_{\mathbb{R}} \phi(x) \phi(x - n) dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{in\theta} \left(\sum_{\ell \in \mathbb{Z}} |\widehat{\phi}(\theta + 2\pi\ell)|^2 \right) d\theta. \end{aligned} \quad (2.10)$$

The periodic function in parentheses therefore has Fourier coefficients δ_{n0} and equals 1 almost everywhere. Assumption (1.5) makes its series uniformly convergent, hence continuous. Equality almost everywhere between continuous functions implies (2.8) everywhere.

⟨1⟩ 2 For each ℓ ,

$$|\theta + 2\pi\ell| \geq d(\theta),$$

so every square-root factor in (2.4) is at least $h_\delta(\theta)$. Multiply by the nonnegative weights in (2.8) and sum. This proves (2.9). $\square$

Remark 2.4 (strict ultraviolet mismatch). For $0 < |\theta| < \pi$ and $\delta \geq 0$,

$$h_\delta(\theta) = \sqrt{\delta^2 + \theta^2} > \sqrt{\delta^2 + 4\sin^2(\theta/2)} = a_\delta(\theta), \quad (2.11)$$

because $2\sin(|\theta|/2) < |\theta|$. No property of a particular Daubechies filter can remove this inequality.

3 A localized high-frequency sequence

Fix open intervals $J \Subset O \Subset \mathbb{T}_L$ and choose real $\eta \in C_c^\infty(J)$ with $\int_{\mathbb{T}_L} \eta(x)^2 dx = 1$. Set

$$\theta_0 = \frac{3\pi}{4}, \quad p_{N,n} = c_N \sqrt{2\varepsilon_N} \eta(\varepsilon_N n) \cos(\theta_0 n), \quad (3.1)$$

where $c_N > 0$ is chosen so that $\sum_n p_{N,n}^2 = 1$. For all sufficiently large N , $2r_N$ is divisible by 8, so the carrier is periodic.

Lemma 3.1 (localized carrier measure). *For every continuous even 2π -periodic function f ,*

$$\lim_{N \rightarrow \infty} \frac{1}{2r_N} \sum_j f(\theta_j) |\tilde{p}_{N,j}|^2 = f(\theta_0). \quad (3.2)$$

Moreover $c_N \rightarrow 1$.

Proof in Lamport form

⟨1⟩ 1 Prove $c_N \rightarrow 1$.

Before multiplication by c_N^2 , the squared norm in (3.1) equals

$$2\varepsilon_N \sum_n \eta(\varepsilon_N n)^2 \cos^2(\theta_0 n) = \varepsilon_N \sum_n \eta(\varepsilon_N n)^2 \quad (2)$$

$$+ \varepsilon_N \sum_n \eta(\varepsilon_N n)^2 \cos(2\theta_0 n). \quad (3.3)$$

The first sum tends to $\int \eta^2 = 1$ by Riemann integration. For the second, partition n into its four residue classes modulo 4. Since $2\theta_0 = 3\pi/2$, each residue-class Riemann sum tends to $\frac{1}{4} \int \eta^2$, while

$$\sum_{a=0}^3 e^{i(3\pi/2)a} = 0. \quad (3.4)$$

Thus the second sum tends to zero. The unnormalized squared norm tends to one, so $c_N \rightarrow 1$.

⟨1⟩ 2 Prove (3.2) first for $f(\theta) = e^{is\theta}$, $s \in \mathbb{Z}$.

By the discrete Fourier inversion formula, the left side of (3.2) is

$$\sum_n p_{N,n} p_{N,n-s}, \quad (3.5)$$

where no wrap-around term occurs for large N because $\text{supp } \eta \subseteq (-L, L)$. The product-to-sum identity turns (3.5) into

$$c_N^2 \varepsilon_N \sum_n \eta(\varepsilon_N n) \eta(\varepsilon_N (n-s)) [\cos(s\theta_0) + \cos(2n\theta_0 - s\theta_0)]. \quad (3.6)$$

The first term tends to $\cos(s\theta_0)$. For the second, split $n = 4q + a$, $a = 0, 1, 2, 3$. On each residue class the phase $e^{i(2n\theta_0 - s\theta_0)}$ equals $e^{-is\theta_0} e^{i(3\pi/2)a}$, while the corresponding Riemann sum tends to $\frac{1}{4} \int \eta(x)^2 dx$ because $\eta(x)\eta(x - \varepsilon_N s) \rightarrow \eta(x)^2$ uniformly. Summing the four limits gives zero by (3.4). Therefore (3.5) tends to

$$\cos(s\theta_0) = \frac{1}{2}(e^{is\theta_0} + e^{-is\theta_0}). \quad (3.7)$$

⟨1⟩ 3 Extend to continuous f .

Equation (3.7) proves weak convergence of the probability measures

$$\nu_N = \frac{1}{2r_N} \sum_j |\tilde{p}_{N,j}|^2 \delta_{\theta_j} \implies \frac{1}{2}(\delta_{\theta_0} + \delta_{-\theta_0}) \quad (3.8)$$

first on trigonometric polynomials. Trigonometric polynomials are uniformly dense in the continuous periodic functions, and both sides of (3.8) have mass one. Hence (3.8) holds for every continuous periodic f . If f is even, its integral against the limiting measure is $f(\theta_0)$, proving (3.2). $\square$

4 The diagonal local-norm no-go theorem

Let $X_N = \{\varepsilon_N n : p_{N,n} \neq 0\}$ and let $\mathcal{W}_N(X_N)$ be the corresponding lattice Weyl subalgebra.

Theorem 4.1 (free quadratic obstruction). *For the MMST diagonal lattice vacua and continuum mass- m vacuum,*

$$\lim_{N \rightarrow \infty} Q_N^{\text{lat}}(p_N) = 2 \sin(3\pi/8) = \sqrt{2 + \sqrt{2}}, \quad (4.1)$$

$$\liminf_{N \rightarrow \infty} Q_N^{\text{ct}}(p_N) \geq 3\pi/4. \quad (4.2)$$

Consequently

$$\liminf_{N \rightarrow \infty} \left\| \omega_N|_{\mathcal{W}_N(X_N)} - (\omega \circ \beta_N)|_{\mathcal{W}_N(X_N)} \right\| \geq c_*, \quad (4.3)$$

$$c_* = e^{-\sqrt{2+\sqrt{2}}/4} - e^{-3\pi/16} > 0. \quad (4.4)$$

Numerically, $c_* = 0.0752053795 \dots$

Proof in Lamport form

$\langle 1 \rangle 1$ Apply Lemma 3.1 to the lattice symbol.

The functions a_δ converge uniformly to $a_0(\theta) = 2|\sin(\theta/2)|$, because

$$0 \leq a_\delta(\theta) - a_0(\theta) \leq \delta. \quad (4.5)$$

Equations (2.1), (3.2), and (4.5) yield

$$Q_N^{\text{lat}}(p_N) \longrightarrow a_0(\theta_0) = 2 \sin(3\pi/8) = \sqrt{2 + \sqrt{2}}. \quad (4.6)$$

$\langle 1 \rangle 2$ Apply the alias lower bound to the continuum symbol.

By (2.9),

$$Q_N^{\text{ct}}(p_N) \geq \frac{1}{2r_N} \sum_j h_{\delta_N}(\theta_j) |\tilde{p}_{N,j}|^2. \quad (4.7)$$

The periodic functions h_δ converge uniformly to $h_0(\theta) = d(\theta)$, because $0 \leq h_\delta - h_0 \leq \delta$. Lemma 3.1 and $0 < \theta_0 < \pi$ therefore give

$$\liminf_N Q_N^{\text{ct}}(p_N) \geq h_0(\theta_0) = 3\pi/4. \quad (4.8)$$

$\langle 1 \rangle 3$ Use the Weyl unitary itself as a norm-one test observable.

By (2.1)–(2.2),

$$|\omega_N(W_N(ip_N)) - \omega(\beta_N(W_N(ip_N)))| = \left| e^{-Q_N^{\text{lat}}(p_N)/4} - e^{-Q_N^{\text{ct}}(p_N)/4} \right|. \quad (4.9)$$

The right side has lower limit at least (4.4), by (4.1)–(4.2) and the strict inequality $\sqrt{2 + \sqrt{2}} < 3\pi/4$. Since $W_N(ip_N)$ is a unitary in $\mathcal{W}_N(X_N)$, the dual norm dominates (4.9). This proves (4.3). $\square$

Corollary 4.2 (no continuum extension repairs the norm). Assume $\text{supp } \phi \subset [0, R_\phi]$. For all sufficiently large N ,

$$\beta_N(\mathcal{W}_N(X_N)) \subset \mathcal{W}(O). \quad (4.10)$$

If $\tilde{\omega}_N$ is any state on the continuum Weyl algebra satisfying $\tilde{\omega}_N \circ \beta_N = \omega_N$, then

$$\liminf_{N \rightarrow \infty} \|(\tilde{\omega}_N - \omega)|_{\mathcal{W}(O)}\| \geq c_*. \quad (4.11)$$

Proof. Choose $a > 0$ with $\text{dist}(\text{supp } \eta, \mathbb{T}_L \setminus O) > a$. Every nonzero coefficient in (3.1) is based at a point of $\text{supp } \eta$, and

$$\text{supp } \phi_{N,n} \subset \varepsilon_N n + [0, R\varepsilon_N]. \quad (4.12)$$

If $R\varepsilon_N < a$, (4.10) follows. On the subalgebra in (4.10), the two functionals in (4.11) restrict to the two functionals in (4.3). Restriction cannot increase a functional norm, so (4.3) implies (4.11). $\square$

Corollary 4.3 (the proposed Part II vertical term is false). There are no constants $C < \infty$ and $\theta > 0$ for which

$$\left\| \omega_N|_{\mathcal{W}_N(X_N)} - (\omega \circ \beta_N)|_{\mathcal{W}_N(X_N)} \right\| \leq C\varepsilon_N^\theta \quad (4.13)$$

holds for all sufficiently large N .

Proof. The right side of (4.13) tends to zero; the lower limit of the left side is at least $c_* > 0$ by Theorem 4.1. $\square$

Remark 4.4 (spatial cutoffs do not evade the obstruction). In the quadratic model, let $g_R \uparrow 1$ and let the one-particle squared energies be

$$A_{N,R} = -\Delta_N + m_0^2 + \lambda g_R, \quad B_N = -\Delta_N + m_0^2 + \lambda.$$

At fixed N , $A_{N,R} \rightarrow B_N$ strongly and $m_0^2 \leq A_{N,R}, B_N \leq 4\varepsilon_N^{-2} + m_0^2 + \lambda$. Continuous functional calculus on this common compact spectral interval gives $A_{N,R}^{\pm 1/2} \rightarrow B_N^{\pm 1/2}$ strongly. Hence the expectation of every finitely supported Weyl operator converges to the full mass- m_1 lattice vacuum, $m_1^2 = m_0^2 + \lambda$. Therefore a putative estimate $\Psi(d_R) + C\varepsilon_N^\theta$, with $\Psi(d_R) \rightarrow 0$, would imply (4.13) after first taking $R \rightarrow \infty$ at fixed N . Theorem 4.1 rules this out.

5 The free facts that do survive (II.2)

Proposition 5.1 (exact gap and group-velocity bounds). *For the MMST free lattice Hamiltonian, after subtraction of the vacuum energy,*

$$\text{gap } H_N = m \quad (5.1)$$

for every N . Its one-particle group velocity satisfies

$$\sup_N \sup_{k \in [-\pi/\varepsilon_N, \pi/\varepsilon_N]} |\partial_k \gamma_N(k)| \leq 1. \quad (5.2)$$

For the quadratic spatial-cutoff model $A_N(g) = -\Delta_N + m_0^2 + \lambda g$, $0 \leq g \leq 1$, the Fock gap is at least m_0 ; on the infinite lattice with compactly supported g , it equals m_0 .

Proof in Lamport form

$\langle 1 \rangle 1$ The one-particle energy is multiplication by $\gamma_N(k)$ and $\gamma_N(k) \geq \gamma_N(0) = m$.

The normal-ordered many-body Hamiltonian is

$$H_N = d\Gamma(\gamma_N).$$

Its nonzero spectrum begins at $\inf \text{spec } \gamma_N = m$, proving (5.1).

$\langle 1 \rangle 2$ Differentiate (1.3).

$$\partial_k \gamma_N(k) = \frac{\sin(\varepsilon_N k)}{\sqrt{\delta_N^2 + 4 \sin^2(\varepsilon_N k/2)}}. \quad (5.3)$$

Since $|\sin \theta| = 2|\sin(\theta/2)\cos(\theta/2)|$, the absolute value of (5.3) is at most $|\cos(\theta/2)| \leq 1$. This proves (5.2).

$\langle 1 \rangle 3$ For a spatial cutoff, $A_N(g) \geq m_0^2$, so $A_N(g)^{1/2} \geq m_0$.

If g is compactly supported on the infinite lattice, multiplication by g is a compact perturbation of $-\Delta_N + m_0^2$. Weyl's theorem gives essential spectrum beginning at m_0^2 . The perturbation is nonnegative, so there is no spectrum below m_0^2 . Thus the bottom is exactly m_0^2 , and the Fock gap is m_0 . $\square$

Proposition 5.2 (exact Wick-constant asymptotic). *The infinite-lattice equal-site field covariance is*

$$c_N = \frac{1}{4\pi} \int_{-\pi}^{\pi} \frac{d\theta}{\sqrt{\delta_N^2 + 4\sin^2(\theta/2)}}. \quad (5.4)$$

As $N \rightarrow \infty$,

$$c_N = \frac{1}{2\pi} \log \frac{8}{m\varepsilon_N} + o(1). \quad (5.5)$$

In particular the coefficient of $\log(1/\varepsilon_N)$ is exactly $1/(2\pi)$.

Proof in Lamport form

$\langle 1 \rangle 1$ Evenness and the substitution $\theta = 2u$ give

$$c_N = \frac{1}{\pi} \int_0^{\pi/2} \frac{du}{\sqrt{\delta_N^2 + 4\sin^2 u}}. \quad (5.6)$$

$\langle 1 \rangle 2$ Add and subtract the model singularity $2u$.

$$c_N = I(\delta_N) + D(\delta_N), \quad (5.7)$$

$$I(\delta) = \frac{1}{\pi} \int_0^{\pi/2} \frac{du}{\sqrt{\delta^2 + 4u^2}} = \frac{1}{2\pi} \operatorname{arsinh} \frac{\pi}{\delta}, \quad (5.8)$$

$$D(\delta) = \frac{1}{\pi} \int_0^{\pi/2} \left[\frac{1}{\sqrt{\delta^2 + 4\sin^2 u}} - \frac{1}{\sqrt{\delta^2 + 4u^2}} \right] du. \quad (5.9)$$

$\langle 1 \rangle 3$ Justify dominated convergence in (5.9).

For $0 < a \leq b$,

$$0 \leq \frac{1}{\sqrt{\delta^2 + a^2}} - \frac{1}{\sqrt{\delta^2 + b^2}} \leq \frac{1}{a} - \frac{1}{b}, \quad (5.10)$$

because the absolute derivative of $(\delta^2 + x^2)^{-1/2}$ is at most x^{-2} . Apply (5.10) with $a = 2\sin u$, $b = 2u$. The majorant

$$\frac{1}{2\sin u} - \frac{1}{2u} \quad (5.11)$$

is integrable at $u = 0$, since it equals $u/12 + O(u^3)$. Therefore

$$\lim_{\delta \downarrow 0} D(\delta) = \frac{1}{2\pi} \int_0^{\pi/2} (\csc u - u^{-1}) du \quad (3)$$

$$= \frac{1}{2\pi} \log \frac{4}{\pi}. \quad (5.12)$$

The last equality follows by integrating from a to $\pi/2$, using $\int \csc u \, du = \log \tan(u/2)$, and then taking $a \downarrow 0$.

⟨1⟩ 4 Since

$$\operatorname{arsinh}(\pi/\delta) = \log(2\pi/\delta) + o(1), \quad (5.13)$$

adding (5.8) and (5.12) yields

$$c_N = \frac{1}{2\pi} \left[\log \frac{2\pi}{\delta_N} + \log \frac{4}{\pi} \right] + o(1) = \frac{1}{2\pi} \log \frac{8}{m\varepsilon_N} + o(1).$$

This proves (5.5). $\square$

Remark 5.3 (what (5.2) does and does not prove). Equation (5.2) is a group-velocity estimate. It is not, by itself, an interacting Lieb–Robinson theorem for unbounded quartic oscillators. MMST obtain a free harmonic Lieb–Robinson estimate separately in their Lemma 4.14 and Proposition 4.15. Part II must import or prove an interacting bound with the exact ε_N scaling; relabeling (5.2) as such a theorem would be invalid.

6 What the cited sources establish

Proposition 6.1 (precise MMST scope). *The following statements, and no stronger diagonal state-norm claim, are supported by the cited MMST paper.*

- (i) *The maps $\alpha_N^{N'}$ form an inductive system of observable algebras, while states at finer scales are pulled back to a fixed coarse algebra.*
- (ii) *Proposition 4.1 embeds the inductive-limit Weyl algebra into the continuum Weyl algebra by β_L .*
- (iii) *Lemmas 4.2–4.3 and Proposition 4.5 prove, for every fixed coarse scale, weak* convergence of the pulled-back free lattice ground states and identify the projective limit with the continuum vacuum on the embedded algebra.*
- (iv) *Theorem 4.11 identifies the resulting time-zero net with the massive free-field net.*
- (v) *Section 6.1 explicitly treats interacting lattice ground-state scaling limits and wavelet interacting dynamics as future generalizations.*

Proof. Items (i)–(iv) are the literal content of MMST equations (2.8)–(2.11), Propositions 4.1 and 4.5, and Theorem 4.11. Item (v) is stated in the first two paragraphs of MMST Section 6.1. Page images of all five locations appear in the separate Part II citation dossier. $\square$

Proposition 6.2 (Battle–Federbush does not supply II.4). *The 1987 Battle–Federbush note does not state $CE1_N/CE2_N$ for the MMST Daubechies OAR lattice, uniformly in N , nor for the mixed interpolation proposed in II.6.*

Proof. The note uses Meyer-type or exponentially localized orthonormal modes ψ_k and expands the Euclidean field with

$$u_k = (-\Delta + M^2)^{-1/2} \psi_k. \quad (6.1)$$

It explicitly singles out the resulting diagonality of the free action in the phase-cell variables as an attractive feature. The MMST finite-scale Hamiltonian instead has the nearest-neighbour symbol (1.3), and its Daubechies functions define symplectic scaling maps between time-zero Weyl algebras. The note contains no theorem with the MMST Hamiltonian, the Wick constant (5.4), a sharp-time GJS observable norm, an N -uniform physical gap, or two mixed vertex species. These missing hypotheses are exactly the conclusion demanded by II.4, so the note cannot be cited as that conclusion. $\square$

7 Why II.6 is not an operator interpolation

Proposition 7.1 (ill-defined mixed Hamiltonian path). *The expression*

$$H_s = (1 - s)H_N(g) + sH(g) \quad (7.1)$$

in the original II.6 program is undefined with the currently specified data.

Proof. $H_N(g)$ is a self-adjoint operator in the lattice ground-state Hilbert space $\mathcal{H}_{N,g}$, whereas $H(g)$ is a self-adjoint operator in the continuum Hilbert space $\mathcal{H}_g$. Addition of unbounded operators is defined only after both are realized as operators or closed forms on one Hilbert space with a specified common form domain. The observable embedding $\beta_N : \mathcal{W}_N \rightarrow \mathcal{W}$ supplies neither:

- (a) it maps C^* -algebra elements, not unbounded Hamiltonians;
- (b) it does not push ω_N to a canonical continuum state;
- (c) its symplectic one-particle map need not intertwine the finite-scale and continuum complex structures, so no stated second-quantized unitary identifies the two vacuum representations;
- (d) no common closed quadratic form for the two Wick-ordered interactions is given.

Therefore (7.1), its derivative $\partial_s H_s$, its spectral gap, and the vectors $D_j \Omega_s$ have no definitions. A mixed-vertex Euclidean measure is not a repair of this operator defect unless an Osterwalder–Schrader construction identifies the entire path on a common Hilbert space and proves common-domain control. $\square$

8 Corrected Part II target and dependency order

The no-go theorem forces a change of topology. It does not merely change the value of the exponent θ .

Definition 8.1 (fixed-coarse comparison). For $M < N$, a lattice state $\omega_{N,g}$ defines a state on the fixed coarse algebra $\mathcal{W}_M$ by

$$\omega_{N,g}^{[M]} := \omega_{N,g} \circ \alpha_M^N. \quad (8.1)$$

The continuum comparison state is

$$\omega_\infty^{[M]} := \omega_\infty \circ \beta_M. \quad (8.2)$$

Both sides of (8.1)–(8.2) are states on exactly the same C^* -algebra.

Definition 8.2 (resolution-separated diagonal). Let $M(N) \leq N$ satisfy

$$M(N) \rightarrow \infty, \quad N - M(N) \rightarrow \infty. \quad (8.3)$$

The first condition makes the tested observables dense in the continuum algebra; the second prevents this comparison class from containing the particular finest-scale $M = N$ carrier used in Theorem 4.1. These two conditions make the topology admissible; they do not by themselves prove convergence.

Definition 8.3 (corrected interacting Part II target). For each fixed M , bounded lattice region $X \subset \Lambda_M$, and $A \in \mathcal{W}_M(X)$,

$$\left| \omega_{N,g}^{[M]}(A) - \omega_\infty^{[M]}(A) \right| \leq \|A\| \Psi_\gamma(d) + E_{M,X,A}(N), \quad (8.4)$$

where $E_{M,X,A}(N) \rightarrow 0$ as $N \rightarrow \infty$, uniformly in the spatial cutoff once $g = 1$ on the required neighbourhood. A stronger uniform estimate may be sought along (8.3), but no supremum over all $M \leq N$ can tend to zero.

Proposition 8.4 (what is proved about the corrected target). *The left side of (8.4) is well-defined. In the free full-space case MMST prove that it tends to zero for each fixed M and A . No estimate (8.4) can hold uniformly over a class containing $M = N$ with $E_{M,X,A}(N) \rightarrow 0$ uniformly over its norm-one local observables.*

Proof. Definitions 1.1 and 8.1 put both functionals on $\mathcal{W}_M$, proving the first assertion. MMST Proposition 4.5 and Lemma 4.3 give the second assertion. For the third, take $M = N$, $X = X_N$, and the norm-one observable $A = W_N(ip_N)$ of Theorem 4.1. The left side then has lower limit at least $c_* > 0$ by (4.3), contradicting any uniform error tending to zero. $\square$

The corrected dependency order is therefore:

- II.0. Choose one lattice model.** Decide between the MMST nearest-neighbour Hamiltonian, a Daubechies Galerkin Hamiltonian, or another action. These choices have different symbols, interactions, counterterms, and locality estimates.
- II.1. Checkpoint completed.** The diagonal local-norm target fails by Theorem 4.1. Fix either the fixed-coarse topology (8.4) or a precise resolution-separated norm before proceeding.
- II.2. Free estimates partially completed.** Equations (5.1)–(5.5) give the MMST gap, group velocity, and Wick asymptotic. A separate interacting Lieb–Robinson theorem is still required.
- II.3. Source audit completed negatively.** Battle–Federbush is mechanism precedent, not $CE1_N/CE2_N$ for the chosen OAR action.
- II.4. Uniform interacting expansion remains open.** State it directly in the fixed-coarse or resolution-separated topology, including counterterms and source norms.
- II.5. LPPL with lattice tails remains open.** It must be formulated on one finite-scale algebra and must include quantitative commutator errors for affiliated unbounded Wick polynomials.
- II.6. Replace the invalid vertical interpolation.** Use a common Euclidean interpolation followed by a proved OS identification, or compare projective states directly. Do not add H_N and H across different Hilbert spaces.

9 Claim ledger

Claim	Status	Exact reason
MMST supplies the free benchmark	PASS	Fixed-coarse weak* projective limit; Proposition 4.5 and Theorem 4.11.
The displayed Part II norm is defined	FAIL	An observable embedding has no canonical push-forward on states.
Diagonal local state norm is $O(\varepsilon_N^\theta)$	FAIL	Explicit local Weyl lower bound (4.3)–(4.4).
N -uniform free gap	PASS	Exact identity gap $H_N = m$.
N -uniform free group velocity	PASS	Exact derivative bound (5.3).
Wick constant coefficient	PASS	Explicit asymptotic (5.5).
Group velocity alone gives interacting LR	FAIL	It controls only the free dispersion; an anharmonic commutator theorem is separate.
Battle supplies $CE1_N/CE2_N$	FAIL	Different variables/action and no theorem in the required norms or mixed family.

Claim	Status	Exact reason
$H_s = (1 - s)H_N + sH$ is defined	FAIL	Operators live on different Hilbert spaces; no common form realization is specified.
Fixed-coarse interacting OAR estimate	CONDITIONAL	Meaningful target; free case known pointwise, interacting uniform expansion still absent.
Full Part II OAR closure as originally written	FAIL	False topology plus undefined interpolation; it cannot be repaired by estimating a different exponent.

10 Reproducibility

The script `m2_ii_checkpoint.py` evaluates the finite-torus quadratic forms in (2.1) and the universal lower form from (2.9), the group-velocity bound (5.3), and the Wick integral (5.4). It asserts convergence to the exact constants in (4.1)–(4.4) and (5.5). The numerical output is diagnostic only; every limiting statement used in the proof was derived analytically above.

References

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- [2] G. Battle and P. Federbush, *Ondelettes and phase cell cluster expansions, a vindication*, Commun. Math. Phys. **109** (1987), 417–419, [doi:10.1007/BF01206144](https://doi.org/10.1007/BF01206144).

Part II static projective compactness

A fixed-coarse interacting scaling state along a cofinal subnet

Lamport-form proof

16 August 2026

Unconditional static conclusion. Once one state has been chosen on every finite-resolution algebra, weak-star compactness produces a cofinal subnet whose pullbacks converge simultaneously on every fixed coarse algebra. The limiting family is projectively consistent and therefore defines a state on the C^* -inductive limit.

The theorem proves neither convergence of the full sequence nor identification with the continuum $P(\phi)_2$ vacuum. It gives no rate, no local-state-norm convergence, no normality, and no limiting dynamics.

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1 Statement

Definition 1.1 (directed algebraic system). For $M \leq N$, let $\mathfrak{A}_M, \mathfrak{A}_N$ be unital C^* -algebras and let

$$\alpha_M^N : \mathfrak{A}_M \longrightarrow \mathfrak{A}_N \tag{1.1}$$

be injective unital $*$ -homomorphisms such that

$$\alpha_N^K \circ \alpha_M^N = \alpha_M^K, \quad M \leq N \leq K. \tag{1.2}$$

Let $(\mathfrak{A}_\infty, \alpha_M^\infty)$ be their C^* -inductive limit, so

$$\alpha_N^\infty \circ \alpha_M^N = \alpha_M^\infty, \quad \overline{\bigcup_M \alpha_M^\infty(\mathfrak{A}_M)}^{\|\cdot\|} = \mathfrak{A}_\infty. \tag{1.3}$$

Theorem 1.2 (cofinal-subnet projective limit). *For every N , let ω_N be a state on $\mathfrak{A}_N$. Then there exist a directed set I , a cofinal subnet map $i \mapsto N(i) \in \mathbb{N}_0$, states $\rho_M \in S(\mathfrak{A}_M)$, and a state*

$\Omega_\infty \in S(\mathfrak{A}_\infty)$ such that

$$\lim_{i \in I} \omega_{N(i)}(\alpha_M^{N(i)}(A)) = \rho_M(A) = \Omega_\infty(\alpha_M^\infty(A)), \quad (1.4)$$

$$\rho_M = \rho_{M'} \circ \alpha_M^{M'}, \quad M \leq M', \quad (1.5)$$

for every fixed M and every $A \in \mathfrak{A}_M$.

2 Proof of Theorem 1.2

⟨1⟩ 1 Prove weak-star compactness at one scale.

The closed unit ball of $\mathfrak{A}_M^*$ is compact in $\sigma(\mathfrak{A}_M^*, \mathfrak{A}_M)$ by Banach–Alaoglu. Within that ball,

$$S(\mathfrak{A}_M) = \{\varphi : \varphi(1) = 1, \varphi(A^*A) \geq 0 \text{ for every } A \in \mathfrak{A}_M\}. \quad (2.1)$$

For fixed A , both maps

$$\varphi \mapsto \varphi(1), \quad \varphi \mapsto \varphi(A^*A) \quad (2.2)$$

are weak-star continuous. Hence every condition in (2.1) is weak-star closed. Therefore $S(\mathfrak{A}_M)$ is weak-star compact.

⟨1⟩ 2 Put all coarse restrictions into one compact space.

Define

$$K := \prod_{M=0}^{\infty} S(\mathfrak{A}_M), \quad (2.3)$$

with the product of the weak-star topologies. Tychonoff’s theorem makes K compact. For every N , define $x_N = (x_N^M)_{M \geq 0} \in K$ by

$$x_N^M := \begin{cases} \omega_N \circ \alpha_M^N, & M \leq N, \\ \omega_M, & M > N. \end{cases} \quad (2.4)$$

The second case only fills coordinates that are eventually irrelevant.

⟨1⟩ 3 Extract a genuinely cofinal subnet.

Every net in a compact space has a convergent subnet. Hence there are a directed set I , a subnet map $N : I \rightarrow \mathbb{N}_0$, and $(\rho_M)_M \in K$ such that

$$x_{N(i)} \longrightarrow (\rho_M)_M \text{ in } K. \quad (2.5)$$

By the definition of a subnet, the map $i \mapsto N(i)$ is cofinal: for every M' , there is i_0 such that $N(i) \geq M'$ whenever $i \geq i_0$. Product convergence in (2.5) means, explicitly,

$$\lim_i x_{N(i)}^M(A) = \rho_M(A), \quad A \in \mathfrak{A}_M. \quad (2.6)$$

⟨1⟩ 4 Prove projective consistency.

Fix $M \leq M'$ and $A \in \mathfrak{A}_M$. Cofinality implies that $N(i) \geq M'$ eventually. For those i , (1.2) gives

$$x_{N(i)}^{M'}(\alpha_M^{M'}(A)) = \omega_{N(i)}(\alpha_{M'}^{N(i)}(\alpha_M^{M'}(A))) \quad (1)$$

$$= \omega_{N(i)}(\alpha_M^{N(i)}(A)) = x_{N(i)}^M(A). \quad (2.7)$$

Take the limit in (2.7) and use (2.6):

$$\rho_{M'}(\alpha_M^{M'}(A)) = \rho_M(A). \quad (2.8)$$

This is exactly (1.5).

⟨1⟩ 5 Define the functional on the algebraic inductive union.

Set

$$\mathfrak{A}_{\text{alg}} := \bigcup_M \alpha_M^\infty(\mathfrak{A}_M) \quad (2.9)$$

and prescribe

$$\Omega_0(\alpha_M^\infty(A)) := \rho_M(A). \quad (2.10)$$

To prove that (2.10) is well-defined, suppose

$$\alpha_M^\infty(A) = \alpha_{M'}^\infty(B). \quad (2.11)$$

Choose $K \geq M, M'$. By (1.3) and injectivity of α_K^∞ ,

$$\alpha_M^K(A) = \alpha_{M'}^K(B). \quad (2.12)$$

Applying ρ_K and then (2.8) gives

$$\rho_M(A) = \rho_K(\alpha_M^K(A)) = \rho_K(\alpha_{M'}^K(B)) = \rho_{M'}(B). \quad (2.13)$$

Thus (2.10) is independent of the representative.

⟨1⟩ 6 Prove positivity, normalization, and boundedness.

Every finite collection of elements of $\mathfrak{A}_{\text{alg}}$ lies in the image of one $\mathfrak{A}_K$. Consequently linearity of Ω_0 follows from linearity of ρ_K . If $C = \alpha_M^\infty(A)$, then

$$\Omega_0(C^*C) = \rho_M(A^*A) \geq 0, \quad \Omega_0(1) = 1. \quad (2.14)$$

For completeness, positivity implies the Cauchy–Schwarz inequality

$$|\rho_M(A)|^2 \leq \rho_M(A^*A)\rho_M(1) \leq \|A\|^2. \quad (2.15)$$

Because every injective $*$ -homomorphism of C^* -algebras is isometric,

$$|\Omega_0(\alpha_M^\infty(A))| \leq \|A\| = \|\alpha_M^\infty(A)\|. \quad (2.16)$$

Thus Ω_0 is bounded with norm one.

⟨1⟩ 7 Extend and identify the coordinate limits.

By (1.3), $\mathfrak{A}_{\text{alg}}$ is norm dense in $\mathfrak{A}_\infty$. Therefore Ω_0 extends uniquely and continuously to a state Ω_∞ on $\mathfrak{A}_\infty$. Equations (2.6) and (2.10) give (1.4). $\square$

3 Spatial-cutoff interface

Corollary 3.1 (iterated cutoff then resolution limit). Suppose that for each N there is a directed cutoff family of states $\{\omega_{N,g}\}_g \subset S(\mathfrak{A}_N)$ satisfying

$$\lim_{g \uparrow 1} \omega_{N,g}(B) = \omega_N(B), \quad B \in \mathfrak{A}_N. \quad (3.1)$$

Then the subnet obtained in Theorem 1.2 satisfies

$$\lim_i \left(\lim_{g \uparrow 1} \omega_{N(i),g}(\alpha_M^{N(i)}(A)) \right) = \Omega_\infty(\alpha_M^\infty(A)) \quad (3.2)$$

for every fixed M and $A \in \mathfrak{A}_M$.

Proof. For fixed i , (3.1), applied to $B = \alpha_M^{N(i)}(A)$, makes the inner limit equal to $\omega_{N(i)}(\alpha_M^{N(i)}(A))$. The outer limit is (1.4). $\square$

Remark 3.2 (no joint-limit inference). Equation (3.2) is an iterated limit. It does not imply a joint limit. The scalar array

$$a_{N,R} := \mathbf{1}_{\{R < N\}} \quad (3.3)$$

satisfies $\lim_{R \rightarrow \infty} a_{N,R} = 0$ for every fixed N , but the cofinal diagonal $R = N - 1$ has $a_{N,N-1} = 1$.

4 Preserved compatible symmetries

Proposition 4.1. *Suppose $\vartheta_N \in \text{Aut}(\mathfrak{A}_N)$ obey*

$$\vartheta_N \alpha_M^N = \alpha_M^N \vartheta_M \quad (4.1)$$

and $\omega_N \vartheta_N = \omega_N$ for every N . Then there is an induced $\vartheta_\infty \in \text{Aut}(\mathfrak{A}_\infty)$, and the state from Theorem 1.2 satisfies

$$\Omega_\infty \vartheta_\infty = \Omega_\infty. \quad (4.2)$$

Proof. Fix M, A . Eventually $N(i) \geq M$, and

$$\omega_{N(i)}(\alpha_M^{N(i)}(\vartheta_M A)) = \omega_{N(i)}(\vartheta_{N(i)}(\alpha_M^{N(i)} A)) = \omega_{N(i)}(\alpha_M^{N(i)} A). \quad (4.3)$$

Taking the limit gives $\rho_M \vartheta_M = \rho_M$. Equation (4.2) first follows on $\mathfrak{A}_{\text{alg}}$ and then on $\mathfrak{A}_\infty$ by norm continuity. $\square$

5 Sharp limitations

Proposition 5.1 (the compactness conclusion cannot be upgraded formally). *The hypotheses of Theorem 1.2 imply none of the following:*

1. *convergence of the full sequence;*
2. *convergence in the norm of state functionals;*
3. *normality of the limiting state in a prescribed representation;*
4. *convergence of dynamics or the ground-state spectral condition.*

Proof. For (1), take $\mathfrak{A}_N = \mathbb{C}^2$, all maps equal to the identity, and alternate the two point-evaluation states.

For (2)–(3), take the constant system $\mathfrak{A}_N = B(\ell^2(\mathbb{N}))$ and $\omega_n(A) = \langle e_n, A e_n \rangle$. Every weak-star cluster state ρ annihilates compact operators because $\langle e_n, K e_n \rangle \rightarrow 0$ for every compact K . Hence ρ is singular. For $P_n = |e_n\rangle\langle e_n|$ and $U_n = 2P_n - I$,

$$\|U_n\| = 1, \quad \omega_n(U_n) = 1, \quad \rho(U_n) = -1. \quad (5.1)$$

Therefore $\|\omega_n - \rho\| = 2$, so no norm convergence occurs.

For (4), the hypotheses mention neither automorphism groups nor Hamiltonians. Consequently no statement involving them is a logical consequence. More concretely, one may put arbitrary, mutually incompatible dynamics on the constant algebra without changing any hypothesis or any step of the proof. $\square$

Remark 5.2 (subnet versus subsequence). No separability assumption was made. Thus weak-star compactness yields a subnet, not necessarily a subsequence. If every $\mathfrak{A}_M$ is separable, the state spaces are weak-star metrizable on bounded sets. Choose successively a convergent subsequence at levels $M = 0, 1, 2, \dots$, each contained in the preceding one, and take the diagonal sequence whose j th term is the j th term of the level- j subsequence. For every fixed M its tail lies in the level- M subsequence, so it converges on $\mathfrak{A}_M$; the consistency proof above then applies unchanged.

Part II free fixed-coarse rate

An explicit $D4$ filter estimate and covariance-symbol proof

Lamport-form proof

16 August 2026

Quantitative free theorem. Fix a coarse scale M , a torus $\mathbb{T}_L$, and a coarse Weyl generator $W_M(\varepsilon_M q + ip)$. For the MMST nearest-neighbour free vacua and the exact Daubechies $D4$ observable embeddings,

$$\left| \omega_{M+r,L}^{(0)}\left(\alpha_M^{M+r}(W_M(\varepsilon_M q + ip))\right) - \omega_L^{(0)}(\beta_M(W_M(\varepsilon_M q + ip))) \right| \leq C_{M,L,m,q,p} 2^{-\theta r}, \quad (\text{R})$$

for all sufficiently large r , where

$$\eta = 2(1 - \log_2 \sqrt{3}) > 0, \quad \theta = \frac{2\eta}{5 + \eta} > 0. \quad (\text{E})$$

Every constant is displayed in Section 4. The theorem holds in the correct fixed-coarse topology and makes no diagonal $M = N$ assertion.

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1 Exact covariance formulas

Fix M , and put $N = M + r$. Use all conventions from the companion model sheet. Write

$$\Gamma_\infty := \frac{\pi}{L}\mathbb{Z}, \quad \gamma(k) := \sqrt{m^2 + k^2}, \quad \gamma_N(k) := \sqrt{m^2 + 4\varepsilon_N^{-2} \sin^2(\varepsilon_N k/2)}. \quad (1.1)$$

Extend the coarse Fourier transforms $\widehat{q}_M, \widehat{p}_M$ periodically from Γ_M to Γ_∞ , with period $2\pi/\varepsilon_M$.

Define the finite and infinite mask products

$$P_r(\ell) := \prod_{j=1}^r m_0(2^{-j}\ell), \quad P_\infty(\ell) := \prod_{j=1}^\infty m_0(2^{-j}\ell) = \widehat{\phi}(\ell). \quad (1.2)$$

The equality in (1.2) is the Fourier form of the $D4$ scaling equation with $\widehat{\phi}(0) = 1$.

Lemma 1.1 (pulled-back fine covariance). *For $\xi = \varepsilon_M q + ip$,*

$$\omega_{M+r,L}^{(0)}(\alpha_M^{M+r}(W_M(\xi))) = e^{-Q_{r,M}(q,p)/4}, \quad (1.3)$$

where

$$Q_{r,M}(q,p) = \frac{\varepsilon_M}{2L} \sum_{k \in \Gamma_\infty} \mathbf{1}_{\Gamma_{M+r}}(k) |P_r(\varepsilon_M k)|^2 \left(\gamma_{M+r}(k)^{-1} |\widehat{q}_M(k)|^2 + \gamma_{M+r}(k) |\widehat{p}_M(k)|^2 \right). \quad (1.4)$$

Proof in Lamport form

⟨1⟩ 1 Iterate the momentum-space refinement map.

The one-step MMST map is multiplication by $\sqrt{2} m_0(\varepsilon_{j+1} k)$, together with periodic extension of the coarse Fourier vector. Therefore

$$R_M^{M+r}(q,p)(k) = 2^{r/2} P_r(\varepsilon_M k)(\widehat{q}_M(k), \widehat{p}_M(k)), \quad k \in \Gamma_{M+r}. \quad (1.5)$$

⟨1⟩ 2 Insert (1.5) in the fine vacuum covariance.

The fine normalized momentum sum has weight

$$\frac{1}{2r_{M+r}} = \frac{2^{-r}}{2r_M} = 2^{-r} \frac{\varepsilon_M}{2L}. \quad (1.6)$$

The squared factor $2^{r/2}$ in (1.5) cancels the 2^{-r} in (1.6). Substitution in the exact free state formula gives (1.3)–(1.4). $\square$

Lemma 1.2 (continuum pullback covariance).

$$\omega_L^{(0)}(\beta_M(W_M(\xi))) = e^{-Q_{\infty,M}(q,p)/4}, \quad (1.7)$$

where

$$Q_{\infty,M}(q,p) = \frac{\varepsilon_M}{2L} \sum_{k \in \Gamma_\infty} |P_\infty(\varepsilon_M k)|^2 \left(\gamma(k)^{-1} |\widehat{q}_M(k)|^2 + \gamma(k) |\widehat{p}_M(k)|^2 \right). \quad (1.8)$$

Proof. The continuum embedding sends q, p to linear combinations of the periodized scaling functions. Their Fourier coefficients equal $\varepsilon_M^{1/2} e^{-ikx} \widehat{\phi}(\varepsilon_M k)$. Parseval's identity on $\mathbb{T}_L$, followed by the continuum free-vacuum formula, gives (1.8). Equation (1.2) then identifies the multiplier with P_∞ . $\square$

2 A uniform algebraic envelope for the exact D4 product

Recall the exact factorization

$$m_0(t) = \left(\frac{1 + e^{-it}}{2} \right)^2 D(t), \quad D(t) = \frac{(1 + \sqrt{3}) + (1 - \sqrt{3})e^{-it}}{2}, \quad (2.1)$$

and

$$|D(t)|^2 = 2 - \cos t. \quad (2.2)$$

Set

$$a := \log_2 \sqrt{3} < 1, \quad \eta := 2 - 2a = 2(1 - \log_2 \sqrt{3}) > 0, \quad C_D := 48\pi^4 e^{1/6}. \quad (2.3)$$

Lemma 2.1 (uniform finite-product decay). *For every $r \geq 0$ and every ℓ satisfying $|\ell| \leq \pi 2^r$,*

$$|P_r(\ell)|^2 \leq C_D (1 + |\ell|)^{-2-\eta}. \quad (2.4)$$

The same estimate holds for P_∞ on $\mathbb{R}$.

Proof in Lamport form

⟨1⟩ 1 Bound the D -factor product.

Let

$$n(\ell) := \lceil \log_2 \max(1, |\ell|) \rceil. \quad (2.5)$$

By (2.2), $|D(t)| \leq \sqrt{3}$. Also

$$|D(t)|^2 = 1 + (1 - \cos t) \leq e^{t^2/2}, \quad |D(t)| \leq e^{t^2/4}. \quad (2.6)$$

For $j > n(\ell)$, use (2.6); for $j \leq n(\ell)$, use $|D| \leq \sqrt{3}$. Since

$$\sum_{j=n+1}^{\infty} 4^{-j} = \frac{4^{-n}}{3}, \quad |\ell|^2 4^{-n(\ell)} \leq 1, \quad (2.7)$$

we obtain, for every finite r ,

$$\prod_{j=1}^r |D(2^{-j}\ell)| \leq (\sqrt{3})^{n(\ell)} \exp\left(\frac{\ell^2}{4} \sum_{j=n(\ell)+1}^{\infty} 4^{-j}\right) \quad (1)$$

$$\leq \sqrt{3} e^{1/12} (1 + |\ell|)^a. \quad (2.8)$$

If $r < n(\ell)$, the omitted tail factor in the first line is absent and the same upper bound remains valid.

⟨1⟩ 2 Bound the cosine product.

For $r \geq 1$, the finite sine-product identity gives

$$\prod_{j=1}^r \left| \cos\left(\frac{\ell}{2^{j+1}}\right) \right| = \frac{|\sin(\ell/2)|}{2^r |\sin(\ell/2^{r+1})|}. \quad (2.9)$$

When $|\ell| \leq \pi 2^r$, the denominator angle has absolute value at most $\pi/2$. The elementary bound $|\sin u| \geq 2|u|/\pi$ on that interval yields

$$\prod_{j=1}^r \left| \cos\left(\frac{\ell}{2^{j+1}}\right) \right| \leq \min\left(1, \frac{\pi |\sin(\ell/2)|}{|\ell|}\right). \quad (2.10)$$

⟨1⟩ 3 Combine the factors.

Equations (2.1), (2.8), and (2.10) imply

$$|P_r(\ell)|^2 \leq 3e^{1/6} (1 + |\ell|)^{2a} \min(1, \pi^4 |\ell|^{-4}). \quad (2.11)$$

For $|\ell| \leq 1$, multiply the desired right side by $(1 + |\ell|)^4 \leq 16$. For $|\ell| \geq 1$, use $(1 + |\ell|)^4 |\ell|^{-4} \leq 16$. In both cases,

$$|P_r(\ell)|^2 \leq 48\pi^4 e^{1/6} (1 + |\ell|)^{-4+2a} = C_D (1 + |\ell|)^{-2-\eta}. \quad (2.12)$$

For $r = 0$, $P_0 = 1$ and $|\ell| \leq \pi$; the claimed bound follows from $C_D \geq (1 + \pi)^{2+\eta}$, which holds for $C_D = 48\pi^4 e^{1/6}$. Finally, take $r \rightarrow \infty$ in (2.12) and use (1.2) to obtain the infinite-product bound.

□

Lemma 2.2 (explicit filter-tail error). *If $|\ell| \leq 2^r$, then*

$$0 \leq |P_r(\ell)|^2 - |P_\infty(\ell)|^2 \leq \frac{\sqrt{3}}{12} \ell^2 4^{-r}. \quad (2.13)$$

Proof in Lamport form

⟨1⟩ 1 Compute the second derivative exactly.

Let $u(t) := |m_0(t)|^2$. From (2.1)–(2.2),

$$u(t) = \frac{2 + 3 \cos t - \cos^3 t}{4}, \quad u'(t) = -\frac{3}{4} \sin^3 t, \quad u''(t) = -\frac{9}{4} \sin^2 t \cos t. \quad (2.14)$$

The maximum of $x(1 - x^2)$ on $[0, 1]$ is $2/(3\sqrt{3})$, attained at $x = 1/\sqrt{3}$. Hence

$$\sup_t |u''(t)| = \frac{\sqrt{3}}{2}. \quad (2.15)$$

Because $u(0) = 1$ and $u'(0) = 0$, Taylor's theorem gives

$$0 \leq 1 - u(t) \leq \frac{\sqrt{3}}{4} t^2. \quad (2.16)$$

The left inequality follows independently from the QMF identity $u(t) + u(t + \pi) = 1$.

⟨1⟩ 2 Estimate the omitted infinite product.

Since every $u(t) \in [0, 1]$,

$$1 - \prod_{j=r+1}^{\infty} u(2^{-j}\ell) \leq \sum_{j=r+1}^{\infty} (1 - u(2^{-j}\ell)). \quad (2.17)$$

For $|\ell| \leq 2^r$, all arguments on the right have absolute value at most $1/2$. Equations (2.16)–(2.17) give

$$1 - \prod_{j=r+1}^{\infty} u(2^{-j}\ell) \leq \frac{\sqrt{3}}{4} \ell^2 \sum_{j=r+1}^{\infty} 4^{-j} = \frac{\sqrt{3}}{12} \ell^2 4^{-r}. \quad (2.18)$$

Multiplication by $|P_r(\ell)|^2 \leq 1$ proves (2.13). $\square$

3 Dispersion error

Lemma 3.1 (nearest-neighbour versus continuum symbol). *If $|\varepsilon_N k| \leq \pi$, then*

$$0 \leq \gamma(k) - \gamma_N(k) \leq \frac{\varepsilon_N^2 k^4}{24m}, \quad (3.1)$$

$$0 \leq \gamma_N(k)^{-1} - \gamma(k)^{-1} \leq \frac{\varepsilon_N^2 k^4}{24m^3}. \quad (3.2)$$

Proof in Lamport form

⟨1⟩ 1 Bound the discrete Laplacian symbol.

For $0 \leq x \leq \pi/2$,

$$\sin x \leq x, \quad \sin x \geq x - \frac{x^3}{6}. \quad (3.3)$$

For the second inequality, the derivative of $\sin x - x + x^3/6$ is nonnegative because $\cos x \geq 1 - x^2/2$. Since $x - x^3/6 \geq 0$ on this interval,

$$x^2 - \frac{x^4}{3} \leq \left(x - \frac{x^3}{6}\right)^2 \leq \sin^2 x \leq x^2. \quad (3.4)$$

Putting $x = |\varepsilon_N k|/2$ gives

$$0 \leq k^2 - 4\varepsilon_N^{-2} \sin^2(\varepsilon_N k/2) \leq \frac{\varepsilon_N^2 k^4}{12}. \quad (3.5)$$

⟨1⟩ 2 Pass to square roots and inverses.

Since both dispersions are at least m ,

$$\gamma - \gamma_N = \frac{\gamma^2 - \gamma_N^2}{\gamma + \gamma_N} \leq \frac{\varepsilon_N^2 k^4}{24m}, \quad (3.6)$$

which is (3.1). Moreover,

$$\gamma_N^{-1} - \gamma^{-1} = \frac{\gamma - \gamma_N}{\gamma\gamma_N} \leq \frac{\varepsilon_N^2 k^4}{24m^3}, \quad (3.7)$$

which is (3.2). $\square$

4 The explicit rate

Put

$$A_q := \sup_{k \in \Gamma_\infty} |\widehat{q}_M(k)|, \quad A_p := \sup_{k \in \Gamma_\infty} |\widehat{p}_M(k)|, \quad (4.1)$$

which are finite because the extensions are periodic. Define

$$\kappa := \frac{\pi}{L}, \quad b := \varepsilon_M \kappa = \frac{\pi}{r_M}, \quad c_0 := \frac{\sqrt{3}}{12}. \quad (4.2)$$

Set

$$C_{\text{low}} := \frac{3\varepsilon_M^3}{2L} \left\{ A_q^2 \left(\frac{\kappa^4}{24m^3} + \frac{c_0 \kappa^2}{m} \right) \right. \quad (2)$$

$$\left. + A_p^2 \left(\frac{\kappa^4}{24m} + c_0(m + \kappa)\kappa^2 \right) \right\}, \quad (4.3)$$

$$C_* := 2C_D \left(\frac{A_q^2}{m} + A_p^2(m + \varepsilon_M^{-1}) \right). \quad (4.4)$$

Finally, let

$$\alpha := \frac{2}{5 + \eta}, \quad \theta := \alpha\eta = \frac{2\eta}{5 + \eta} = 2 - 5\alpha. \quad (4.5)$$

Theorem 4.1 (fixed-coarse power rate). *Let r_* be the least integer satisfying*

$$2^{\alpha r_*} \geq \max(1, 2b). \quad (4.6)$$

For every $r \geq r_$,*

$$|Q_{r,M}(q, p) - Q_{\infty,M}(q, p)| \leq C_Q 2^{-\theta r}, \quad (4.7)$$

where

$$C_Q := C_{\text{low}}(1 + b^{-1})^5 + \frac{2^\eta C_*}{\pi\eta}. \quad (4.8)$$

Consequently,

$$\left| \omega_{M+r,L}^{(0)}(\alpha_M^{M+r}(W_M(\varepsilon_M q + ip))) - \omega_L^{(0)}(\beta_M(W_M(\varepsilon_M q + ip))) \right| \leq \frac{C_Q}{4} 2^{-\theta r}. \quad (4.9)$$

Proof in Lamport form

⟨1⟩ 1 Choose an integer momentum split.

For $r \geq r_*$, put

$$J_r := \left\lfloor b^{-1} 2^{\alpha r} \right\rfloor. \quad (4.10)$$

Then $J_r \geq 1$,

$$bJ_r \leq 2^{\alpha r} \leq 2^r, \quad bJ_r \geq 2^{\alpha r} - b \geq \frac{1}{2} 2^{\alpha r}, \quad (4.11)$$

and every mode $k_n = \kappa n$ with $|n| \leq J_r$ lies in Γ_{M+r} . The first inequality in (4.11) also permits use of Lemma 2.2.

⟨1⟩ 2 Bound the low-momentum part.

For $|n| \leq J_r$, split the momentum integrand as

$$|\gamma_N |P_r|^2 - \gamma |P_\infty|^2| \leq |\gamma_N - \gamma| |P_r|^2 + \gamma ||P_r|^2 - |P_\infty|^2|, \quad (4.12)$$

$$|\gamma_N^{-1} |P_r|^2 - \gamma^{-1} |P_\infty|^2| \leq |\gamma_N^{-1} - \gamma^{-1}| |P_r|^2 + \gamma^{-1} ||P_r|^2 - |P_\infty|^2|. \quad (4.13)$$

Here every mask argument is $\ell = \varepsilon_M k_n$. Lemmas 2.2 and 3.1, together with $\varepsilon_{M+r} = 2^{-r} \varepsilon_M$, give

$$q\text{-integrand error} \leq 4^{-r} \varepsilon_M^2 A_q^2 \left(\frac{k_n^4}{24m^3} + \frac{c_0 k_n^2}{m} \right), \quad (4.14)$$

$$p\text{-integrand error} \leq 4^{-r} \varepsilon_M^2 A_p^2 \left(\frac{k_n^4}{24m} + c_0 \gamma(k_n) k_n^2 \right). \quad (4.15)$$

Because

$$2J + 1 \leq 3(1 + J), \quad |k_n| \leq \kappa(1 + J), \quad \gamma(k_n) \leq m + |k_n| \leq (m + \kappa)(1 + J), \quad (4.16)$$

the definition (4.3) yields

$$E_{\text{low}}(r) \leq 4^{-r} C_{\text{low}}(1 + J_r)^5 \leq C_{\text{low}}(1 + b^{-1})^5 2^{-(2-5\alpha)r}. \quad (4.17)$$

⟨1⟩ 3 Bound both high-momentum tails.

Lemma 2.1 and

$$\gamma_{M+r}(k) \leq \gamma(k) \leq m + |k| \leq (m + \varepsilon_M^{-1})(1 + \varepsilon_M |k|) \quad (4.18)$$

show that the sum of the absolute fine and continuum integrands at $k_n = \kappa n$ is at most

$$C_*(1 + b|n|)^{-1-\eta}. \quad (4.19)$$

The fine term is declared zero outside Γ_{M+r} , exactly as in (1.4). Since $x \mapsto (1 + bx)^{-1-\eta}$ is decreasing,

$$\sum_{|n| > J} (1 + b|n|)^{-1-\eta} \leq 2 \int_J^\infty (1 + bx)^{-1-\eta} dx = \frac{2}{b\eta} (1 + bJ)^{-\eta}. \quad (4.20)$$

Multiplication by the covariance weight $\varepsilon_M/(2L)$, followed by $b = \pi \varepsilon_M/L$ and (4.11), gives

$$E_{\text{high}}(r) \leq \frac{C_*}{\pi\eta} (1 + bJ_r)^{-\eta} \leq \frac{2^\eta C_*}{\pi\eta} 2^{-\alpha\eta r}. \quad (4.21)$$

⟨1⟩ 4 Match the exponents.

By (4.5),

$$2 - 5\alpha = \alpha\eta = \theta. \quad (4.22)$$

Adding (4.17) and (4.21) proves (4.7)–(4.8).

⟨1⟩ 5 Pass from covariance forms to states.

For $x, y \geq 0$, the mean-value theorem gives

$$|e^{-x/4} - e^{-y/4}| = \frac{1}{4}e^{-\zeta/4}|x - y| \leq \frac{1}{4}|x - y| \quad (4.23)$$

for some ζ between x and y . Apply (4.23) to $Q_{r,M}$ and $Q_{\infty,M}$ and use (4.7). This is (4.9). $\square$

Corollary 4.2 (lattice-spacing form). Because $\varepsilon_{M+r}/\varepsilon_M = 2^{-r}$, (4.9) is equivalently

$$|\omega_{M+r,L}^{(0)} \circ \alpha_M^{M+r}(W_M(\xi)) - \omega_L^{(0)} \circ \beta_M(W_M(\xi))| \leq \frac{C_Q}{4} \left(\frac{\varepsilon_{M+r}}{\varepsilon_M} \right)^\theta. \quad (4.24)$$

5 Claim ledger

Claim	Status	Exact reason
Fixed-coarse free convergence	PASS	The covariance formulas are (1.4) and (1.8), and Theorem 4.1 supplies a power rate.
Uniform finite $D4$ filter tail	PASS	Lemmas 2.1 and 2.2 prove both the global envelope and the low-frequency truncation error from the exact four mask coefficients.
Finest-scale local state norm	FAIL	This theorem fixes M and sends $r = N - M \rightarrow \infty$; it does not include $M = N$.
Interacting covariance reduction	FAIL	Interacting ground states are not Gaussian, so no quadratic covariance symbol determines their Weyl expectations.

References

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Part II interacting Weyl reduction

The exact scalar convergence criterion and the excluded quartic LR citation

Lamport-form proof and proof-obligation ledger

16 August 2026

Reduction theorem. To obtain a full fixed-coarse weak-star scaling limit, it is sufficient to prove convergence of the single scalar characteristic functional

$$\chi_{N,M}(\xi) := \omega_N(\alpha_M^N(W_M(\xi)))$$

for every fixed M and every coarse one-particle vector ξ . Positivity, boundedness, extension from Weyl polynomials, projective consistency, and continuum identification are then automatic by the proved steps below.

This static reduction requires neither an interacting Lieb–Robinson bound nor a uniform spectral gap. Those inputs are needed only for a quantitative spatial-cutoff or dynamical theorem.

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1 Characteristic-functional criterion

Let $(\mathcal{W}_N, \alpha_M^N)$ be the Weyl C^* -inductive system fixed in the model sheet. Let

$$\mathcal{D}_M := \text{span}\{W_M(\xi) : \xi \in h_{M,L}\}. \quad (1.1)$$

By the definition of the Weyl C^* -algebra,

$$\overline{\mathcal{D}_M}^{\|\cdot\|} = \mathcal{W}_M. \quad (1.2)$$

Theorem 1.1 (full-sequence Weyl criterion). *For every N , let $\omega_N \in S(\mathcal{W}_N)$. Assume that, for every fixed M and every $\xi \in h_{M,L}$, the limit*

$$\chi_M(\xi) := \lim_{N \rightarrow \infty, N \geq M} \omega_N(\alpha_M^N(W_M(\xi))) \quad (1.3)$$

exists. Then:

1. *there is a unique state $\rho_M \in S(\mathcal{W}_M)$ satisfying $\rho_M(W_M(\xi)) = \chi_M(\xi)$;*
2. *for every $A \in \mathcal{W}_M$,*

$$\lim_{N \rightarrow \infty} \omega_N(\alpha_M^N(A)) = \rho_M(A); \quad (1.4)$$

3. *the family is projectively consistent,*

$$\rho_M = \rho_{M'} \circ \alpha_M^{M'}, \quad M \leq M'; \quad (1.5)$$

4. *there is a unique state $\Omega_\infty \in S(\mathcal{W}_\infty)$ with*

$$\Omega_\infty \circ \alpha_M^\infty = \rho_M. \quad (1.6)$$

2 Proof of Theorem 1.1

⟨1⟩ 1 Define the limit on finite Weyl polynomials.

For

$$D = \sum_{j=1}^s c_j W_M(\xi_j) \in \mathcal{D}_M, \quad (2.1)$$

set

$$\rho_M^{(0)}(D) := \sum_{j=1}^s c_j \chi_M(\xi_j). \quad (2.2)$$

This is independent of the representation (2.1). Indeed, if $D = 0$ in $\mathcal{W}_M$, then $\alpha_M^N(D) = 0$ for every N , and hence

$$\sum_j c_j \chi_M(\xi_j) = \lim_N \omega_N(\alpha_M^N(D)) = 0. \quad (2.3)$$

If $D = \sum_j c_j W_M(\xi_j)$ and $E = \sum_k d_k W_M(\zeta_k)$, concatenate the two finite sums in (2.2). For $a, b \in \mathbb{C}$ this gives $\rho_M^{(0)}(aD + bE) = a\rho_M^{(0)}(D) + b\rho_M^{(0)}(E)$; representation independence just proved makes the equality intrinsic.

⟨1⟩ 2 Prove boundedness and positivity before extending.

For $D \in \mathcal{D}_M$, (1.3) and linearity of the finite sum give

$$\rho_M^{(0)}(D) = \lim_N \omega_N(\alpha_M^N(D)). \quad (2.4)$$

Because α_M^N is injective and therefore isometric,

$$|\rho_M^{(0)}(D)| \leq \limsup_N \|\alpha_M^N(D)\| = \|D\|. \quad (2.5)$$

If $D \in \mathcal{D}_M$, then $D^*D \in \mathcal{D}_M$, and

$$\rho_M^{(0)}(D^*D) = \lim_N \omega_N(\alpha_M^N(D)^* \alpha_M^N(D)) \geq 0. \quad (2.6)$$

Also $\rho_M^{(0)}(I) = 1$.

⟨1⟩ 3 Extend to the coarse C^* -algebra.

Equations (1.2) and (2.5) give a unique bounded extension $\rho_M : \mathcal{W}_M \rightarrow \mathbb{C}$. Positivity persists under norm limits: if $A \in \mathcal{W}_M$ and $D_j \rightarrow A$, then $D_j^* D_j \rightarrow A^* A$, so

$$\rho_M(A^* A) = \lim_j \rho_M^{(0)}(D_j^* D_j) \geq 0. \quad (2.7)$$

Together with $\rho_M(I) = 1$, this makes ρ_M a state.

⟨1⟩ 4 Upgrade convergence from Weyl polynomials to every coarse observable.

Fix $A \in \mathcal{W}_M$ and $\delta > 0$. By (1.2), choose $D \in \mathcal{D}_M$ with $\|A - D\| < \delta$. Then

$$|\omega_N(\alpha_M^N(A)) - \rho_M(A)| \leq |\omega_N(\alpha_M^N(A - D))| \quad (1)$$

$$+ |\omega_N(\alpha_M^N(D)) - \rho_M(D)| + |\rho_M(D - A)| \quad (2)$$

$$\leq 2\delta + |\omega_N(\alpha_M^N(D)) - \rho_M(D)|. \quad (2.8)$$

The last term tends to zero by (2.4). Therefore

$$\limsup_N |\omega_N(\alpha_M^N(A)) - \rho_M(A)| \leq 2\delta. \quad (2.9)$$

Letting $\delta \downarrow 0$ proves (1.4).

⟨1⟩ 5 Prove projective consistency.

Fix $M \leq M'$ and $A \in \mathcal{W}_M$. For every $N \geq M'$,

$$\alpha_{M'}^N(\alpha_M^{M'}(A)) = \alpha_M^N(A). \quad (2.10)$$

Apply ω_N , take $N \rightarrow \infty$, and use (1.4) at scales M, M' :

$$\rho_{M'}(\alpha_M^{M'}(A)) = \rho_M(A). \quad (2.11)$$

This is (1.5).

⟨1⟩ 6 Construct the inductive-limit state.

On the algebraic union prescribe

$$\Omega_0(\alpha_M^\infty(A)) := \rho_M(A). \quad (2.12)$$

Equation (2.11) makes this well-defined. It is positive, unital, and bounded by one on each embedded algebra. It therefore extends uniquely to the norm completion $\mathcal{W}_\infty$, proving (1.6). $\square$

3 Exact continuum-identification criterion

Corollary 3.1 (identification on one common algebra). Let ω_{ct} be a state on a continuum Weyl algebra $\mathcal{W}_{\text{ct}}$, and let

$$\beta : \mathcal{W}_\infty \longrightarrow \mathcal{W}_{\text{ct}} \quad (3.1)$$

be an injective $*$ -homomorphism satisfying $\beta \alpha_M^\infty = \beta_M$. If

$$\chi_M(\xi) = \omega_{\text{ct}}(\beta_M(W_M(\xi))) \quad (3.2)$$

for every M, ξ , then

$$\Omega_\infty = \omega_{\text{ct}} \circ \beta \quad \text{on } \mathcal{W}_\infty. \quad (3.3)$$

Proof. Equation (3.2) and Theorem 1.1 give equality on every Weyl generator, hence on $\mathcal{D}_M$, hence by norm continuity on every $\mathcal{W}_M$. The union of the embedded $\mathcal{W}_M$ is norm dense in $\mathcal{W}_\infty$, so the equality extends to all of $\mathcal{W}_\infty$. $\square$

Remark 3.2 (the exact remaining interacting statement). For the finite-torus Wick-quartic states from the model sheet, the unresolved scalar statement is

$$\lim_{N \rightarrow \infty} \omega_{N,L,\lambda,1}(\alpha_M^N(W_M(\xi))) = \omega_{P(\phi)_2,L}(\beta_M(W_M(\xi))) \quad (3.4)$$

for every fixed M and ξ . Proving (3.4) closes the static, full-sequence, identified fixed-coarse theorem by Corollary 3.1. Neither Theorem 1.1 nor Corollary 3.1 asserts (3.4).

4 Compatible symmetry

Proposition 4.1. *Suppose $\vartheta_N \in \text{Aut}(\mathcal{W}_N)$ satisfy*

$$\vartheta_N \alpha_M^N = \alpha_M^N \vartheta_M, \quad \omega_N \vartheta_N = \omega_N. \quad (4.1)$$

Then every ρ_M and Ω_∞ constructed above is invariant under the induced symmetry.

Proof. For $A \in \mathcal{W}_M$,

$$\rho_M(\vartheta_M A) = \lim_N \omega_N(\alpha_M^N(\vartheta_M A)) = \lim_N \omega_N(\vartheta_N(\alpha_M^N A)) = \rho_M(A). \quad (4.2)$$

The inductive-limit assertion follows first on the algebraic union and then by norm continuity. $\square$

5 Why the cited anharmonic LR theorem does not cover the Wick quartic

Nachtergaele–Raz–Schlein–Sims, Theorem 4.1, assumes for the on-site perturbation V that

$$V \in C^1(\mathbb{R}), \quad V' \in L^1(\mathbb{R}), \quad \kappa_V := \int_{\mathbb{R}} |w| |\widehat{V'}(w)| dw < \infty. \quad (5.1)$$

Proposition 5.1 (exact hypothesis failure). *For every $\lambda > 0$ and $c \geq 0$, the Wick quartic*

$$V_c(u) := \lambda(u^4 - 6cu^2 + 3c^2) \quad (5.2)$$

does not satisfy $V'_c \in L^1(\mathbb{R})$. Hence the cited theorem cannot be applied to this interaction.

Proof in Lamport form

$\langle 1 \rangle 1$ Differentiate the stated polynomial.

$$V'_c(u) = 4\lambda u^3 - 12\lambda c u = 4\lambda u(u^2 - 3c). \quad (5.3)$$

$\langle 1 \rangle 2$ Obtain a lower bound on the tails.

If $|u| \geq \sqrt{6c}$, then

$$|u^2 - 3c| \geq \frac{1}{2}u^2, \quad (5.4)$$

and therefore

$$|V'_c(u)| \geq 2\lambda|u|^3. \quad (5.5)$$

⟨1⟩ 3 Integrate.

For every $R \geq \sqrt{6c}$,

$$\int_R^\infty |V'_c(u)| du \geq 2\lambda \int_R^\infty u^3 du = \infty. \quad (5.6)$$

Thus $V'_c \notin L^1(\mathbb{R})$, so the first analytic hypothesis in (5.1) already fails. $\square$

Remark 5.2. Proposition 5.1 excludes only this citation route. It is not a nonexistence theorem for quartic Lieb–Robinson bounds. A replacement would require a polynomial-specific energy-weighted argument, with constants controlled uniformly in the resolution.

6 Revised proof obligations

Item	Status	Exact content
Projective cofinal-subnet existence	PASS	Proved independently in the compactness companion.
Full-sequence reduction to Weyl generators	PASS	Theorem 1.1.
Continuum identification once generator limits are known	PASS	Corollary 3.1.
Interacting characteristic limit (3.4)	FAIL	This is the remaining static finite-torus theorem. It requires a uniform Euclidean or cluster-expansion comparison.
N -uniform gap and local second moment	FAIL	Not needed for the static weak-star criterion, but still needed by the quantitative LPPL route.
Quartic interacting LR estimate	FAIL	The cited theorem assumes (5.1), which the Wick quartic violates by Proposition 5.1.
Local-state-norm rate	FAIL	Pointwise convergence on a norm-dense set does not imply uniform convergence over the norm-one ball.

References

- [1] B. Nachtergaele, H. Raz, B. Schlein, R. Sims, *Lieb–Robinson bounds for harmonic and anharmonic lattice systems*, Commun. Math. Phys. **286** (2009), 1073–1098.

Part II gate scoping in Lamport notation

Minimal logic, momentum-source obstruction, and executed route

Fresh reconstruction of `part_ii/gate_II11_scoping.tex`

16 August 2026

This module records the exact logical gate and the obstruction that dictates the thermal mean-shift method. The two transfer lemmas proposed by the scoping document are not left as promises: their complete proofs appear as Steps $\langle 1 \rangle 10$ and $\langle 1 \rangle 13$ of the V4 Lamport module.

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1	The gate and what is logically sufficient	1
2	Why momentum cannot be inserted as a sharp-time Gaussian source	2
3	The executed thermal two-torus route	3
4	Scope conclusion	4

1 The gate and what is logically sufficient

For fixed coarse scale M , let α_M^N be the D4 lattice refinement and β_M the continuum embedding. The gate is

$$\lim_{N \rightarrow \infty} \omega_N(\alpha_M^N(W_M(\xi))) = \omega_{L,\lambda}^{\text{ct}}(\beta_M(W_M(\xi))) \quad (\xi \in h_{M,L}). \quad (\text{G})$$

$\langle 1 \rangle 1$ Identify the minimum statement that must be proved.

$\langle 2 \rangle 1$ It is enough to prove (G) pointwise for every pair (M, ξ) .

Because: The Weyl-reduction module proves that the finite linear span of Weyl operators is norm dense in each coarse Weyl algebra. If the state values converge on each generator, linearity gives convergence on that span. Given A and an approximant A_0 in the span with $\|A - A_0\| < \delta$, the state norm-one bound gives

$$|\omega_N(\alpha_M^N(A)) - \rho_M(A)| \leq 2\delta + |\omega_N(\alpha_M^N(A_0)) - \rho_M(A_0)|.$$

Let $N \rightarrow \infty$, then $\delta \downarrow 0$. Positivity, normalization, projective consistency, and continuum identification follow in that module from the same pointwise limits.

$\langle 2 \rangle 2$ Uniformity in ξ on bounded sets is therefore not a hypothesis of the gate.

Because: Step $\langle 2 \rangle 1$ uses only one fixed approximant, hence finitely many fixed Weyl symbols. Quantitative uniform estimates are useful outputs but not logical prerequisites.

⟨2⟩ 3 An N -uniform interacting gap is not an input.

Because: V4 first derives eigenvalue convergence from normal-ordered partition functions and only then obtains $E_1(N) - E_0(N) \rightarrow \gamma_L > 0$. The eventual gap is an output used in the final thermal-to-ground interchange.

⟨2⟩ 4 The right side of (G) denotes the ground state of the specific K_L constructed as the V3 strong semigroup limit.

Because: V3 identifies $H_0 + V_L$ on the cylinder core but does not prove essential self-adjointness of that core restriction. Part II neither needs nor proves uniqueness among other possible realizations. The gate is an identification with the constructed object, in convention $(a, b) = (0, 0)$.

⟨2⟩ 5 This proves ⟨1⟩1. □

2 Why momentum cannot be inserted as a sharp-time Gaussian source

⟨1⟩ 2 Prove the sharp-time momentum-source obstruction.

Let $0 \neq h \in C^\infty(\mathbb{T}_L)$ be real and $C = (-\partial_t^2 - \partial_x^2 + m^2)^{-1}$ on $\mathbb{R}_t \times \mathbb{T}_L$. The formal momentum insertion would use $J = \delta'_0 \otimes h$.

⟨2⟩ 1 The Fourier transform of J is

$$\hat{J}(k_0, k) = ik_0 \hat{h}(k). \quad (2.1)$$

Because: Distributional differentiation in time multiplies the Fourier transform by ik_0 ; the spatial tensor factor supplies $\hat{h}(k)$.

⟨2⟩ 2 Its covariance norm is

$$\|J\|_{H^{-1}}^2 = \frac{1}{2L} \sum_{k \in (\pi/L)\mathbb{Z}} |\hat{h}(k)|^2 \int_{\mathbb{R}} \frac{dk_0}{2\pi} \frac{k_0^2}{k_0^2 + m^2 + k^2}. \quad (2.2)$$

Because: The first-chaos norm of a source is $\langle J, CJ \rangle$; in Fourier variables, C multiplies by $(k_0^2 + m^2 + k^2)^{-1}$. Insert (2.1) and Parseval's factor $1/(2L)$.

⟨2⟩ 3 The right side of (2.2) is infinite.

Because: Choose k_* with $\hat{h}(k_*) \neq 0$. For $|k_0| \geq \sqrt{m^2 + k_*^2}$,

$$\frac{k_0^2}{k_0^2 + m^2 + k_*^2} \geq \frac{1}{2}.$$

The corresponding k_0 integral dominates the integral of $1/2$ over two unbounded rays and therefore diverges. All summands are nonnegative, so one divergent term makes the sum infinite.

⟨2⟩ 4 The same obstruction holds for every spatial lattice.

Because: Only the spatial momentum set and the value of the positive dispersion change. Time remains continuous, and for each nonzero spatial coefficient $k_0^2/(k_0^2 + \gamma_N(k)^2) \rightarrow 1$ as $|k_0| \rightarrow \infty$.

⟨2⟩ 5 Hence $\Phi(\delta'_0 \otimes h)$ is not a Gaussian first-chaos random variable and $e^{i\Pi(h)}$ cannot be represented as $e^{i\Phi(J)}$ with an H^{-1} source.

Because: A Gaussian source functional exists in L^2 precisely when its covariance norm is finite; (2.2) violates that condition.

⟨2⟩ 6 This proves ⟨1⟩2. □

⟨1⟩ 3 Replace the impossible momentum source by the exact jump mean shift.

⟨2⟩ 1 At the free thermal level, compute momentum insertions by the oscillator trace formula.

Because: V1 derives the one-mode expression $\text{Tr}(e^{-tH} D(z))/\text{Tr} e^{-tH} = e^{-|z|^2 \coth(t\gamma/2)/2}$ and takes its finite or trace-dominated infinite product. No path-space derivative source is introduced.

⟨2⟩ 2 At the interacting level, let $e^{i\Pi(p)}$ translate the marked configuration and shift the Gaussian bridge by the unique harmonic path H_p .

Because: V1 solves $-\ddot{H} + \gamma^2 H = 0$, $H(t) - H(0) = \rho$, $\dot{H}(t) = \dot{H}(0)$ and proves the exact Mehler-chain identity. The shifted interaction is the finite Appell sum

$$\mathcal{U}(\varphi + H) - \mathcal{U}(\varphi) = \lambda \int [4H:\varphi^3: + 6H^2:\varphi^2: + 4H^3\varphi + H^4]. \quad (2.3)$$

Thus the momentum variable enters only through a deterministic, bounded harmonic interpolant.

⟨2⟩ 3 This proves ⟨1⟩3. □

3 The executed thermal two-torus route

⟨1⟩ 4 Reduce the gate to eight explicit packages.

Item	Required statement	Executed proof
A1	Continuum K_L , trace class, simple ground, positive gap	V3 Steps ⟨1⟩9–⟨1⟩16
A2	Lattice and continuum marked-slice mean-shift traces	V1 Step ⟨1⟩6; V3 Step ⟨1⟩15
A3	Exact free thermal forms and fixed-coarse rate	V1 Steps ⟨1⟩3, ⟨1⟩7–⟨1⟩8
A4	Same-noise interaction comparison, Wick dictionary, shifted terms	repaired V2 Steps ⟨1⟩1–⟨1⟩8
A5	Uniform plain and shifted exponential integrability	repaired V2 Step ⟨1⟩9; V3 Steps ⟨1⟩5–⟨1⟩6
A6	Fixed- t thermal convergence	V4 Steps ⟨1⟩1–⟨1⟩7
A7	Normal-ordered partition convergence, Laplace spectral transfer, derived gap	V4 Steps ⟨1⟩8–⟨1⟩11
A8	Thermal-to-ground interchange and C^* assembly	V4 Steps ⟨1⟩12–⟨1⟩16

⟨2⟩ 1 A1–A5 imply A6.

Because: A2 writes both thermal functionals on the common A4 probability space. A3 compares deterministic free factors; A4 compares all random exponents in L^2 ; A5 gives uniform integrability; A1 supplies the valid continuum trace identity. V4 performs the three-term quotient subtraction explicitly.

⟨2⟩ 2 A5 and the sourceless part of A6 imply the partition convergence in A7 after subtracting the lattice zero-point energy.

Because: The free normal-ordered product converges by the dispersion estimate, and the interacting factor is $\mathbb{E}e^{-\mathcal{U}_N}$. The unshifted free product with its zero-point factor has no nonzero continuum limit; V4 uses only $\prod_k(1 - e^{-t\gamma_N(k)})^{-1}$.

⟨2⟩ 3 A7 and the A1 continuum gap imply the uniform spectral-tail input for A8.

Because: The exact Laplace-transfer proof gives $E_1(N) - E_0(N) \rightarrow \gamma_L > 0$ and $D_N(t_*) \rightarrow D_\infty(t_*)$. For $t \geq t_*$,

$$D_N(t) \leq e^{-(t-t_*)[E_1(N)-E_0(N)]} D_N(t_*).$$

⟨2⟩ 4 A6 plus this tail input permits the order of limits to be interchanged, and the Weyl criterion then proves (G) on the whole inductive-limit algebra.

Because: V4 Step ⟨1⟩13 writes the three-term triangle inequality, first chooses t uniformly large by the derived gap, and only then sends $N \rightarrow \infty$ using fixed- t convergence. Its final two steps invoke the proved Weyl-reduction module.

⟨2⟩ 5 This proves ⟨1⟩4. □

⟨1⟩ 5 Explain why the slab-diagonal fallback is not load bearing.

⟨2⟩ 1 A slab approximation at time length T has an error controlled by the individual lattice gap $\gamma_{N,L}$.

Because: Spectral expansion gives an excited-state tail with leading factor $e^{-T\gamma_{N,L}}$.

⟨2⟩ 2 Interchanging $T \rightarrow \infty$ and $N \rightarrow \infty$ on that route would require an independent eventual uniform lower bound on $\gamma_{N,L}$.

Because: Without such a bound, no single T makes $e^{-T\gamma_{N,L}}$ small for all sufficiently large N .

⟨2⟩ 3 The thermal route is stronger because the trace partition functions derive precisely that gap through Laplace transfer.

Because: The compact time circle gives trace identities, and convergence of all large-time Laplace transforms determines the ordered atoms of the spectrum. The slab route has no corresponding normalized trace input. It is therefore a fallback, not part of the executed proof chain.

⟨2⟩ 4 This proves ⟨1⟩5. □

4 Scope conclusion

The scoping document contributes one substantive negative result, Step ⟨1⟩2, and the dependency architecture in Step ⟨1⟩4. Its tentative A4 coupling and tentative counterterm were superseded: the executed coupling is V2’s edge-symmetrized same-noise construction, and the equal-point variance offset cancels the tadpole offset, leaving $(a, b) = (0, 0)$. The repaired V2 module further supplies the four-edge term and the fundamental-band/alias partition required for A4 to be true at proof level.

Part II V1 in Lamport notation

Thermal Weyl functionals, Mehler bridges, and the exact mean shift

Fresh reconstruction of `part_ii/v1_thermal_representations.tex`

16 August 2026

All finite-dimensional identities in V1 are reconstructed below. The reference measure in the mean-shift formula is the *free periodic Gaussian* measure. The continuum interacting marked-slice identity is not assumed here; it is proved by the V3 reconstruction and consumed only afterward.

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2	Structured proof	1
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1 Objects and statement

Fix $L, m > 0$, $\varepsilon_N = L/r_N$, the spatial lattice Λ_N with $2r_N$ sites, and

$$\Gamma_N = \{\pi j/L : -r_N \leq j < r_N\}, \quad \gamma_N(k)^2 = m^2 + 4\varepsilon_N^{-2} \sin^2(\varepsilon_N k/2).$$

On $L^2(\mathbb{R}^{\Lambda_N}, d\boldsymbol{\varphi})$ let $\Phi_N(x) = \varphi_x$ and $\Pi_N(x) = -i\varepsilon_N^{-1} \partial_{\varphi_x}$. For real q, p set

$$\Phi_N(q) = \varepsilon_N \sum_x q_x \Phi_N(x), \quad \Pi_N(p) = \varepsilon_N \sum_x p_x \Pi_N(x), \quad W_N(\varepsilon_N q + ip) = e^{i(\Phi_N(q) + \Pi_N(p))}.$$

Let $H_N^{(0)}$ be the nearest-neighbour oscillator Hamiltonian and

$$K_N = H_N^{(0)} + V_N, \quad V_N = \lambda \varepsilon_N \sum_x :\Phi_N(x)^4:_{c_N, L}.$$

The model-sheet module proves that K_N is self-adjoint, bounded below, has compact resolvent, and has a simple ground state.

2 Structured proof

$\langle 1 \rangle$ 1 Derive the exact Schrödinger action of the Weyl operator.

$\langle 2 \rangle$ 1 For $s \in \mathbb{R}$ define

$$(U_s \psi)(\boldsymbol{\varphi}) = e^{is\varepsilon_N q \cdot \boldsymbol{\varphi} + \frac{i}{2} s^2 \varepsilon_N q \cdot p} \psi(\boldsymbol{\varphi} + sp).$$

⟨3⟩ 1 U_s is unitary.

Because: Translation preserves Lebesgue measure and the prefactor has modulus one.

⟨3⟩ 2 $U_{s+s'} = U_s U_{s'}$.

Because: Direct substitution gives the exponent

$$s\varepsilon_{Nq} \cdot \varphi + s'\varepsilon_{Nq} \cdot (\varphi + sp) + \frac{1}{2}(s^2 + s'^2)\varepsilon_{Nq} \cdot p = (s + s')\varepsilon_{Nq} \cdot \varphi + \frac{1}{2}(s + s')^2\varepsilon_{Nq} \cdot p.$$

⟨3⟩ 3 $s \mapsto U_s$ is a strongly continuous unitary group and its generator extends $\Phi_N(q) + \Pi_N(p)$ on C_c^∞ .

Because: Strong continuity first holds on C_c^∞ by dominated convergence and then on all L^2 by density and unitarity. On the core,

$$\left. \frac{d}{ds} U_s \psi \right|_{s=0} = (i\varepsilon_{Nq} \cdot \varphi + p \cdot \nabla) \psi = i(\Phi_N(q) + \Pi_N(p)) \psi.$$

Stone's theorem gives a self-adjoint generator extending the displayed differential operator. Moreover $U_s C_c^\infty \subset C_c^\infty$ for every s ; Nelson's invariant-domain criterion therefore makes C_c^∞ a core for that generator. Thus the differential operator is essentially self-adjoint there and its closure generates U_s .

⟨2⟩ 2 At $s = 1$,

$$(W_N(\varepsilon_{Nq} + ip)\psi)(\varphi) = e^{i\varepsilon_{Nq} \cdot \varphi + \frac{i}{2}\varepsilon_{Nq} \cdot p} \psi(\varphi + p), \quad (1.1)$$

$$W_N(\varepsilon_{Nq} + ip) = e^{\frac{i}{2}\varepsilon_{Nq} \cdot p} e^{i\Phi_N(q)} e^{i\Pi_N(p)}. \quad (1.2)$$

⟨2⟩ 3 This proves ⟨1⟩1. □

⟨1⟩ 2 Prove trace class and the marked-slice kernel identity.

⟨2⟩ 1 K_N has compact resolvent and is bounded below; hence e^{-tK_N} is trace class for $t > 0$.

Because: The model sheet proves $K_N \geq a(-\Delta_\varphi + |\varphi|^2) - b$ in quadratic-form sense for some $a > 0, b < \infty$. The comparison oscillator has compact resolvent and summable heat eigenvalues; the min–max principle yields $\text{Tr } e^{-tK_N} < \infty$.

⟨2⟩ 2 Let $k_t(\mathbf{x}, \mathbf{y})$ be the continuous kernel of e^{-tK_N} . For bounded multiplication f and translation $T_p \psi(x) = \psi(x + p)$,

$$\text{Tr}(e^{-tK_N} M_f T_p) = \int_{\mathbb{R}^N} k_t(\mathbf{x} + p, \mathbf{x}) f(\mathbf{x}) d\mathbf{x}. \quad (2.1)$$

Because: Expand the trace in a real eigenbasis $K_N u_j = E_j u_j$. Its scalar series is absolutely convergent because $M_f T_p$ is bounded. To justify exchanging the series and spatial integration, put $D(x) = e^{-(t/2)K_N}(x, x)$ and use Cauchy–Schwarz twice:

$$\begin{aligned} \sum_j e^{-tE_j} |u_j(x) u_j(x + p)| &\leq \left(\sum_j e^{-tE_j/2} \right) D(x)^{1/2} D(x + p)^{1/2}, \\ \int D(x)^{1/2} D(x + p)^{1/2} dx &\leq \left(\int D(x) dx \right)^{1/2} \left(\int D(x + p) dx \right)^{1/2} = \text{Tr } e^{-(t/2)K_N} < \infty. \end{aligned}$$

Thus Fubini applies. The kernel series is the continuous heat kernel, and the change of variable in the trace gives (2.1) with $f(x)$ (not $f(x + p)$).

⟨2⟩ 3 With $f(x) = e^{i\varepsilon_N q \cdot x}$, (1.1) and (2.1) give

$$\mathrm{Tr}(e^{-tK_N} W_N(\varepsilon_N q + ip)) = e^{\frac{i}{2}\varepsilon_N q \cdot p} \int e^{i\varepsilon_N q \cdot x} k_t(x + p, x) dx. \quad (2.2)$$

⟨2⟩ 4 This proves ⟨1⟩2. □

⟨1⟩ 3 Diagonalize the free system and compute its thermal Weyl functional.

⟨2⟩ 1 Set $X_x = \varepsilon_N^{1/2} \Phi_N(x)$ and $P_x = \varepsilon_N^{1/2} \Pi_N(x)$. The unitary discrete Fourier transform gives

$$H_N^{(0)} = \sum_{k \in \Gamma_N} \gamma_N(k) (a_k^* a_k + \tfrac{1}{2}),$$

$$\tilde{X}(k) = (2\gamma_N(k))^{-1/2} (a_k + a_{-k}^*), \quad \tilde{P}(k) = -i(\gamma_N(k)/2)^{1/2} (a_k - a_{-k}^*). \quad (3.1)$$

Because: Fourier transformation diagonalizes the circulant nearest-neighbour matrix; substitution of (3.1) and $[a_k, a_{k'}^*] = \delta_{kk'}$ cancels the aa and a^*a^* terms and leaves the displayed oscillator sum.

⟨2⟩ 2 For real q, p ,

$$\Phi_N(q) + \Pi_N(p) = \sum_k (\bar{z}_k a_k + z_k a_k^*), \quad z_k = \varepsilon_N^{1/2} \left[\frac{\tilde{q}(k)}{\sqrt{2\gamma_N(k)}} + i \sqrt{\frac{\gamma_N(k)}{2}} \tilde{p}(k) \right]. \quad (3.2)$$

Moreover

$$\sum_k |z_k|^2 = \frac{1}{2(2r_N)} \sum_k \left(\gamma_N(k)^{-1} |\hat{q}(k)|^2 + \gamma_N(k) |\hat{p}(k)|^2 \right). \quad (3.3)$$

Because: Parseval gives (3.2). In $|z_k|^2$ the cross term is $\mathrm{Im}(\tilde{q}(k) \overline{\tilde{p}(k)})$; it is odd under $k \mapsto -k$ and cancels in the sum. The relation between the two Fourier normalizations gives (3.3).

⟨2⟩ 3 For one oscillator $H = \gamma(a^*a + 1/2)$ and $D(z) = e^{za^* - \bar{z}a}$,

$$\frac{\mathrm{Tr}(e^{-tH} D(z))}{\mathrm{Tr} e^{-tH}} = \exp[-\tfrac{1}{2}|z|^2 \coth(t\gamma/2)]. \quad (3.4)$$

Because: BCH gives $D(z) = e^{-|z|^2/2} e^{za^*} e^{-\bar{z}a}$. In the number basis,

$$\langle n | e^{za^*} e^{-\bar{z}a} | n \rangle = L_n(|z|^2), \quad \sum_{n \geq 0} w^n L_n(x) = \frac{e^{-xw/(1-w)}}{1-w}.$$

The first identity follows by expanding both exponentials and matching the number change; the second follows from $\sum_{n \geq j} \binom{n}{j} w^n = w^j (1-w)^{-j-1}$. Put $w = e^{-t\gamma}$ and use $(1+w)/(1-w) = \coth(t\gamma/2)$.

⟨2⟩ 4 The finite tensor-product trace therefore equals

$$F_N^{(0)}(t; \varepsilon_N q + ip) = \exp[-\tfrac{1}{4} Q_N^{(t)}(q, p)], \quad (3.5)$$

$$Q_N^{(t)}(q, p) = \frac{1}{2r_N} \sum_{k \in \Gamma_N} \coth(t\gamma_N(k)/2) \left(\gamma_N(k)^{-1} |\hat{q}(k)|^2 + \gamma_N(k) |\hat{p}(k)|^2 \right). \quad (3.6)$$

Also $\mathrm{Tr} e^{-tH_N^{(0)}} = \prod_k [2 \sinh(t\gamma_N(k)/2)]^{-1}$.

Because: Apply (3.4) to each commuting oscillator and insert (3.3); the zero-point geometric sum is $[2 \sinh(t\gamma/2)]^{-1}$.

⟨2⟩ 5 This proves ⟨1⟩3. □

⟨1⟩ 4 Verify the Mehler kernel and the unique harmonic jump interpolant.

⟨2⟩ 1 For $H_\omega = \frac{1}{2}(-\partial_x^2 + \omega^2 x^2)$, set

$$k_\Delta^\omega(x, y) = \left(\frac{\omega}{2\pi \sinh(\omega\Delta)} \right)^{1/2} \exp \left[-\frac{\omega \coth(\omega\Delta)}{2}(x^2 + y^2) + \frac{\omega xy}{\sinh(\omega\Delta)} \right]. \quad (4.1)$$

⟨3⟩ 1 $\partial_\Delta k_\Delta^\omega = -H_\omega k_\Delta^\omega$.

Because: Differentiate the prefactor and exponent, using $(\omega \coth \omega\Delta)' = -\omega^2 \operatorname{csch}^2(\omega\Delta)$ and $(\omega \operatorname{csch} \omega\Delta)' = -\omega^2 \coth(\omega\Delta) \operatorname{csch}(\omega\Delta)$; the coefficients of $1, x^2, xy, y^2$ agree separately.

⟨3⟩ 2 $k_\Delta^\omega \rightarrow \delta_{x=y}$ as $\Delta \downarrow 0$.

Because: In variables $u = (x + y)/\sqrt{2}$, $v = (x - y)/\sqrt{2}$, the quadratic form has coefficients $\omega \tanh(\omega\Delta/2)$ and $\omega \coth(\omega\Delta/2)$; the latter tends to infinity with the normalized Gaussian mass, while the former tends to zero.

⟨3⟩ 3 Thus (4.1) is the heat kernel.

Because: For compactly supported smooth initial data, both convolution with (4.1) and $e^{-\Delta H_\omega}$ solve the same L^2 Cauchy problem. Their difference v obeys $d\|v\|_2^2/d\Delta = -2\langle v, H_\omega v \rangle \leq 0$ and starts at zero.

⟨2⟩ 2 For a prescribed jump ρ , the boundary-value problem

$$-\ddot{H} + \omega^2 H = 0, \quad H(t) - H(0) = \rho, \quad \dot{H}(t) = \dot{H}(0)$$

has the unique solution

$$H(s) = \frac{\rho \sinh(\omega(s - t/2))}{2 \sinh(\omega t/2)}. \quad (4.2)$$

Because: The general solution is $A \cosh(\omega s) + B \sinh(\omega s)$. Solving the two linear boundary equations gives $A = -\rho/2$ and $B = (\rho/2) \coth(\omega t/2)$, which is (4.2) by the subtraction formula. The homogeneous system has only $A = B = 0$.

⟨2⟩ 3 The interpolant satisfies

$$|H(s)| \leq |\rho|/2, \quad S(H) := \frac{1}{2} \int_0^t (\dot{H}^2 + \omega^2 H^2) ds = \frac{\omega \rho^2}{4} \coth(\omega t/2). \quad (4.3)$$

Because: $|s - t/2| \leq t/2$ and $|\sinh|$ is increasing on $[0, \infty)$. Integration by parts and the ODE give $S(H) = \frac{1}{2}[H\dot{H}]_0^t = \frac{1}{2}\rho\dot{H}(0)$; differentiating (4.2) at 0 gives the last expression.

⟨2⟩ 4 This proves ⟨1⟩4. □

⟨1⟩ 5 Prove the exact discrete Gaussian shift identity.

Fix the equal partition $s_i = i\Delta$, $\Delta = t/M$, and let $H_i = H(s_i)$. Define the periodic Mehler-chain integral

$$J_0[f_0, \dots, f_{M-1}] = \int \prod_{i=0}^{M-1} k_\Delta^\omega(\eta_{i+1}, \eta_i) f_i(\eta_i) \prod_i d\eta_i, \quad \eta_M = \eta_0,$$

and the jump chain J_ρ by replacing the closing condition by $x_M = x_0 + \rho$.

⟨2⟩ 1 Under $x_i = \eta_i + H_i$, every interior Mehler exponent has zero linear term in η_i .

Because: Apart from its normalization, the negative logarithm of one kernel is

$$Q_\Delta(y, x) = \frac{\omega}{2 \sinh(\omega \Delta)} [\cosh(\omega \Delta)(x^2 + y^2) - 2xy].$$

For $1 \leq i < M$, the coefficient of η_i in $\sum_i Q_\Delta(\eta_{i+1} + H_{i+1}, \eta_i + H_i)$ is proportional to

$$2 \cosh(\omega \Delta) H_i - H_{i-1} - H_{i+1} = 0,$$

the exact three-term propagation law of every solution of $-\ddot{H} + \omega^2 H = 0$. The combined coefficient of the periodic variable $\eta_0 = \eta_M$ is the difference of the endpoint classical momenta and vanishes because $\dot{H}(t) = \dot{H}(0)$. Hence all linear terms, including the closing one, vanish.

$\langle 2 \rangle 2$ The constant part of the exponent is $S(H)$, independent of the mesh.

Because: $Q_\Delta(H_{i+1}, H_i)$ is exactly the classical action $\frac{1}{2} \int_{s_i}^{s_{i+1}} (\dot{H}^2 + \omega^2 H^2) ds$ of the unique harmonic path joining the two endpoints. The restriction of (4.2) is that path. Hence

$$\sum_i Q_\Delta(H_{i+1}, H_i) = \frac{1}{2} \int_0^t (\dot{H}^2 + \omega^2 H^2) ds = \frac{1}{2} [H \dot{H}]_0^t = S(H).$$

$\langle 2 \rangle 3$ The endpoint conditions transform correctly: $x_M = x_0 + \rho$ becomes $\eta_M = \eta_0$ because $H(t) - H(0) = \rho$.

$\langle 2 \rangle 4$ Therefore

$$J_\rho[f_0, \dots, f_{M-1}] = e^{-S(H)} J_0[f_0(\cdot + H_0), \dots, f_{M-1}(\cdot + H_{M-1})]. \quad (5.1)$$

The multimode identity is the product of (5.1).

$\langle 2 \rangle 5$ This proves $\langle 1 \rangle 5$. □

$\langle 1 \rangle 6$ Derive the interacting marked-slice mean-shift formula.

$\langle 2 \rangle 1$ The harmonic-reference FKN formula gives

$$k_t^K(x, y) = k_t^{(0)}(x, y) \mathbb{E}_{x,y}^{t,0} \exp \left[- \int_0^t V_N(B_s) ds \right]. \quad (6.1)$$

Because: For bounded truncations of V_N this is the Trotter product formula with the Mehler kernel. The Wick quartic is continuous and bounded below; truncate it from above, apply monotone/dominated convergence to the positive kernels, and use the closed lower-bounded form limit. This proves (6.1) without importing an interacting scaling limit.

$\langle 2 \rangle 2$ Insert (6.1) into (2.2) and approximate the time integral by Riemann sums.

Because: Each bridge path is continuous and V_N is continuous, so the Riemann sums converge pointwise. If $V_N \geq -c_N$, every exponential is bounded by e^{tc_N} ; the remaining Mehler chain has finite mass. Dominated convergence applies.

$\langle 2 \rangle 3$ Apply (5.1) mode by mode with jump $\rho_k = \varepsilon_N^{1/2} \tilde{p}(k)$.

Because: The position translation in (1.1) is p , hence in the rescaled mode coordinate it is exactly ρ_k . The product Jacobian is one.

$\langle 2 \rangle 4$ The phase from (1.1) cancels the value of the marked position source on $H(0) = -p/2$.

Because: The inserted factor becomes

$$e^{\frac{i}{2}\varepsilon_N q \cdot p} e^{i\varepsilon_N q \cdot (\eta(0) - p/2)} = e^{i\varepsilon_N q \cdot \eta(0)}.$$

$\langle 2 \rangle 5$ Let $\mu_{N,t}^{\text{per}}$ be the free periodic Gaussian field with mode covariance

$$c_\omega(s) = \frac{\cosh(\omega(t/2 - |s|_{\mathbb{T}_t}))}{2\omega \sinh(\omega t/2)}. \quad (6.2)$$

Then

$$\frac{\text{Tr}(e^{-tK_N} W_N(\varepsilon_N q + ip))}{\text{Tr} e^{-tK_N}} = e^{-S_N(H_p)} \frac{\mathbb{E}_{\mu_{N,t}^{\text{per}}}[e^{i\varepsilon_N q \cdot \eta(0)} e^{-\mathcal{U}_N(\eta + H_p)}]}{\mathbb{E}_{\mu_{N,t}^{\text{per}}}[e^{-\mathcal{U}_N(\eta)}]}. \quad (6.3)$$

Because: The periodic Mehler chain is centered Gaussian and its two-point function is (6.2), obtained either by inverting $-\partial_s^2 + \omega^2$ on the time circle or by the oscillator trace. Normalize the periodic chain by the zero-source partition function. Steps $\langle 2 \rangle 2$ – $\langle 2 \rangle 4$ give exactly the numerator; $q = p = 0$ gives the denominator.

$\langle 2 \rangle 6$ The shifted Wick quartic is exactly

$$:(\eta + H)^4:_c - :\eta^4:_c = 4H:\eta^3:_c + 6H^2:\eta^2:_c + 4H^3\eta + H^4. \quad (6.4)$$

Because: Expand $(\eta + H)^4 - 6c(\eta + H)^2 + 3c^2$ and use $\eta^3 = :\eta^3:_c + 3c\eta$ and $\eta^2 = :\eta^2:_c + c$; all additional c terms cancel.

$\langle 2 \rangle 7$ This proves $\langle 1 \rangle 6$. □

$\langle 1 \rangle 7$ Construct the continuum free thermal functional.

Let $\gamma = (-\partial_x^2 + m^2)^{1/2}$ on $\mathbb{T}_L$ and

$$h_{\text{ct}} = H^{-1/2}(\mathbb{T}_L) \oplus iH^{1/2}(\mathbb{T}_L).$$

$\langle 2 \rangle 1$ For $\zeta = q + ip \in h_{\text{ct}}$, the symplectic form $\text{Im} \int \bar{\zeta} \zeta'$ is finite and defines the continuum Weyl operator $W_{\text{ct}}(\zeta)$.

Because: Sobolev duality pairs $H^{-1/2}$ with $H^{1/2}$. The covariance norm $\langle q, \gamma^{-1}q \rangle + \langle p, \gamma p \rangle$ is finite by definition.

$\langle 2 \rangle 2$ $e^{-t d\Gamma(\gamma)}$ is trace class and

$$\text{Tr} e^{-t d\Gamma(\gamma)} = \prod_{k \in (\pi/L)\mathbb{Z}} (1 - e^{-t\gamma(k)})^{-1}.$$

Because: The occupation basis gives the product of geometric series. It converges because $\sum_k e^{-t\gamma(k)} < \infty$.

$\langle 2 \rangle 3$ For square-summable displacement amplitudes, absolute convergence permits factorization of the trace into one-mode traces. Hence

$$F_\infty^{(0)}(t; q + ip) = \exp \left[-\frac{1}{4} \sum_k \coth(t\gamma(k)/2) \left(\gamma(k)^{-1} |\hat{q}(k)|^2 + \gamma(k) |\hat{p}(k)|^2 \right) \right], \quad (7.1)$$

with the Fourier normalization fixed by the model sheet.

Because: $|\langle n, D(z)n \rangle| \leq 1$ dominates the trace by the partition function; Fubini applies, and (3.4) evaluates every factor.

⟨2⟩ 4 The D4 image of every fixed coarse symbol belongs to h_{ct} .

Because: Its Fourier coefficients contain $P_\infty(\varepsilon_M k)$ with $|P_\infty(\ell)|^2 \leq C_D(1 + |\ell|)^{-2-\eta}$, $\eta > 0$. The momentum sum is bounded by $\sum_k |k|(1 + \varepsilon_M |k|)^{-2-\eta} < \infty$, and the position sum is easier.

⟨2⟩ 5 This proves ⟨1⟩7. □

⟨1⟩ 8 Prove the free thermal fixed-coarse rate uniformly for $t \geq t_0 > 0$.

⟨2⟩ 1 For $w_t(\gamma) = \coth(t\gamma/2)$ and $\bar{w} = \coth(t_0 m/2)$,

$$\begin{aligned} 1 &\leq w_t(\gamma) \leq \bar{w}, \\ |(\gamma w_t(\gamma))'| &\leq \bar{w} + s_*, \\ |(\gamma^{-1} w_t(\gamma))'| &\leq (\bar{w} + s_*)m^{-2}, \end{aligned}$$

where $s_* = \sup_{x \geq t_0 m/2} x / \sinh^2 x < \infty$.

Because: Differentiate $\coth(t\gamma/2)$ and use $\gamma \geq m$.

⟨2⟩ 2 On low momenta, split each weighted multiplier difference into a dispersion term and a D4-filter term.

Because: For the momentum multiplier,

$$|\gamma_N w_t(\gamma_N)|P_r|^2 - \gamma w_t(\gamma)|P_\infty|^2| \leq |\gamma_N w_t(\gamma_N) - \gamma w_t(\gamma)| |P_r|^2 + \gamma w_t(\gamma) ||P_r|^2 - |P_\infty|^2|.$$

The first term is at most $(\bar{w} + s_*)\varepsilon_N^2 k^4 / (24m)$ times $|P_r|^2$; the position multiplier is the same with the derivative bound $m^{-2}(\bar{w} + s_*)$. The second terms use the explicit finite-product filter tail.

⟨2⟩ 3 On high momenta, multiply the vacuum D4 envelope estimate by at most $\bar{w}$.

Because: $1 \leq w_t \leq \bar{w}$ uniformly in $t \geq t_0$; no derivative is used in the tail.

⟨2⟩ 4 Choose the same split exponent as the free-rate module, $2 - 5\alpha = \alpha\eta = \theta = 2\eta/(5 + \eta)$. Then

$$|Q_{M+r}^{(t)}(\xi) - Q_\infty^{(t)}(\xi)| \leq C_{M,\xi,m,L,t_0} 2^{-\theta r}, \quad t \in [t_0, \infty]. \quad (8.1)$$

Because: The low sum is $O(4^{-r}(1 + J_r)^5)$ and the high tail is $O(J_r^{-\eta})$. Put $J_r \asymp 2^{\alpha r}$; the two exponents become equal. Every sum and counting constant is the explicit one proved in the free-rate module.

⟨2⟩ 5 The characteristic functionals obey the same rate up to a factor $1/4$.

Because: For $x, y \geq 0$, $|e^{-x/4} - e^{-y/4}| \leq |x - y|/4$.

⟨2⟩ 6 This proves ⟨1⟩8. □

3 V1 conclusion and dependency interface

The proved outputs are the exact lattice Weyl phase, trace-kernel formula, free thermal closed forms, Mehler/jump shift, interacting finite-lattice representation (6.3), and uniform free fixed-coarse rate (8.1). Formula (6.3) uses only a finite-dimensional FKN argument. The analogous continuum interacting formula requires the continuum Hamiltonian/periodic FKN construction; the V3 module proves it before V4 uses it.

Part II V2 in Lamport notation

Same-noise interaction comparison, edge correction, and shifted repair

Fresh reconstruction of `part_ii/v2_interaction_comparison.tex`

16 August 2026

This reconstruction corrects the false statement that four edge modes cannot conserve momentum and replaces the incomplete proof of the shifted comparison by an exact fundamental-band/alias partition. The repaired estimates have the rates claimed in V2. The original V2 file is not valid without these additions.

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1 Definitions

Fix $t \geq t_0 > 0$, $L, m > 0$, and assume $t_0 L \geq 1$. Put

$$D_N = \frac{2\pi}{t} \mathbb{Z} \times \Gamma_N, \quad D_\infty = \frac{2\pi}{t} \mathbb{Z} \times \frac{\pi}{L} \mathbb{Z}, \quad R_N = \frac{\pi}{\varepsilon_N}.$$

For $\kappa = (k_0, k)$ define

$$c_N(\kappa) = \frac{1}{k_0^2 + \gamma_N(k)^2}, \quad c_\infty(\kappa) = \frac{1}{k_0^2 + m^2 + k^2}, \quad c_\times(\kappa) = \sqrt{c_N(\kappa)c_\infty(\kappa)}. \quad (0.1)$$

Let $k_e = -R_N$. On one real Gaussian Fourier family $G(-\kappa) = \overline{G(\kappa)}$ set

$$G_N(k_0, k) = G(k_0, k) \quad (k \neq k_e), \quad G_N(k_0, k_e) = 2^{-1/2} [G(k_0, k_e) + G(k_0, -k_e)]. \quad (0.2)$$

Then

$$\eta_N(z) = (2Lt)^{-1/2} \sum_{D_N} G_N(\kappa) c_N(\kappa)^{1/2} e^{i\kappa z}, \quad \eta(z) = (2Lt)^{-1/2} \sum_{D_\infty} G(\kappa) c_\infty(\kappa)^{1/2} e^{i\kappa z}. \quad (0.3)$$

2 Structured proof

$\langle 1 \rangle$ 1 Determine the laws and the edge covariance exactly.

$\langle 2 \rangle$ 1 η_N is real.

Because: Away from the edge, the terms at κ and $-\kappa$ are conjugate. On the edge, the lattice characters at $\pm R_N$ agree, and

$$G_N(-k_0, k_e) = 2^{-1/2}[G(-k_0, k_e) + G(-k_0, -k_e)] = \overline{G_N(k_0, k_e)}.$$

$\langle 2 \rangle 2$ The G_N are orthonormal on D_N and have the real-field pairing.

Because: Off the edge this is the covariance of G . On the edge,

$$\mathbb{E}[G_N(k_0, k_e)\overline{G_N(k'_0, k_e)}] = \frac{1}{2}(1+1)\delta_{k_0 k'_0},$$

while cross terms between $+R_N$ and $-R_N$ noises vanish. Expanding the two two-term averages without conjugation leaves exactly the pairings with $k'_0 = -k_0$, and therefore $\mathbb{E}[G_N(k_0, k_e)G_N(k'_0, k_e)] = \delta_{k_0, -k'_0}$.

$\langle 2 \rangle 3$ Therefore

$$\mathbb{E}[\eta_N(s, x)\eta_N(s', y)] = \frac{1}{2L} \sum_{k \in \Gamma_N} e^{ik(x-y)} c_{\gamma_N(k)}(s - s'), \quad (1.1)$$

$$c_\omega(\tau) = \frac{\cosh(\omega(t/2 - |\tau|_{\mathbb{T}_t}))}{2\omega \sinh(\omega t/2)}.$$

Because: Fourier pairing gives the sum, and

$$\frac{1}{t} \sum_{k_0 \in (2\pi/t)\mathbb{Z}} \frac{e^{ik_0\tau}}{k_0^2 + \omega^2} = c_\omega(\tau).$$

Both sides solve $(-\partial_\tau^2 + \omega^2)c = \delta_{\mathbb{T}_t}$; the derivative jump at zero is -1 , which fixes the Fourier coefficients and normalization.

$\langle 2 \rangle 4$ The mixed covariance is

$$\mathbb{E}[\eta_N(z)\eta(z')] = C_\times(z - z') + \mathcal{R}_N(z, z'), \quad (1.2)$$

where $C_\times$ is the D_N Fourier sum with symbol $c_\times$ and

$$\mathcal{R}_N = (2Lt)^{-1} \sum_{k_0} c_\times(k_0, k_e) e^{ik_0(s-s')} \left[2^{-1/2}(e^{ik_e(x-y)} + e^{ik_e(x+y)}) - e^{ik_e(x-y)} \right]. \quad (1.3)$$

Because: Only (0.2) differs from the common-noise pairing. Pairing its two summands with the continuum noises at $\pm R_N$ gives the two characters in (1.3); subtracting the surrogate D_N term gives the bracket.

$\langle 2 \rangle 5$ For $\varepsilon_N \leq 1/2$,

$$\|\mathcal{R}_N\|_\infty \leq \frac{3 \coth(2t_0)}{8L} \varepsilon_N. \quad (1.4)$$

Because: The bracket is at most 3, $c_\times(k_0, k_e) \leq (k_0^2 + \gamma_e^2)^{-1}$, $\gamma_e \geq 2/\varepsilon_N$, and

$$\sum_{k_0} \frac{1}{k_0^2 + \gamma_e^2} = \frac{t}{2\gamma_e} \coth(t\gamma_e/2).$$

$\langle 2 \rangle 6$ Equivalently, the true mixed kernel is the Fourier sum over the symmetric band $\tilde{\Gamma}_N = \Gamma_N \cup \{+R_N\}$ with edge weights $\chi(\pm R_N) = 2^{-1/2}$ and $\chi = 1$ otherwise.

Because: Combine the two edge characters in (1.3) with the original edge term. On lattice points $e^{iR_N x} = e^{-iR_N x}$, so both continuum partners represent the same lattice degree of freedom.

⟨2⟩ 7 This proves ⟨1⟩1. □

⟨1⟩ 2 Establish Wick recombination and existence of the continuum integrated chaoses.

⟨2⟩ 1 For a centered Gaussian X of variance v and any c ,

$$:X^4:_c = :X^4:_v + 6(v - c):X^2:_v + 3(v - c)^2. \quad (2.1)$$

Because: Expand both sides using $:X^4:_a = X^4 - 6aX^2 + 3a^2$ and $:X^2:_a = X^2 - a$; coefficients of $X^4, X^2, 1$ agree.

⟨2⟩ 2 With lattice thermal variance $v_N(t)$ and vacuum Wick constant $c_{N,L}$,

$$d_N(t) := v_N(t) - c_{N,L} = \frac{1}{2L} \sum_{k \in \Gamma_N} \frac{1}{\gamma_N(k)(e^{t\gamma_N(k)} - 1)}. \quad (2.2)$$

Moreover $0 \leq d_N \leq \bar{d} < \infty$ and $|d_N - d_\infty| \leq C\varepsilon_N^2$ uniformly for $t \geq t_0$.

Because: $\coth(y/2) - 1 = 2/(e^y - 1)$. On the band, $\gamma_N(k) \geq \max(m, 2|k|/\pi)$; the positive majorant decays exponentially. For common modes apply the mean value theorem to $f_t(y) = [y(e^{ty} - 1)]^{-1}$ and $0 \leq \gamma(k) - \gamma_N(k) \leq \varepsilon_N^2 k^4/(24m)$. The derivative is bounded by $C(m, t_0)(1 + t)e^{-t\gamma_-(k)/2}$, whose k^4 -weighted sum is uniform in $t \geq t_0$. Missing spatial modes contribute $O(e^{-2t_0/\varepsilon_N})$.

⟨2⟩ 3 For jointly centered Gaussian X, Y ,

$$\mathbb{E}[:X^j:_v :Y^\ell:_v] = \delta_{j\ell} j! \mathbb{E}[XY]^j. \quad (2.3)$$

Because: Multiply the generating functions $e^{\alpha X - \alpha^2 v_X/2}$ and $e^{\beta Y - \beta^2 v_Y/2}$; their expectation is $e^{\alpha\beta \mathbb{E}[XY]}$. Compare the absolutely convergent power-series coefficients.

⟨2⟩ 4 Let $Y_j^{(K)} = \int_{\mathbb{T}_t \times \mathbb{T}_L} : \eta_K^j :$ for $j = 2, 4$. Then

$$\mathbb{E}[Y_j^{(K)} Y_j^{(K')}] = j!(2Lt)^{-(j-2)} \sum_{\substack{\kappa_i \in D_\infty^{\leq K \wedge K'} \\ \sum_i \kappa_i = 0}} \prod_i c_\infty(\kappa_i). \quad (2.4)$$

Because: Apply (2.3) pointwise, expand the covariance Fourier series, and integrate over the two spacetime variables. Each integration supplies $2Lt$ if the total momentum is zero and zero otherwise.

⟨2⟩ 5 The sums in (2.4) increase with K and are bounded; hence $Y_j^{(K)}$ is Cauchy in L^2 and defines Y_j .

Because: Put $c_{\text{env}}(\kappa) = (\pi^2/4)(m^2 + |\kappa|^2)^{-1}$. Step ⟨1⟩3 proves the convolution bounds that make $\sum c_{\text{env}}(\kappa)(c_{\text{env}}^*)(-\kappa)$ and $\sum c_{\text{env}}^2$ finite. Positivity gives monotonicity; the polarization identity gives the Cauchy norm.

⟨2⟩ 6 Define

$$\begin{aligned} \mathcal{U}_N &= \lambda[X_4 + 6d_N X_2 + 3d_N^2(2Lt)], \\ \mathcal{U} &= \lambda[Y_4 + 6d_\infty Y_2 + 3d_\infty^2(2Lt)]. \end{aligned} \quad (2.5)$$

Then $\mathcal{U}_N$ is exactly the lattice-vacuum Wick quartic integrated over spacetime.

Because: Apply (2.1) at every spacetime point with $v = v_N$ and $c = c_{N,L}$, then sum and integrate.

⟨2⟩ 7 This proves ⟨1⟩2. □

⟨1⟩ 3 Prove the covariance envelope and convolution estimates.

⟨2⟩ 1 $c_N, c_\times, c_\infty \leq c_{\text{env}}$.

Because: On $|\varepsilon_N k| \leq \pi$, $\sin(|\varepsilon_N k|/2) \geq |\varepsilon_N k|/\pi$, so $k_0^2 + \gamma_N(k)^2 \geq (4/\pi^2)(m^2 + k_0^2 + k^2)$. By its definition, $c_\infty = (m^2 + k_0^2 + k^2)^{-1} \leq c_{\text{env}}$; hence $c_\times = (c_N c_\infty)^{1/2} \leq c_{\text{env}}$ as well.

⟨2⟩ 2 For $R \geq 0$, the anisotropic grid satisfies

$$\sum_{|\kappa| \leq R} c_{\text{env}}(\kappa) \leq C(m, t_0) tL [1 + \log(2 + R)], \quad (3.1)$$

$$\sum_{|\kappa| > R} \frac{1 + \log(2 + |\kappa|)}{(m^2 + |\kappa|^2)^2} \leq C(m, t_0) tL \frac{1 + \log(2 + R)}{m^2 + R^2}. \quad (3.2)$$

Because: Cover each grid point by its rectangle of side lengths $2\pi/t$ and π/L . Outside a fixed disk the radial functions vary by a bounded factor on one rectangle and the sum is bounded by the grid density $tL/(2\pi^2)$ times the radial integral. Inside the disk there are finitely many rectangles; the isolated maximum is absorbed because $t \geq t_0$ and $tL \geq 1$. The radial integrals are $\int_0^R r(m^2 + r^2)^{-1} dr = O(1 + \log(2 + R))$ and $\int_R^\infty r(1 + \log(2 + r))(m^2 + r^2)^{-2} dr = O((1 + \log(2 + R))/(m^2 + R^2))$.

⟨2⟩ 3 For $F_2 = c_{\text{env}} * c_{\text{env}}$,

$$F_2(\chi) \leq C(m, t_0) tL \frac{1 + \log(2 + |\chi|)}{m^2 + |\chi|^2}. \quad (3.3)$$

Because: Split the summation into $|\kappa| \leq |\chi|/2$, $|\chi - \kappa| \leq |\chi|/2$, and the complement. On the first region the second factor is $O((m^2 + |\chi|^2)^{-1})$ and (3.1) sums the first; the second region is symmetric. On the complement both arguments exceed $|\chi|/2$; replace the product by twice the square of the factor with smaller argument and apply (3.2) without its logarithm.

⟨2⟩ 4 For $F_3 = c_{\text{env}} * F_2$,

$$F_3(\chi) \leq C(m, t_0) (tL)^2 \frac{[1 + \log(2 + |\chi|)]^2}{m^2 + |\chi|^2}. \quad (3.4)$$

Because: Use the same three regions. In the first, sum F_2 up to radius $|\chi|/2$; (3.3) and a radial integral produce two logarithms. In the second, bound F_2 at radius $|\chi|/2$ and sum c_{env} by (3.1). In the complement, the product is bounded by $CtL(1 + \log(2 + |u|))(m^2 + |u|^2)^{-2}$ at the smaller argument; apply (3.2).

⟨2⟩ 5 The dispersion-symbol difference obeys

$$|c_N - c_\infty|(\kappa) \leq \frac{\varepsilon_N^2 k^4}{12} c_N(\kappa) c_\infty(\kappa), \quad |c_\times - c_\infty| \leq |c_N - c_\infty|. \quad (3.5)$$

Because: Subtract the two resolvents and use $0 \leq k^2 - 4\varepsilon_N^{-2} \sin^2(\varepsilon_N k/2) \leq \varepsilon_N^2 k^4/12$. For the geometric mean, $|\sqrt{a} - \sqrt{b}| = |a - b|/(\sqrt{a} + \sqrt{b})$.

⟨2⟩ 6 This proves ⟨1⟩3. □

⟨1⟩ 4 Prove the unshifted L^2 interaction comparison, including every edge sector.

⟨2⟩ 1 Wick–Isserlis and Fourier orthogonality give

$$\mathbb{E}[X_4^2] = 4!(2Lt)^{-2} \sum_{\substack{\kappa_i \in D_N \\ \sum k_{0,i}=0, \sum \kappa_i=0}} \prod_i c_N(\kappa_i), \quad (4.1)$$

$$\mathbb{E}[Y_4^2] = 4!(2Lt)^{-2} \sum_{\substack{\kappa_i \in D_\infty \\ \sum \kappa_i=0}} \prod_i c_\infty(\kappa_i), \quad (4.2)$$

$$\mathbb{E}[X_4 Y_4] = 4!(2Lt)^{-2} \sum_{\substack{\kappa_i \in \tilde{D}_N \\ \sum \kappa_i=0}} \prod_i \chi(k_i) c_\times(\kappa_i). \quad (4.3)$$

Because: The lattice spatial sum enforces conservation modulo $2R_N$; the continuum spatial integral enforces exact conservation. Step ⟨1⟩1⟨2⟩6 gives the symmetric band and weights in (4.3).

⟨2⟩ 2 The four-edge sector omitted in V2 is

$$\Delta_{4e} = 4!(2Lt)^{-2} \frac{3}{2} \sum_{k_{0,1}+\dots+k_{0,4}=0} \prod_i f(k_{0,i}), \quad f(k_0) = c_\times(k_0, R_N). \quad (4.4)$$

Because: Exact spatial conservation with four edge inputs requires two $+R_N$ and two $-R_N$. There are $\binom{4}{2} = 6$ orders and their weight is $(2^{-1/2})^4 = 1/4$, hence $6/4 = 3/2$. Configurations with exactly three edge inputs are impossible, but configurations with four are not.

⟨2⟩ 3 The correction satisfies

$$0 \leq \Delta_{4e} \leq \frac{9t}{256L^2} \coth(2t_0)^3 \varepsilon_N^5. \quad (4.5)$$

Because: Let $A = \sum f$ and $B = \sum f^2$. From $f \leq (k_0^2 + \gamma_e^2)^{-1}$ and $\gamma_e \geq 2/\varepsilon_N$,

$$A \leq \frac{t\varepsilon_N}{4} \coth(2t_0), \quad B \leq \gamma_e^{-2} A \leq \frac{t\varepsilon_N^3}{16} \coth(2t_0).$$

Furthermore $\sum \sum_{k_{0,i}=0} \prod f = \|f * f\|_2^2 \leq \|f * f\|_\infty \|f * f\|_1 \leq BA^2$. Insert this into (4.4).

⟨2⟩ 4 The remaining one-edge and two-edge differences between the symmetric and half-open descriptions are $O(t\varepsilon_N^3 \log_N^2)$.

Because: For one edge, sum its symbol to obtain $O(t\varepsilon_N)$ and bound the remaining three-fold constrained sum at spatial momentum R_N by (3.4), namely $O((tL)^2 \varepsilon_N^2 \log_N^2)$; the prefactor gives the claimed order. For two edges, put the temporal edge sequence in ℓ^2 and use $\|f * f\|_\infty \leq \sum f^2 = O(t\varepsilon_N^3)$; the remaining two-fold sum is bounded by (3.3). There are finitely many placements. Step ⟨2⟩3 covers four edges and is smaller.

⟨2⟩ 5 The lattice alias contribution in (4.1) obeys

$$Al_N \leq C(m, t_0) t L \varepsilon_N^2 \log_N^3. \quad (4.6)$$

Because: A nonzero alias has total spatial momentum $\sigma \in \{\pm 2R_N, -4R_N\}$. Fix κ_1 and sum the other three factors by (3.4) at $\sigma - \kappa_1$, whose spatial modulus is at least R_N ; this is at most $C(tL)^2 \varepsilon_N^2 \log_N^2$. Sum the remaining envelope over the band by (3.1), giving $CtL \log_N$, and multiply by $4!(2Lt)^{-2}$.

⟨2⟩ 6 For the principal fourth-chaos difference, telescope the four products. Then

$$\|X_4 - Y_4\|_2^2 \leq C(m, t_0) t L \varepsilon_N^2 \log_N^3. \quad (4.7)$$

Because: Each telescoped common-mode term is bounded by

$$\sum_{\kappa \in D_N} |c_N - c_\infty|(\kappa) F_3(-\kappa).$$

Use (3.5), (3.4), sum k_0 in $(m^2 + k_0^2 + k^2)^{-3}$, and then sum $k^4 \log^2(2 + |k|)(m^2 + k^2)^{-5/2}$ up to R_N ; it is $O(tL \log_N^3)$. The common-mode result is $O((tL)^3 \varepsilon_N^2 \log_N^3)$ before the prefactor. Missing continuum modes use $c_{\text{env}} F_3$ and the $|k| \geq R_N$ tail, giving one fewer logarithm. Add (4.5), the one/two-edge bound, and (4.6).

⟨2⟩ 7 The second-chaos difference satisfies

$$\|X_2 - Y_2\|_2^2 = 2 \sum_{D_N} (c_N - c_\infty)^2 + 2 \sum_{D_\infty \setminus D_N} c_\infty^2 \leq C(m, t_0) t L \varepsilon_N^2. \quad (4.8)$$

Because: The two symmetric edge pairings have weights $1/2$ and restore full weight; the lattice half-open alias replaces the missing opposite edge, so no residual edge term remains. Square (3.5), sum k_0 with exponent 4, and note the remaining band sum is $O(LR_N^2)$, giving $O(tL \varepsilon_N^2)$. The continuum tail of the squared resolvent has the same order.

⟨2⟩ 8 Combining (2.2), (2.5), (4.7), and (4.8),

$$\|\mathcal{U}_N - \mathcal{U}\|_2 \leq C(m, \lambda, t_0)(1 + t)(1 + L)\varepsilon_N \log_N^{3/2}. \quad (4.9)$$

Because: Apply the triangle inequality to the fourth chaos, the $6dX_2$ term, the coefficient difference $(d_N - d_\infty)Y_2$, and the scalar difference $3(d_N^2 - d_\infty^2)2Lt$. Use $d_N, d_\infty \leq \bar{d}$ and $\|Y_2\|_2^2 \leq CtL$.

⟨2⟩ 9 This proves ⟨1⟩4. □

⟨1⟩ 5 Verify the Wick dictionary and absence of a continuum counterterm.

⟨2⟩ 1 Let

$$\delta_N = \frac{1}{4L} \sum_{k \in \Gamma_N} \left(\frac{1}{\gamma_N(k)} - \frac{1}{\gamma(k)} \right).$$

With $u_j = \pi j / (2r_N)$ and $\tilde{\delta} = m\varepsilon_N / 2$, pair the nonedge $\pm j$ terms to obtain

$$\delta_N = \frac{1}{2\pi} \sum_{j=1}^{r_N-1} g_{\tilde{\delta}}(u_j) \Delta u + E_{\text{edge}}, \quad g_{\tilde{\delta}}(u) = \frac{1}{\sqrt{\tilde{\delta}^2 + \sin^2 u}} - \frac{1}{\sqrt{\tilde{\delta}^2 + u^2}}, \quad (5.1)$$

where $0 \leq E_{\text{edge}} \leq \varepsilon_N / (8L)$.

Because: Substitute the two exact dispersions, use $\Delta u = \pi / (2r_N)$, and separate the unpaired $j = -r_N$ term.

⟨2⟩ 2 $|g'_{\tilde{\delta}}(u)| \leq C_0 < 6$ uniformly on $[0, \pi/2]$ and $\tilde{\delta} \geq 0$.

Because: Write $g_{\tilde{\delta}}(u) = \psi(\sin^2 u) - \psi(u^2)$ with $\psi(a) = (\delta^2 + a)^{-1/2}$. Use $u^2 - \sin^2 u \leq u^4/3$, $\sin u \geq 2u/\pi$, $|\sin 2u - 2u| \leq (2u)^3/6$, and $\psi''(a) = 3(\delta^2 + a)^{-5/2}/4$; the two derivative terms are bounded by $\frac{1}{2}(\pi/2)^5$ and $2/3$.

⟨2⟩ 3 The Riemann sum in (5.1) differs from its integral by $O(\varepsilon_N)$, and replacing $\tilde{\delta}$ by 0 changes the integral by $O(\varepsilon_N^2 \log(1/\varepsilon_N))$.

Because: The first claim is the Lipschitz Riemann bound. For the second, split the integral at $u = \tilde{\delta}$; the difference is at most Cu below the split and $C\tilde{\delta}^2/u$ above it.

⟨2⟩ 4 The limiting integral is

$$\frac{1}{2\pi} \int_0^{\pi/2} (\csc u - u^{-1}) du = \frac{1}{2\pi} \log \frac{4}{\pi}. \quad (5.2)$$

Because: The antiderivative difference is $\log \tan(u/2) - \log u$; its lower endpoint is $-\log 2$ and its upper endpoint is $-\log(\pi/2)$.

⟨2⟩ 5 Thus

$$\left| \delta_N - \frac{1}{2\pi} \log \frac{4}{\pi} \right| \leq C(m, L) \varepsilon_N. \quad (5.3)$$

⟨2⟩ 6 No quadratic or scalar counterterm survives in $\mathcal{U}$.

Because: At continuum truncation K , the field's thermal variance is $v_K = c_{L,K}^{\text{ct}} + d_K$. Recombining its own-variance Wick chaos by (2.1) with coefficient d_K gives exactly the vacuum-truncation Wick quartic. Since $d_K \rightarrow d_\infty$ exponentially, (2.5) is the $K \rightarrow \infty$ truncation-convention quartic. The finite offset (5.2) occurs simultaneously in the lattice field variance and in its Wick subtraction, so it cancels rather than becoming a mass term.

⟨2⟩ 7 This proves ⟨1⟩5. □

⟨1⟩ 6 Establish the shift-field estimates.

Fix a coarse level M , $N = M + r$, and a real coarse momentum symbol p_M . Define

$$h_N(s, x) = \frac{1}{2L} \sum_{k \in \Gamma_N} a_N(k, s) e^{ikx}, \quad a_N = \frac{\varepsilon_M^{1/2}}{2} P_r(\varepsilon_M k) \hat{p}_M(k) \frac{\sinh(\gamma_N(k)(s - t/2))}{\sinh(\gamma_N(k)t/2)}, \quad (6.1)$$

and h_∞ by replacing P_r, γ_N, Γ_N with $P_\infty, \gamma, \Gamma_\infty$.

⟨2⟩ 1 h_N, h_∞ are real and $\|h_N\|_\infty, \|h_\infty\|_\infty \leq \Sigma_M(p)$.

Because: Hermitian Fourier coefficients pair except at the half-open edge. For $r \geq 1$, $P_r(\pm\pi 2^r) = 0$ because the product contains $m_0(\pi) = 0$; for $r = 0$ the edge coefficient is real. The hyperbolic ratio in (6.1) has modulus at most one. Sum the D4 envelope $|P_r(\ell)| \leq C_D^{1/2} (1 + |\ell|)^{-1-\eta/2}$ by an integral comparison; the result is the explicit finite $\Sigma_M(p)$ of V2.

⟨2⟩ 2 For every integer $n \geq 1$,

$$\varepsilon_N \sum_x \int_0^t |h_N|^n \leq 2Lt \Sigma_M(p)^n, \quad \int_{\mathbb{T}_L} \int_0^t |h_\infty|^n \leq 2Lt \Sigma_M(p)^n. \quad (6.2)$$

Because: The spatial volumes are both $2L$; integrate the pointwise bound.

⟨2⟩ 3 With $\theta_H = (1 + \eta)/2$,

$$\sup_s \frac{1}{2L} \left[\sum_{k \in \Gamma_N} |a_N - a_\infty|^2 + \sum_{|k| \geq R_N} |a_\infty|^2 \right] \leq C_H 2^{-r\theta_H}. \quad (6.3)$$

Because: Split $a_N - a_\infty$ into filter and dispersion differences. On the band,

$$|P_r - P_\infty|(\ell) \leq C 2^{-r} |\ell| (1 + |\ell|)^{-1-\eta/2}.$$

Its squared sum is $O(2^{-r(1+\eta)})$. For the dispersion term use the γ derivative of the hyperbolic ratio, bounded by $t \coth(t_0 m/2)$, and $|\gamma_N - \gamma| \leq \varepsilon_N^2 k^4 / (24m)$. Split at $|\varepsilon_M k| = 2^{r/2}$: use this Taylor bound below and the trivial bound 2 above. Both pieces are $O(2^{-r(1+\eta)/2})$. The missing-mode D4 tail is $O(2^{-r(1+\eta)})$. The slowest exponent is θ_H .

⟨2⟩ 4 This proves ⟨1⟩6. □

⟨1⟩ 7 Prove the repaired normalized power-symbol estimates.

Let $B_N = \Gamma_N$, let $[k]_N$ be the representative modulo $2R_N$ in B_N , and, for $j = 1, 2, 3$, define

$$K_{N,j}^{\text{fold}}(p) = (2Lt)^{-(j-1)} \sum_{\substack{\kappa_i \in D_N \\ \sum k_{0,i}=p_0, [\sum k_i]_N=p}} \prod_i c_N(\kappa_i), \quad (7.1)$$

$$K_{\times,j}(\kappa) = (2Lt)^{-(j-1)} \sum_{\substack{\kappa_i \in D_N \\ \sum \kappa_i = \kappa}} \prod_i c_{\times}(\kappa_i), \quad (7.2)$$

$$K_{\infty,j}(\kappa) = (2Lt)^{-(j-1)} \sum_{\sum \kappa_i = \kappa} \prod_i c_{\infty}(\kappa_i). \quad (7.3)$$

⟨2⟩ 1 $\sup K_{a,j} \leq B_j(m, t_0)$ for all three families.

Because: For $j = 1$ use Step ⟨1⟩3⟨2⟩1. For $j = 2, 3$, dominate by F_2, F_3 in (3.3)–(3.4); the factors $(tL)^{j-1}$ cancel the normalization $(2Lt)^{j-1}$. Folded aliases are a finite sum of shifted exact outputs and satisfy the same supremum bound.

⟨2⟩ 2 On the fundamental band,

$$\sup_p |K_{N,j}^{\text{fold}}(p) - K_{\infty,j}(p)| \leq C\varepsilon_N^2 \log_N^2, \quad (7.4)$$

$$\sup_p |K_{\times,j}(p) - K_{\infty,j}(p)| \leq C\varepsilon_N^2 \log_N^2. \quad (7.5)$$

Because: Partition (7.1) into exact output and the finitely many nonzero shifts $2R_N n$. On exact common configurations telescope the j factors and use the flat consequence of (3.5), $|c_N - c_{\infty}| \leq C\varepsilon_N^2$, followed by the band sums of the remaining $j - 1$ envelope convolution; these are $O((tL)^{j-1} \log_N^{j-1})$. For a nonzero output shift, one convolution argument has spatial modulus at least R_N and (3.3)–(3.4) gives $O(\varepsilon_N^2 \log_N^{j-1})$ after normalization. For continuum configurations with one input outside the band, fix that input and use the same tail convolution; the result is $O(\varepsilon_N^2 \log_N)$. This proves (7.4). The proof of (7.5) is identical without folded output and using the second inequality in (3.5).

⟨2⟩ 3 For every exact output with $|k| \geq R_N$,

$$K_{\times,j}(k_0, k) + K_{\infty,j}(k_0, k) \leq C\varepsilon_N^2 \log_N^{j-1}. \quad (7.6)$$

Because: Apply (3.3) or (3.4) at spatial distance at least R_N and divide by $(2Lt)^{j-1}$.

⟨2⟩ 4 This proves ⟨1⟩7. □

⟨1⟩ 8 Prove the shifted-interaction comparison with no undefined remainder.

For $j = 1, 2, 3$, put $g_N = h_N^{4-j}$, $g_{\infty} = h_{\infty}^{4-j}$ and

$$T_{N,j} = \varepsilon_N \sum_x \int g_N : \eta_N^j :, \quad T_{\infty,j} = \int \int g_{\infty} : \eta^j :.$$

⟨2⟩ 1 The true mixed covariance-power contribution differs from the translation-invariant surrogate by at most

$$C(2Lt)^2 \Sigma_M(p)^{2(4-j)} \varepsilon_N \log_N^{j-1}. \quad (8.1)$$

Because: By (1.4), $\|\mathcal{R}_N\|_\infty \leq C\varepsilon_N$, while $\|C_\times\|_\infty \leq C \log_N$. Hence $|(C_\times + \mathcal{R}_N)^j - C_\times^j| \leq C\varepsilon_N \log_N^{j-1}$. Integrate against the pointwise bounds from (6.2).

⟨2⟩ 2 Write $b_N(p, k_0)$ and $b_\infty(k, k_0)$ for the spacetime Fourier coefficients of g_N, g_∞ . Extend b_N by zero outside B_N and set $\delta b = b_N - b_\infty$ on B_N , $\delta b = -b_\infty$ outside. Then the surrogate quadratic difference has the exact decomposition

$$\begin{aligned} Q_{N,j} - 2Q_{\times,j} + Q_{\infty,j} &= \langle \delta b, K_{\infty,j} \delta b \rangle + \langle b_N, (K_{N,j}^{\text{fold}} - K_{\infty,j}) b_N \rangle \\ &\quad - 2\langle b_N, (K_{\times,j} - K_{\infty,j}) b_{\infty, B_N} \rangle - 2A_{\times,j}, \end{aligned} \quad (8.2)$$

where

$$A_{\times,j} = \frac{1}{2Lt} \sum_{p \in B_N, n \neq 0} K_{\times,j}(p_0, p + 2R_N n) \overline{b_N(-p, -p_0)} b_\infty(p + 2R_N n, p_0). \quad (8.3)$$

Because: Expand all three Fourier pairings. The lattice–lattice spatial sums fold the exact output modulo $2R_N$, producing (7.1). In the mixed pairing, continuum integration fixes the exact output $p + 2R_N n$, while the lattice sum sees its representative p ; split $n = 0$ and $n \neq 0$. Add and subtract the continuum symbol on the fundamental band. Expanding $\langle \delta b, K_{\infty,j} \delta b \rangle$ gives the common-band continuum terms plus the entire continuum tail, leaving exactly (8.2)–(8.3). No term remains unnamed.

⟨2⟩ 3 The mixed alias-output term obeys

$$|A_{\times,j}| \leq C\varepsilon_N^2 \log_N^2 \|b_N\|_2 \|b_\infty\|_2. \quad (8.4)$$

Because: Only finitely many n occur: the support of a j -fold output lies in $[-jR_N, jR_N]$. For $n \neq 0$, (7.6) bounds the symbol. Apply Cauchy–Schwarz to the finite repeated coefficient sums; each coefficient occurs at most $2j + 1$ times.

⟨2⟩ 4 The coefficient norms satisfy

$$\|b_N\|_2^2 + \|b_\infty\|_2^2 \leq Ct \Sigma_M(p)^{2(4-j)}, \quad \|b_N - b_\infty\|_2^2 \leq Ct 2^{-r\theta_H}. \quad (8.5)$$

Because: Parseval and (6.2) give the first inequality. For the second, write the $(4 - j)$ -fold convolution difference as

$$a_N^{*n} - a_\infty^{*n} = \sum_{\ell=0}^{n-1} a_N^{*\ell} * (a_N - a_\infty) * a_\infty^{*(n-1-\ell)}.$$

Use $\ell^2 * \ell^1 \rightarrow \ell^2$, (6.3), and the uniform ℓ^1 bound from the D4 envelope. For a lattice alias, $|k_1 + \dots + k_n| \geq R_N$ forces one $|k_i| \geq R_N/n$; place that factor in the D4 ℓ^2 tail and the remaining factors in ℓ^1 . Missing continuum modes are identical. Integrate the uniform in s estimate over $[0, t]$.

⟨2⟩ 5 Equations (7.4)–(7.6), (8.2)–(8.5), and the Schur bound yield

$$\mathbb{E}[(T_{N,j} - T_{\infty,j})^2] \leq C \left[t 2^{-r\theta_H} + t \varepsilon_N^2 \log_N^2 + t^2 \varepsilon_N \log_N^{j-1} \right]. \quad (8.6)$$

Because: The first term of (8.2) is at most $B_j \|\delta b\|_2^2$. The next two use (7.4)–(7.5) times the coefficient norms. The last uses (8.4). Add the physical edge remainder (8.1), with the harmless volume factors absorbed in the fixed parameter constant.

⟨2⟩ 6 Since $\varepsilon_N = \varepsilon_M 2^{-r}$ and $\theta_H < 1$,

$$\varepsilon_N \log_N^2 \leq C(\varepsilon_M, \eta) 2^{-r\theta_H}. \quad (8.7)$$

Because: Divide by $2^{-r\theta_H}$; the ratio is $\varepsilon_M 2^{-r(1-\theta_H)}(1 + \log(1/\varepsilon_M) + r \log 2)^2$, whose supremum over $r \geq 0$ is finite because an exponential with positive exponent dominates a polynomial.

⟨2⟩ 7 The deterministic $j = 0$ term has absolute difference $O(t 2^{-r\theta_H/2})$.

Because: It is the zero Fourier coefficient of $h_N^4 - h_\infty^4$. Use the same four-factor convolution telescope and alias/missing-mode estimate as in (8.5), followed by Cauchy–Schwarz.

⟨2⟩ 8 After Wick-binomial coefficients and the $d_N - d_\infty$ recombination errors are included,

$$\|[\mathcal{U}_N(\eta_N + h_N) - \mathcal{U}_N(\eta_N)] - [\mathcal{U}(\eta + h_\infty) - \mathcal{U}(\eta)]\|_2 \leq C(1+t)^{5/2} \left[2^{-r(1+\eta)/4} + \varepsilon_N \log_N \right]. \quad (8.8)$$

Because: Take square roots in (8.6), use $\theta_H/2 = (1+\eta)/4$, sum $j = 0, 1, 2, 3$ with coefficients 1, 4, 6, 4, and use $|d_N - d_\infty| = O(\varepsilon_N^2)$. The largest volume power is bounded by $(1+t)^{5/2}$.

⟨2⟩ 9 This proves ⟨1⟩ 8. □

⟨1⟩ 9 Prove the uniform Nelson exponential bounds.

⟨2⟩ 1 Pointwise, for real u ,

$$:u^4:_c = (u^2 - 3c)^2 - 6c^2 \geq -6c^2.$$

Hence

$$\mathcal{U}_N(\eta_N + h) \geq -c_1 t \log_N^2 \quad (9.1)$$

for every deterministic real shift h .

Because: Integrate the pointwise inequality over volume $2Lt$ and use $c_{N,L} \leq C(m, L) \log_N$. Replacing u by the real number $u + h$ changes nothing.

⟨2⟩ 2 For $b \leq c_1 t \log_N^2$, choose the largest dyadic level $N_b \leq N$ with $1 + \log(1/\varepsilon_{N_b}) \leq L_b = (b/(2c_1 t))^{1/2}$. Then

$$\{\mathcal{U}_N \leq -b\} \subseteq \{\mathcal{U}_N - \mathcal{U}_{N_b} \leq -b/2\}. \quad (9.2)$$

Because: Dyadic log increments are exactly $\log 2$, so for sufficiently large b the level exists and its logarithm is at least $L_b/2$. By (9.1), $\mathcal{U}_{N_b} \geq -c_1 t L_b^2 = -b/2$.

⟨2⟩ 3 The comparison estimates imply

$$\rho_b := \|\mathcal{U}_N - \mathcal{U}_{N_b}\|_2 \leq C L_b^{3/2} e^{-c L_b}. \quad (9.3)$$

Because: Apply (4.9) at levels N, N_b through the common continuum interaction; $\varepsilon_{N_b} \leq e^{1-L_b/2}$. In the shifted case also use (8.8); its dyadic term is $O(e^{-(1+\eta)L_b/8})$. Take the smaller positive exponent c .

⟨2⟩ 4 If Z is a sum of Gaussian chaoses of order at most four, then for $q \geq 1$,

$$\|Z\|_{2q} \leq (2q-1)^2 \|Z\|_2. \quad (9.4)$$

Because: Write $Z = \sum_{j=0}^4 Z_j$ orthogonally. For $\tau = \frac{1}{2} \log(2q-1)$, set $Y = \sum e^{j\tau} Z_j$; then $Z = \Gamma(e^{-\tau})Y$. Nelson hypercontractivity gives $\|Z\|_{2q} \leq \|Y\|_2 \leq e^{4\tau} \|Z\|_2 = (2q-1)^2 \|Z\|_2$.

⟨2⟩ 5 Chebyshev with $(2q)^2 = e^{cL_b/2}$ gives

$$\mathbb{P}(\mathcal{U}_N \leq -b) \leq \exp[-e^{cL_b/4}] \quad (9.5)$$

for all sufficiently large b , uniformly in N ; the same holds for shifted interactions on bounded coarse-symbol sets.

Because: By (9.2)–(9.4),

$$\mathbb{P}(\mathcal{U}_N \leq -b) \leq \left[\frac{2}{b} (2q)^2 \rho_b \right]^{2q}.$$

Insert (9.3); for large b the bracket is at most e^{-1} , and $2q = e^{cL_b/4}$.

⟨2⟩ 6 For every $p < \infty$ and $t \in [t_0, T]$,

$$\sup_N \mathbb{E}[e^{-p\mathcal{U}_N}] < \infty, \quad \sup_N \mathbb{E}[e^{-p\mathcal{U}_N(\eta_N + h_N)}] < \infty. \quad (9.6)$$

Because: The layer-cake formula integrates $pe^{pb}\mathbb{P}(\mathcal{U}_N \leq -b)$. The double-exponential tail (9.5) dominates e^{pb} ; the bounded b interval contributes a finite uniform constant.

⟨2⟩ 7 This proves ⟨1⟩9. □

3 V2 conclusion

The repaired V2 outputs are (4.9), the Wick dictionary (5.3) with no residual counterterm, the shift-field convergence (6.3), the fully accounted shifted comparison (8.8), and the unshifted/shifted Nelson bounds (9.6). The original four-edge assertion and the original R_{tail} line must not be used; equations (4.4)–(4.5) and (8.2)–(8.4) replace them.

Part II V3 in Lamport notation

Construction of the continuum torus Hamiltonian, trace class, periodic Feynman–Kac formula, and the gap

Fresh reconstruction of `part_ii/v3_continuum_package.tex`

16 August 2026

The continuum package is reconstructed below without importing a pre-existing $P(\varphi)_2$ Hamiltonian. The only printed input is Simon’s four-result second-quantization cluster, checked in the accompanying citation-evidence dossier. The periodic cutoff estimate that V3 states only by analogy is supplied explicitly in Steps $\langle 3 \rangle 8$ – $\langle 3 \rangle 10$.

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1 Data, exact target, and imports

Fix $L, m > 0$, $\lambda \geq 0$, and $a, b \in \mathbb{R}$. Put

$$\Gamma_\infty = \frac{\pi}{L}\mathbb{Z}, \quad \gamma(k) = (m^2 + k^2)^{1/2}, \quad \mathbb{T}_L = \mathbb{R}/(2L\mathbb{Z}). \quad (0.1)$$

Whenever a periodic time interval is used, fix $t \in [t_0, T]$ and impose

$$t_0 L \geq 1. \quad (0.2)$$

The gate ultimately uses $(a, b) = (0, 0)$. Allowing arbitrary (a, b) here makes explicit that no continuum counterterm is being hidden.

Let $(e_j)_{j \geq 1}$ be the real orthonormal Fourier basis

$$e_0 = (2L)^{-1/2}, \quad e_k^c = L^{-1/2} \cos(kx), \quad e_k^s = L^{-1/2} \sin(kx) \quad (k > 0), \quad (0.3)$$

and let γ_j be the corresponding value of γ . On

$$(Q, \mu) = \left(\mathbb{R}^{\mathbb{N}}, \bigotimes_{j \geq 1} \mathcal{N}(0, 1) \right) \quad (0.4)$$

write the coordinate variables as ξ_j and define

$$\varphi^{(K)}(x) = \sum_{|k_j| \leq K} \frac{\xi_j}{\sqrt{2\gamma_j}} e_j(x), \quad c_K = \mathbb{E}_\mu[\varphi^{(K)}(x)^2] = \frac{1}{2L} \sum_{|k| \leq K} \frac{1}{2\gamma(k)}. \quad (0.5)$$

The free Hamiltonian on $L^2(Q, \mu)$ is

$$H_0 = d\Gamma(\gamma), \quad e^{-sH_0} = \Gamma(e^{-s\gamma}), \quad Z_0(s) = \text{Tr } e^{-sH_0} = \prod_{k \in \Gamma_\infty} (1 - e^{-s\gamma(k)})^{-1}. \quad (0.6)$$

Its bounded smooth cylinder core is

$$\mathcal{D} = \{F(\xi_1, \dots, \xi_n) : n < \infty, F \in C_b^\infty(\mathbb{R}^n)\}. \quad (0.7)$$

The following are the only external statements used.

Imported assumption 1.1 (Simon second-quantization cluster). For a real-Hilbert-space contraction A , the Gaussian second quantization $\Gamma(A)$ is positivity preserving; it is an L^p contraction for every $1 \leq p \leq \infty$; it is positivity improving if $\|A\| < 1$; and

$$\|\Gamma(A)f\|_q \leq \|f\|_p \quad \text{if } 1 < p \leq q < \infty, \quad \|A\| \leq \sqrt{\frac{p-1}{q-1}}. \quad (0.8)$$

These are Simon, Theorem I.12, Corollary I.15, Theorems I.16 and I.17. The exact pages are reproduced in the evidence dossier.

The elementary bounded-perturbation Trotter product formula and the spectral theorem are used in the forms proved in the model-sheet reconstruction. All remaining inputs below are results already proved in the V1 and repaired V2 Lamport modules.

2 Interaction and sharp-time convergence

$\langle 1 \rangle$ 1 Bound the cutoff variance.

$\langle 2 \rangle$ 1 For $K \geq m$,

$$c_K \leq C_c(m, L) [1 + \log(1 + K/m)]. \quad (1.1)$$

Because: Pair the sine and cosine modes and compare the decreasing summand with the integral:

$$\sum_{0 < k \leq K} \frac{1}{\sqrt{m^2 + k^2}} \leq \frac{L}{\pi} \int_0^K \frac{du}{\sqrt{m^2 + u^2}} + \frac{1}{m} = \frac{L}{\pi} \text{arsinh}(K/m) + \frac{1}{m}.$$

Since $\operatorname{arsinh} x \leq 1 + \log(1 + x)$, (1.1) follows.

$\langle 2 \rangle 2$ This proves $\langle 1 \rangle 1$. □

$\langle 1 \rangle 2$ Define and bound the mode-cutoff interaction.

For $K \in \Gamma_\infty$, set

$$V^{(K)} = \lambda \int_{\mathbb{T}_L} \left[:\varphi^{(K)}(x)^4:_{c_K} + a:\varphi^{(K)}(x)^2:_{c_K} + b \right] dx. \quad (1.2)$$

$\langle 2 \rangle 1$ For $u \in \mathbb{R}$ and $c \geq 0$,

$$:u^4:_c + a:u^2:_c + b = (u^2 - 3c)^2 + a(u^2 - 3c) - 6c^2 + 2ac + b. \quad (1.3)$$

Because: Expanding the right side gives $u^4 + (a - 6c)u^2 + 3c^2 - ac + b$, which is $(u^4 - 6cu^2 + 3c^2) + a(u^2 - c) + b$.

$\langle 2 \rangle 2$ Consequently,

$$:u^4:_c + a:u^2:_c + b \geq - \left(6c^2 + 2|a|c + \frac{a^2}{4} + |b| \right). \quad (1.4)$$

Because: For $v = u^2 - 3c$, $v^2 + av = (v + a/2)^2 - a^2/4 \geq -a^2/4$; $2ac + b \geq -2|a|c - |b|$.

$\langle 2 \rangle 3$ Thus

$$V^{(K)} \geq -b_K, \quad b_K := 2L\lambda \left(6c_K^2 + 2|a|c_K + \frac{a^2}{4} + |b| \right) \leq C[1 + \log(1 + K/m)]^2. \quad (1.5)$$

$\langle 2 \rangle 4$ If $\lambda > 0$, $V^{(K)}(\xi) \rightarrow +\infty$ as $|\xi| \rightarrow \infty$ in the finitely many retained coordinates.

Because: Cauchy–Schwarz on a space of volume $2L$ and Parseval give

$$\int_{\mathbb{T}_L} (\varphi^{(K)})^4 dx \geq \frac{1}{2L} \left(\int_{\mathbb{T}_L} (\varphi^{(K)})^2 dx \right)^2, \quad \int_{\mathbb{T}_L} (\varphi^{(K)})^2 dx = \sum_{|k_j| \leq K} \frac{\xi_j^2}{2\gamma_j}. \quad (1.6)$$

All Wick-subtraction and a terms are affine in the last sum. A positive quadratic polynomial in that sum therefore dominates them at infinity. When $\lambda = 0$, the free harmonic operator supplies coercivity.

$\langle 2 \rangle 5$ This proves $\langle 1 \rangle 2$. □

$\langle 1 \rangle 3$ Construct the limiting sharp-time interaction in $L^2(\mu)$.

$\langle 2 \rangle 1$ Own-variance Wick ordering gives the exact chaos decomposition

$$V^{(K)} = \lambda I_4(f_K) + \lambda a I_2(g_K) + 2L\lambda b, \quad (1.7)$$

where f_K, g_K are restrictions of fixed infinite-mode kernels.

Because: $:X^r:_{\mathbb{E}X^2}$ is the r th homogeneous Wiener chaos. The integral imposes spatial momentum conservation and introduces no lower-chaos terms because the Wick constant in (1.2) is exactly the variance in (0.5).

⟨2⟩ 2 For $K' \geq K$, the quartic tail satisfies

$$\|I_4(f_{K'}) - I_4(f_K)\|_2^2 = 4!(2L)^{-2} \sum_{\substack{k_1+\dots+k_4=0 \\ \max_i |k_i| \in (K, K']}} \prod_{i=1}^4 \frac{1}{2\gamma(k_i)}. \quad (1.8)$$

Because: The Wick isometry contracts only equal chaos orders. Expanding four Fourier sums and integrating $x, y \in \mathbb{T}_L$ supplies two Kronecker constraints; one is eliminated, leaving (1.8), including the factor $4!$.

⟨2⟩ 3 Let $g(k) = (m + |k|)^{-1}$. Then $(2\gamma(k))^{-1} \leq g(k)$ and

$$(g * g)(k) \leq CL \frac{1 + \log(2 + |k|/m)}{m + |k|}, \quad (g * g * g)(k) \leq CL^2 \frac{[1 + \log(2 + |k|/m)]^2}{m + |k|}. \quad (1.9)$$

Because: $m + |k| \leq 2\sqrt{m^2 + k^2}$. For the first convolution split the summation into $|u| \leq |k|/2$, $|k - u| \leq |k|/2$, and the complement. In the first region the second denominator is at least $(m + |k|)/2$ and $\sum_{|u| \leq R} (m + |u|)^{-1} \leq CL[1 + \log(1 + R/m)]$; the second region is symmetric. In the complement both denominators are at least a fixed multiple of $m + |k|$, and the remaining square tail is $O(L/(m + |k|))$. Repeat the same three-region split after convolving the first estimate with g ; the partial sum of $(1 + \log)/(m + |u|)$ contributes the second logarithm. These are finite sums on the grid $(\pi/L)\mathbb{Z}$, so no continuum-limit interchange occurs.

⟨2⟩ 4 The part of (1.8) with $|k_1| > K$ is at most

$$CL^3 \sum_{|k_1| > K} \frac{[1 + \log(2 + |k_1|/m)]^2}{(m + |k_1|)^2} \leq CL^3 \frac{[1 + \log(1 + K/m)]^2}{K/m}. \quad (1.10)$$

Because: Sum the remaining three momenta by (1.9). Integral comparison of the decreasing tail $\log^2 u/u^2$ gives the final bound. The union over the four choices of the large momentum changes only C .

⟨2⟩ 5 The quadratic tail has variance

$$\|I_2(g_{K'}) - I_2(g_K)\|_2^2 = \frac{1}{2} \sum_{K < |k| \leq K'} \gamma(k)^{-2} \leq \frac{CL}{K}. \quad (1.11)$$

Because: In the real sine/cosine basis it is the sum of independent multiples of $\xi_j^2 - 1$, whose variance is 2. The mode normalizations give the factor $1/2$; compare the remaining sum with $(2L/\pi) \int_K^\infty (m^2 + u^2)^{-1} du$.

⟨2⟩ 6 There is a chaos- ≤ 4 random variable $V_L \in L^2(\mu)$ such that

$$\|V^{(K)} - V^{(K')}\|_2 + \|V^{(K)} - V_L\|_2 \leq C_V \frac{1 + \log(1 + K/m)}{(K/m)^{1/2}}. \quad (1.12)$$

Because: Combine (1.10), (1.11), orthogonality of distinct chaoses, and completeness of $L^2(\mu)$. The constant chaos cancels. The limiting kernels are precisely the truncation-convention objects denoted $\lambda f(\cdot; \varphi^4_{:C_L} + a; \varphi^2_{:C_L} + b) dx$.

⟨2⟩ 7 $V^{(K)} \rightarrow V_L$ in every finite $L^p(\mu)$.

Because: If X is a sum of chaoses of orders at most four, (0.8) applied with $A = e^{-\tau} \mathbf{1}$, $q = 1 + e^{2\tau}$, and the orthogonal chaos decomposition gives

$$\|X\|_q \leq (q - 1)^2 \|X\|_2 \quad (q \geq 2). \quad (1.13)$$

Use (1.13) for the difference and interpolate for $p < 2$.

⟨2⟩ 8 This proves ⟨1⟩3. □

3 The three Gaussian path ensembles and uniform exponential bounds

⟨1⟩ 4 Fix the three ensembles and their covariance identities.

⟨2⟩ 1 The vacuum ensemble is (Q, μ) of (0.4).

⟨2⟩ 2 The periodic ensemble $\mu_{t,L}^{\text{per}}$ is the centered field on $\mathbb{T}_t \times \mathbb{T}_L$ with per-mode covariance

$$c_\omega(s - s') = \frac{\cosh(\omega(t/2 - |s - s'|_{\mathbb{T}_t}))}{2\omega \sinh(\omega t/2)}, \quad \omega = \gamma(k). \quad (2.1)$$

⟨2⟩ 3 The stationary Ornstein–Uhlenbeck ensemble is the Gaussian process $X_s = (\xi_j(s))_j$ with

$$\mathbb{E}[\xi_j(s)\xi_j(s')] = e^{-\gamma_j|s-s'|}. \quad (2.2)$$

For bounded cylinder F, G ,

$$\mathbb{E}[\overline{F(X_0)}G(X_s)] = \langle F, \Gamma(e^{-s\gamma})G \rangle_{L^2(\mu)}. \quad (2.3)$$

Because: Both sides of (2.3) are finite-dimensional Gaussian integrals with unit marginals and cross covariance $e^{-s\gamma_j}$ in coordinate j . Equality for cylinders follows from their characteristic functions; L^2 density extends it.

⟨2⟩ 4 This proves ⟨1⟩4. □

⟨1⟩ 5 Prove the abstract uniform Nelson lemma used at every cutoff.

Let A_K be chaos- ≤ 4 random variables indexed by a full tail $\mathcal{K} \subset \Gamma_\infty$. Define $\ell_K = 1 + \log(1 + K/m)$ and assume

$$\ell_{K^+} - \ell_K \leq \log 2, \quad (2.4)$$

$$A_K \geq -c_1 \ell_K^2, \quad (2.5)$$

$$\|A_K - A_{K'}\|_2 \leq C_A \ell_K (K/m)^{-\kappa} \quad (K' \geq K), \quad \kappa > 0. \quad (2.6)$$

⟨2⟩ 1 For b with $L_b = \sqrt{b/(2c_1)}$ sufficiently large and $b \leq c_1 \ell_K^2$, choose the largest $K_b \leq K$ for which $\ell_{K_b} \leq L_b$. Then

$$\frac{L_b}{2} < \ell_{K_b} \leq L_b. \quad (2.7)$$

Because: Maximality gives $\ell_{K_b^+} > L_b$ and (2.4) gives $\ell_{K_b} > L_b - \log 2 \geq L_b/2$. If $b > c_1 \ell_K^2$, (2.5) makes $\mathbb{P}(A_K \leq -b) = 0$.

⟨2⟩ 2 $A_{K_b} \geq -b/2$, hence

$$\{A_K \leq -b\} \subset \{A_K - A_{K_b} \leq -b/2\}. \quad (2.8)$$

⟨2⟩ 3 The L^2 scale of the difference obeys

$$\rho_b := \|A_K - A_{K_b}\|_2 \leq 2C_A L_b e^{-\kappa(L_b/2-1)}. \quad (2.9)$$

Because: $K_b/m = e^{\ell_{K_b}-1} - 1$. For large L_b , this is at least $\frac{1}{2}e^{\ell_{K_b}-1}$; insert (2.7) into (2.6).

⟨2⟩ 4 For every integer $q \geq 1$,

$$\mathbb{P}(A_K \leq -b) \leq \left[\frac{2(2q-1)^2 \rho_b}{b} \right]^{2q}. \quad (2.10)$$

Because: Apply Markov to $|A_K - A_{K_b}|^{2q}$ in (2.8) and apply (1.13) with exponent $2q$ to the chaos- ≤ 4 difference.

⟨2⟩ 5 Choose q so that $(2q)^2 \leq e^{\kappa L_b/4} < (2q+2)^2$. After increasing the fixed threshold b_1 ,

$$\mathbb{P}(A_K \leq -b) \leq \exp[-e^{\kappa L_b/8}] \quad (b \geq b_1), \quad (2.11)$$

uniformly in K .

Because: Substitute (2.9) into (2.10). The logarithm of the bracket is at most $-\kappa L_b/8$ after the polynomial factors in L_b are absorbed; $2q \geq \frac{1}{2}e^{\kappa L_b/8}$ gives (2.11), after another harmless increase of b_1 .

⟨2⟩ 6 For every $p < \infty$,

$$\sup_K \mathbb{E}[e^{-pA_K}] < \infty. \quad (2.12)$$

Because: Layer cake gives

$$\mathbb{E}[e^{-pA_K}] \leq e^{pb_1} + p \int_{b_1}^{\infty} e^{pb} \mathbb{P}(A_K \leq -b) db.$$

Since L_b is a positive constant times $\sqrt{b}$, the double-exponential decay in (2.11) dominates e^{pb} and makes the integral finite uniformly in K . If $A_K \rightarrow A$ in L^2 , take an a.s. convergent subsequence and apply Fatou to obtain the same bound for e^{-A} .

⟨2⟩ 7 The bounded successor-gap hypothesis (2.4) holds on every full lattice tail.

Because: For $K^+ = K + \pi/L$ and $K \geq \pi/L$,

$$\ell_{K^+} - \ell_K = \log \left(1 + \frac{\pi/L}{m+K} \right) < \log 2.$$

This condition is essential to the scale selection (2.7); (2.5)–(2.6) alone do not imply that selection for a tower-sparse index set.

⟨2⟩ 8 This proves ⟨1⟩5. □

⟨1⟩ 6 Instantiate the Nelson lemma in all three ensembles.

⟨2⟩ 1 For $0 < \sigma \leq \sigma_0$,

$$\sup_K \|e^{-\sigma V^{(K)}}\|_{L^p(\mu)} < \infty, \quad e^{-\sigma V_L} \in L^p(\mu). \quad (2.13)$$

Because: Conditions (2.5) and (2.6) are respectively (1.5) and (1.12), with $\kappa = 1/2$, after multiplying constants by σ_0 .

⟨2⟩ 2 For

$$\mathcal{W}_K(t) = \int_0^t V^{(K)}(X_s) ds, \quad \mathcal{W}(t) = \int_0^t V_L(X_s) ds, \quad (2.14)$$

one has, uniformly for $0 < t \leq T$,

$$\|\mathcal{W}_K - \mathcal{W}_{K'}\|_2 \leq t \|V^{(K)} - V^{(K')}\|_2, \quad \mathcal{W}_K \geq -TC\ell_K^2, \quad \sup_K \|e^{-\mathcal{W}_K}\|_p < \infty. \quad (2.15)$$

Because: Stationarity and Cauchy–Schwarz give

$$\|\mathcal{W}_K - \mathcal{W}_{K'}\|_2^2 = \int_0^t \int_0^t \mathbb{E}[\Delta V(X_s) \Delta V(X_u)] ds du \leq t^2 \|\Delta V\|_2^2.$$

The pathwise lower bound is the integral of (1.5). Time integration of a global Gaussian chaos remains in chaos orders at most four, so Step $\langle 1 \rangle 5$ applies.

$\langle 2 \rangle 3$ Let $\mathcal{U}_t^{(K)}$ be the spatial-mode truncation of the periodic interaction. Then

$$\mathcal{U}_t^{(K)} \rightarrow \mathcal{U}_t \quad \text{in } L^2(\mu_{t,L}^{\text{per}}), \quad \sup_K \|e^{-\mathcal{U}_t^{(K)}}\|_p < \infty. \quad (2.16)$$

$\langle 3 \rangle 1$ The pathwise lower bound is $\mathcal{U}_t^{(K)} \geq -Ct\ell_K^2$.

Because: Apply (1.4) at every spacetime point using the vacuum cutoff constant c_K , which obeys (1.1), and integrate over $\mathbb{T}_t \times \mathbb{T}_L$.

$\langle 3 \rangle 2$ The quadratic chaos tail satisfies

$$\|Y_2^{(K')} - Y_2^{(K)}\|_2^2 \leq C \sum_{|k| > K} \sum_{k_0 \in (2\pi/t)\mathbb{Z}} (m^2 + k_0^2 + k^2)^{-2} \leq C(t, L, m)K^{-2}. \quad (2.17)$$

Because: The first inequality is the Wick isometry and the continuum covariance envelope. Summing k_0 by comparison with its integral gives $\sum_{k_0} (A^2 + k_0^2)^{-2} \leq C(t_0)(A^{-4} + tA^{-3})$; the remaining one-dimensional spatial tail is $O(tLK^{-2})$ (and the A^{-4} term is smaller under $tL \geq 1$).

$\langle 3 \rangle 3$ The quartic chaos tail satisfies the explicit bound

$$\|Y_4^{(K')} - Y_4^{(K)}\|_2^2 \leq C(t, L, m) \sum_{\substack{|k| > K \\ k_0 \in (2\pi/t)\mathbb{Z}}} \frac{[1 + \log(2 + |(k_0, k)|)]^2}{(m^2 + k_0^2 + k^2)^2} \leq C(t, L, m)K^{-2}[1 + \log K]^2. \quad (2.18)$$

Because: Select one of the four momenta whose spatial component exceeds K ; there are four choices. Sum the other three constrained covariance factors using the two-dimensional convolution estimate already proved in the repaired V2 module: $c_{\text{env}}^{*3}(-\kappa) \leq C(tL)^2[1 + \log(2 + |\kappa|)]^2/(m^2 + |\kappa|^2)$. Multiplication by the selected factor gives the first sum. Cover the anisotropic grid by its fundamental rectangles. Its radial majorant is decreasing outside a fixed disk, so the sum is bounded by its density times

$$\int_{r > K} \frac{r[1 + \log(2 + r)]^2}{(m^2 + r^2)^2} dr \leq C \frac{[1 + \log K]^2}{m^2 + K^2}.$$

This is the missing “same tail counts” calculation in the source V3 document.

$\langle 3 \rangle 4$ The thermal recombination coefficient tail is exponentially small.

Because:

$$d^{(K)}(t) = \frac{1}{2L} \sum_{|k| \leq K} \frac{1}{\gamma(k)(e^{t\gamma(k)} - 1)},$$

and for $t \geq t_0$ the omitted summand is bounded by $Ce^{-t_0|k|}/(m + |k|)$. Hence $|d^{(K)} - d^{(K')}| \leq Ce^{-t_0K/2}$ after enlarging C .

$\langle 3 \rangle 5$ Equations (2.17)–(2.18), the finite Wick recombination identity, and boundedness of $d^{(K)}$ imply

$$\|\mathcal{U}_t^{(K)} - \mathcal{U}_t^{(K')}\|_2 \leq C[1 + \log(1 + K/m)]^2(K/m)^{-1/2}. \quad (2.19)$$

Because: The actual chaos tails from (2.17)–(2.18) are at least as small as $K^{-1} \log K$ after taking square roots. The displayed $K^{-1/2} \log^2 K$ rate is a weaker common bound that also covers the coefficient cross terms. Each cross term is a bounded coefficient difference times a bounded lower-chaos norm.

⟨3⟩ 6 Absorb one logarithm by $\ell_K \leq C_\varsigma(K/m)^\varsigma$, choose $0 < \varsigma < 1/2$, and apply Step ⟨1⟩5 with $\kappa = 1/2 - \varsigma$. This proves (2.16).

⟨2⟩ 4 The deterministically shifted periodic interactions obey the same Cauchy and exponential-integrability conclusions, uniformly for the bounded coarse-data class used in V4.

Because: Substitute $u = \eta^{(K)}(s, x) + H^{(K)}(s, x)$ in (1.4). Since (1.4) holds for every real u , integration gives the same lower bound $\mathcal{U}_t^{(K)}(\eta + H^{(K)}) \geq -Ct\ell_K^2$. Expand each shifted Wick monomial by the Appell identity (3.12) and its degree-two analogue. The repaired V2 theorem, whose fundamental-band and alias sums are displayed in its Steps ⟨1⟩7–⟨1⟩8, bounds the resulting random-chaos tails and deterministic coefficient differences by the right side of (2.19), uniformly over the stated coarse-data bounds. These are precisely the lower-tail and L^2 -increment hypotheses of Step ⟨1⟩5; applying that step proves the asserted uniform exponential integrability and L^2 convergence.

⟨2⟩ 5 This proves ⟨1⟩6. □

⟨1⟩ 7 Derive uniform spectral lower bounds for the finite-cutoff operators.

Let $K^{(K)}$ be defined in the next section. Its finite-mode periodic trace identity will be proved independently of this step.

⟨2⟩ 1 For a fixed admissible t_0 ,

$$e^{-t_0 \inf \text{spec } K^{(K)}} \leq \text{Tr } e^{-t_0 K^{(K)}} = Z_0(t_0) \mathbb{E}[e^{-\mathcal{U}_{t_0}^{(K)}}] \leq Z_0(t_0) B. \quad (2.20)$$

Because: The first inequality selects the lowest heat eigenvalue; the identity is proved in Step ⟨1⟩8 below; the last inequality is (2.16) at $p = 1$. No uniform lower bound is used in proving that finite-mode identity, so this is not circular.

⟨2⟩ 2 Taking logarithms gives a K -independent constant c_λ with

$$K^{(K)} \geq -c_\lambda. \quad (2.21)$$

⟨2⟩ 3 For $0 < \delta < 1$, apply (2.21) to the amplified coupling $\lambda/(1 - \delta)$ to obtain

$$K^{(K)} = \delta H_0 + (1 - \delta) [H_0 + (1 - \delta)^{-1} V^{(K)}] \geq \delta H_0 - c_\delta \quad (2.22)$$

as quadratic forms.

⟨2⟩ 4 This proves ⟨1⟩7. □

4 Finite-mode systems and their exact Feynman–Kac formulas

For fixed K , decompose $L^2(Q, \mu) = L^2(\mu_{\leq K}) \otimes L^2(\mu_{> K})$ and set

$$K^{(K)} = (H_0^{\leq K} + V^{(K)}) \otimes \mathbf{1} + \mathbf{1} \otimes H_0^{> K}, \quad T_K(s) = e^{-sK^{(K)}}. \quad (3.1)$$

⟨1⟩ 8 Construct each finite-mode system and prove every representation used later.

⟨2⟩ 1 The form of $H_0^{\leq K} + V^{(K)}$ is densely defined, lower bounded, and closed.

Because: Choose $C > b_K$ and write

$$q_K(f) = \sum_{|k_j| \leq K} \gamma_j \|\partial_j f\|_2^2 + \int (V^{(K)} + C) |f|^2 d\mu_{\leq K}. \quad (3.2)$$

Both summands are nonnegative and their domains contain $C_c^\infty(\mathbb{R}^{n_K})$. If f_n is Cauchy in the norm $\|f\|_2^2 + q_K(f)$, it converges in L^2 and in each weighted weak derivative. A subsequence converges pointwise a.e.; Fatou for the multiplication term and closedness of weak derivatives identify the limits. Thus the form norm is complete.

⟨2⟩ 2 The finite-mode operator has compact resolvent.

Because: On every Euclidean ball the Gaussian density is bounded above and below by positive constants, so the weighted form norm controls the ordinary H^1 norm there; Rellich makes the restricted embedding compact. Outside a sufficiently large ball, coercivity from (1.6) gives $(V^{(K)} + C) \geq R$, hence $\int_{|\xi| > R_0} |f|^2 d\mu \leq R^{-1} q_K(f)$. First take R large and then use Rellich on the ball. The unit ball of the form domain is therefore relatively compact in L^2 .

⟨2⟩ 3 For one retained coordinate, the free Mehler kernel with respect to $\mathcal{N}(0, 1)$ is

$$M_s^{(j)}(x, y) = (1 - r_j^2)^{-1/2} \exp \left[-\frac{r_j^2(x^2 + y^2) - 2r_j xy}{2(1 - r_j^2)} \right], \quad r_j = e^{-s\gamma_j}. \quad (3.3)$$

Because: Completing the square shows $\int M_s^{(j)}(x, y) d\mu(y) = 1$; symmetry is evident. Applying the Gaussian generating function to Hermite polynomials gives eigenvalue r_j^n in degree n . Hence the product over retained coordinates is exactly $\Gamma(e^{-s\gamma})$ on that finite Gaussian space.

⟨2⟩ 4 For bounded f, g on the retained coordinates,

$$\langle f, e^{-s(H_0^{\leq K} + V^{(K)})} g \rangle = \mathbb{E} \left[\overline{f(X_0^{\leq K})} g(X_s^{\leq K}) e^{-\int_0^s V^{(K)}(X_u) du} \right]. \quad (3.4)$$

⟨3⟩ 1 First replace $V^{(K)}$ by $V_R = V^{(K)} \wedge R$, with $R > 0$.

Because: By (1.5), $-b_K \leq V_R \leq R$, so multiplication by V_R is bounded and self-adjoint.

⟨3⟩ 2 The bounded-perturbation Trotter formula gives

$$e^{-s(H_0 + V_R)} = \lim_{n \rightarrow \infty} (e^{-(s/n)H_0} e^{-(s/n)V_R})^n. \quad (3.5)$$

⟨3⟩ 3 Insertion of (3.3) between the multiplication factors turns the matrix element of the product into the OU chain integral with weight

$$\exp \left[-\frac{s}{n} \sum_{j=1}^n V_R(X_{js/n}) \right].$$

⟨3⟩ 4 As $n \rightarrow \infty$ this weight converges a.s. to $e^{-\int_0^s V_R(X_u) du}$ and is bounded by e^{sb_K} .

Because: The finite-dimensional OU process has continuous paths and V_R is bounded continuous. The displayed sum is therefore an ordinary Riemann sum; dominated convergence applies.

⟨3⟩ 5 As $R \rightarrow \infty$, the operators and path integrals converge to (3.4).

Because: $V_R \uparrow V^{(K)}$, so the associated closed forms increase and monotone form convergence gives strong semigroup convergence. On each continuous path, $e^{-\int V_R} \downarrow e^{-\int V^{(K)}}$; it is bounded by e^{sb_K} , so dominated convergence applies. This proves (3.4).

⟨2⟩ 5 The kernel in (3.4) is jointly continuous and strictly positive.

Because: Condition the OU law on its two endpoints. The product Mehler density is continuous and strictly positive. The bridge expectation of the weight lies in $(0, e^{sb_K}]$. Continuity follows by dominated convergence after realizing the finite-dimensional Gaussian bridge as an affine continuous function of its endpoints plus an endpoint-independent bridge.

⟨2⟩ 6 Tensoring with the independent free tail gives, for bounded cylinders f, g ,

$$\langle f, T_K(s)g \rangle = \mathbb{E} \left[\overline{f(X_0)} g(X_s) e^{-\mathcal{W}_K(s)} \right]. \quad (3.6)$$

Because: The retained and omitted OU coordinates are independent; apply (3.4) to the first factor and (2.3) to the second.

⟨2⟩ 7 $T_K(s)$ is trace class and

$$\mathrm{Tr} T_K(s) \leq e^{sb_K} Z_0(s). \quad (3.7)$$

Because: The min–max principle and $V^{(K)} \geq -b_K$ imply that every ordered eigenvalue of the retained interacting operator is at least the corresponding free eigenvalue minus b_K . Tensoring with the free tail and summing all occupation numbers gives (3.7). In particular, the full tensor operator has compact resolvent.

⟨2⟩ 8 The periodic trace formula is

$$\mathrm{Tr} T_K(s) = Z_0(s) \mathbb{E}_{\mu_{s,L}^{\mathrm{per}}} [e^{-\mathcal{U}_s^{(K)}}]. \quad (3.8)$$

Because: Apply the diagonal trace to the strictly positive kernel from (3.4), normalize the product of cyclic Mehler kernels by its free trace, and integrate the intermediate points. The normalized cyclic Gaussian chain has covariance (2.1): for one mode, direct inversion of the cyclic tridiagonal quadratic form gives $\cosh(\omega(s/2 - |u - v|)) / (2\omega \sinh(\omega s/2))$. The equal-time variance differs from c_K by $d^{(K)}(s)$; the exact Wick-recombination identity

$$:X^4_{\cdot c_K} = :X^4_{\cdot v_K} + 6(v_K - c_K):X^2_{\cdot v_K} + 3(v_K - c_K)^2 \quad (3.9)$$

identifies the path action with $\mathcal{U}_s^{(K)}$. Passing from equal cyclic partitions to the continuous bridge uses the bounded-truncation and dominated limits already justified in Steps ⟨3⟩1–⟨3⟩5.

⟨2⟩ 9 For a Weyl symbol whose position and momentum components are supported in $|k| \leq K$,

$$\frac{\mathrm{Tr}(T_K(s)W_{\mathrm{ct}}(\xi))}{\mathrm{Tr} T_K(s)} = e^{-S^{(K)}(H_p)} \frac{\mathbb{E}[e^{i\Phi(J_q)} e^{-\mathcal{U}_s^{(K)}(\eta + H_p^{(K)})}]}{\mathbb{E}[e^{-\mathcal{U}_s^{(K)}}]}. \quad (3.10)$$

Here

$$S^{(K)}(H_p) = \frac{1}{8L} \sum_{|k| \leq K} \gamma(k) \coth(s\gamma(k)/2) |b_\infty(k)|^2. \quad (3.11)$$

Because: Apply the V1 proof mode by mode: the Weyl operator translates the endpoint by its momentum component and multiplies by its position phase; the unique solution of $(-\partial_u^2 + \gamma(k)^2)H = 0$ off the marked slice with the prescribed jump removes the endpoint translation; completing the Gaussian square produces (3.11) and the sharp-time phase. The Wick Appell identity

$$:(X + H)^4_{\cdot c} = :X^4_{\cdot c} + 4H:X^3_{\cdot c} + 6H^2:X^2_{\cdot c} + 4H^3X + H^4 \quad (3.12)$$

gives the shifted action without an extra $-6c(2HX + H^2)$ term. To remove the interaction cutoff, replace $V^{(K)}$ by $V_R = V^{(K)} \wedge R$ exactly as in Steps $\langle 3 \rangle 1$ – $\langle 3 \rangle 5$: the Weyl insertion is unitary, the truncated path weight is bounded by e^{sb_K} , and dominated convergence first passes the Trotter mesh and then $R \rightarrow \infty$. All retained spaces are finite-dimensional. Tensoring with the omitted free modes produces the same free factor in numerator and denominator, so that factor cancels.

$\langle 2 \rangle 10$ The kernel is positivity improving, so the finite-mode lowest eigenvalue is simple.

Because: If $f, g \geq 0$ are nonzero, the endpoint Mehler density is strictly positive and the Feynman–Kac weight is strictly positive on every finite path; therefore $\langle f, e^{-s(H_0^{\leq K} + V^{(K)})} g \rangle > 0$. Let T denote this compact self-adjoint integral operator and let $Th = \|T\|h$. Its strictly positive kernel gives $|Th(x)| \leq T|h|(x)$; the variational bound $\langle |h|, T|h| \rangle \leq \|T\| \|h\|^2$ forces equality. Equality in the integral triangle inequality for a strictly positive kernel forces $h(y)$ to have one common phase for almost every y . Thus every top eigenvector is a scalar multiple of one strictly positive vector, so the top eigenvalue is simple. Tensoring with the free tail preserves a unique total ground vector because each strictly positive tail frequency has the unique oscillator vacuum.

$\langle 2 \rangle 11$ This proves $\langle 1 \rangle 8$ and supplies the identity used in (2.20). $\square$

5 Strong semigroup limit and the continuum generator

$\langle 1 \rangle 9$ Prove that $T_K(s)$ converges strongly to one self-adjoint semigroup.

$\langle 2 \rangle 1$ For a bounded cylinder ψ supported in modes at most K_0 and $K, K' \geq K_0$,

$$\|(T_K(s) - T_{K'}(s))\psi\|_2 \leq \|\psi\|_\infty \|e^{-\mathcal{W}_K(s)} - e^{-\mathcal{W}_{K'}(s)}\|_2. \quad (4.1)$$

Because: Subtract the two conditional-expectation versions of (3.6), condition on X_0 , square, integrate, and apply conditional Jensen.

$\langle 2 \rangle 2$ For real x, y ,

$$|e^{-x} - e^{-y}| \leq |x - y|(e^{-x} + e^{-y}). \quad (4.2)$$

Because: Integrate the derivative of $e^{-[(1-u)x+uy]}$ over $u \in [0, 1]$ and bound every intermediate exponential by $e^{-x} + e^{-y}$.

$\langle 2 \rangle 3$ Equations (1.13), (2.13)–(2.15), and Hölder give

$$\|e^{-\mathcal{W}_K} - e^{-\mathcal{W}_{K'}}\|_2 \leq C_T s \|V^{(K)} - V^{(K')}\|_2. \quad (4.3)$$

Because: Apply (4.2), then $\|XY\|_2 \leq \|X\|_4 \|Y\|_4$ with $X = \mathcal{W}_K - \mathcal{W}_{K'}$ and $Y = e^{-\mathcal{W}_K} + e^{-\mathcal{W}_{K'}}$. The first norm is at most $9s\|V^{(K)} - V^{(K')}\|_2$ by (1.13), (2.15); the second is uniformly bounded by (2.13)–(2.15).

$\langle 2 \rangle 4$ $T_K(s)$ is strongly Cauchy on $L^2(\mu)$, uniformly for s in a compact interval.

Because: Equations (4.1), (4.3), and (1.12) prove this on the dense bounded cylinder core. The uniform operator bound $\|T_K(s)\| \leq e^{sc_\lambda}$ from (2.21) extends it to all L^2 . Call the strong limit $T(s)$.

$\langle 2 \rangle 5$ $T(s)$ is a self-adjoint semigroup with $\|T(s)\| \leq e^{sc_\lambda}$.

Because: Strong limits preserve the adjoint identity on matrix elements. For the product, insert and subtract $T_K(s)T_K(u)$ and use uniform boundedness to pass both factors strongly: $T(s)T(u) = T(s+u)$. The norm bound passes by lower semicontinuity.

⟨2⟩ 6 $T(s)$ is strongly continuous at $s = 0$.

Because: For a bounded cylinder ψ and fixed K containing its modes,

$$\|T(s)\psi - \psi\| \leq \|T(s)\psi - T_K(s)\psi\| + \|T_K(s)\psi - \psi\|.$$

By (4.1)–(4.3), the first term is at most $Cs\|\psi\|_\infty \sup_{K' \geq K} \|V^{(K')} - V^{(K)}\|_2$; the second tends to zero for fixed K . First let $s \downarrow 0$, then $K \rightarrow \infty$. Density and the uniform norm bound finish.

⟨2⟩ 7 There is a unique self-adjoint $K_L \geq -c_\lambda$ such that

$$T(s) = e^{-sK_L}. \quad (4.4)$$

Because: Apply the spectral-generator theorem for a strongly continuous self-adjoint semigroup bounded by e^{sc_λ} . The semigroup uniquely determines its generator.

⟨2⟩ 8 For bounded measurable f, g ,

$$\langle f, T(s)g \rangle = \mathbb{E}[\overline{f(X_0)}g(X_s)e^{-\mathcal{W}(s)}]. \quad (4.5)$$

Because: For bounded cylinders, pass $K \rightarrow \infty$ in (3.6) by (4.3) and strong convergence. Approximate bounded measurable functions by their conditional expectations on the increasing mode sigma-fields. Martingale convergence gives a.e. and L^2 convergence; dominated convergence on the right uses $e^{-\mathcal{W}} \in L^1$, while strong continuity of $T(s)$ handles the left.

⟨2⟩ 9 The uniform form bound passes to the limit:

$$K_L \geq \tfrac{1}{2}H_0 - c_{1/2}. \quad (4.6)$$

⟨3⟩ 1 Choose $M > c_{1/2} \vee c_\lambda$. From (2.22),

$$(K^{(K)} + M)^{-1} \leq (\tfrac{1}{2}H_0 + M - c_{1/2})^{-1}. \quad (4.7)$$

Because: For strictly positive self-adjoint S , spectral calculus gives

$$q_S(\psi) = \sup_{\phi} [2 \operatorname{Re} \langle \psi, \phi \rangle - ip\phi S^{-1}\phi], \quad \langle \phi, S^{-1}\phi \rangle = \sup_{\psi} [2 \operatorname{Re} \langle \psi, \phi \rangle - q_S(\psi)]. \quad (4.8)$$

Thus form order and the reverse resolvent order are equivalent.

⟨3⟩ 2 The resolvents converge strongly:

$$(K^{(K)} + M)^{-1} o(K_L + M)^{-1}. \quad (4.9)$$

Because: Use the Laplace representation $S^{-1} = \int_0^\infty e^{-Ms} e^{-s(S-M)} ds$ with the semigroups T_K, T ; $M > c_\lambda$ supplies an integrable uniform majorant.

⟨3⟩ 3 Pass to the limit in (4.7) on every vector and use (4.8) in the reverse direction. This gives (4.6) and $Q(K_L) \subset Q(H_0)$.

⟨2⟩ 10 This proves ⟨1⟩ 9. □

⟨1⟩ 10 Identify the generator and its form on the cylinder core.

⟨2⟩ 1 For $\psi \in \mathcal{D}$,

$$\frac{(1 - T(s))\psi}{s} = \frac{(1 - e^{-sH_0})\psi}{s} - \frac{1}{s} \mathbb{E}[\psi(X_s)(e^{-\mathcal{W}(s)} - 1) \mid X_0]. \quad (4.10)$$

Because: This is (4.5) written as a conditional expectation, with the free conditional expectation subtracted and added.

⟨2⟩ 2 The first term in (4.10) converges in L^2 to $H_0\psi$.

Because: Let $d\mu_\psi(\lambda) = d\langle\psi, E_{H_0}(\lambda)\psi\rangle$. The spectral theorem gives

$$\left\| \frac{1 - e^{-sH_0}}{s} \psi - H_0\psi \right\|_2^2 = \int_0^\infty \left| \frac{1 - e^{-s\lambda}}{s} - \lambda \right|^2 d\mu_\psi(\lambda).$$

The integrand converges pointwise to zero and is bounded by λ^2 , because $0 \leq (1 - e^{-s\lambda})/s \leq \lambda$. Since $\psi \in D(H_0)$, $\int \lambda^2 d\mu_\psi < \infty$; dominated convergence proves the claim.

⟨2⟩ 3 Write $e^{-x} - 1 = -x + \rho(x)$, where

$$|\rho(x)| \leq \frac{1}{2}x^2 e^{x-}, \quad x_- = \max(-x, 0). \quad (4.11)$$

Because: Taylor's formula with integral remainder gives $\rho(x) = x^2 \int_0^1 (1-u)e^{-ux} du$; bound $e^{-ux} \leq e^{x-}$.

⟨2⟩ 4 The linear part satisfies

$$\frac{1}{s} \mathbb{E}[\psi(X_s) \mathcal{W}(s) \mid X_0] = \frac{1}{s} \int_0^s e^{-uH_0} (V_L e^{-(s-u)H_0} \psi) du \longrightarrow V_L \psi \quad (4.12)$$

in L^2 .

Because: The Markov property gives the identity. Since $V_L \in L^4$ and ψ is bounded,

$$\|V_L(e^{-vH_0}\psi - \psi)\|_2 \leq \|V_L\|_4 \|e^{-vH_0}\psi - \psi\|_4.$$

For $\psi = F(\xi_1, \dots, \xi_n)$, the Mehler formula and bounded derivatives of F give

$$\sup_{0 \leq v \leq s} \|e^{-vH_0}\psi - \psi\|_4 \leq C_F \sum_{j \leq n} (1 - e^{-s\gamma_j})^{1/2} \rightarrow 0.$$

Also e^{-uH_0} is strongly continuous on $V_L\psi \in L^2$. Both errors are uniform for $0 \leq u \leq s$, proving (4.12).

⟨2⟩ 5 The remainder in (4.11), divided by s , tends to zero in L^2 .

Because: Conditional Jensen, Hölder, (1.13), and (2.15) give

$$\frac{1}{s} \|\mathbb{E}[\psi(X_s) \rho(\mathcal{W}) \mid X_0]\|_2 \leq \frac{\|\psi\|_\infty}{2s} \|\mathcal{W}\|_8^2 \|e^{\mathcal{W}-}\|_4 \leq Cs. \quad (4.13)$$

Indeed $\|\mathcal{W}\|_8 \leq 49s\|V_L\|_2$ and $e^{\mathcal{W}-} \leq 1 + e^{-\mathcal{W}}$, whose L^4 norm is uniformly bounded by Step ⟨1⟩6.

⟨2⟩ 6 Therefore

$$\mathcal{D} \subset D(K_L), \quad K_L\psi = H_0\psi + V_L\psi \quad (\psi \in \mathcal{D}). \quad (4.14)$$

Because: Combine (4.10)–(4.13) with the generator characterization $K_L\psi = \lim_{s \downarrow 0} s^{-1}(1 - e^{-sK_L})\psi$.

⟨2⟩ 7 For $\psi \in \mathcal{D}$,

$$q_{K_L}(\psi) = \langle\psi, H_0\psi\rangle + \int V_L|\psi|^2 d\mu. \quad (4.15)$$

Because: Spectral calculus gives $s^{-1}\langle\psi, (1 - e^{-sK_L})\psi\rangle \rightarrow q_{K_L}(\psi)$ after shifting by the lower spectral bound. Equation (4.14) identifies the same limit with the right side. This proves equality on $\mathcal{D}$ only; no unproved claim that $\mathcal{D}$ is a form core is made.

⟨2⟩ 8 This proves ⟨1⟩10. □

6 Uniform smoothing, trace class, and trace-norm approximation

⟨1⟩ 11 Prove the cutoff-uniform $L^2 \rightarrow L^4$ smoothing estimate.

Put $s_4 = (\log 3)/m$.

⟨2⟩ 1 By self-adjoint duality it suffices to prove

$$\sup_K \|T_K(s_4)\|_{4/3 \rightarrow 2} < \infty. \quad (5.1)$$

Because: $\|T_K(s_4)\|_{2 \rightarrow 4} = \|T_K(s_4)\|_{4/3 \rightarrow 2}$. For $s > s_4$, compose with $\|T_K(s - s_4)\|_{2 \rightarrow 2} \leq e^{c\lambda(s-s_4)}$.

⟨2⟩ 2 Replace $W = V^{(K)}$ by $W_R = W \wedge R$ and use Trotter with $\tau = s_4/n$. It is enough to bound

$$(e^{-\tau W_R} e^{-\tau H_0})^n : L^{4/3} \rightarrow L^2 \quad (5.2)$$

uniformly in n, K, R .

Because: The bounded Trotter products converge strongly. The form semigroups for $W_R \uparrow W$ converge strongly. An a.e. convergent subsequence and Fatou pass any common L^2 bound through both limits.

⟨2⟩ 3 For $j = 0, \dots, n$, define

$$p_j = 1 + \frac{1}{3}e^{j\tau m}, \quad P_j = 1 + (p_j - 1)e^{2\tau m}. \quad (5.3)$$

Then $p_0 = 4/3$, $p_n = 2$, and

$$\|e^{-\tau H_0}\|_{p_j \rightarrow P_j} \leq 1. \quad (5.4)$$

Because: $\|e^{-\tau \gamma}\| = e^{-\tau m}$ and $e^{-\tau m} = \sqrt{(p_j - 1)/(P_j - 1)}$; apply (0.8).

⟨2⟩ 4 Define r_j by

$$\frac{1}{p_{j+1}} = \frac{1}{r_j} + \frac{1}{P_j}. \quad (5.5)$$

Then multiplication gives

$$\|e^{-\tau W_R} u\|_{p_{j+1}} \leq \|e^{-\tau W_R}\|_{r_j} \|u\|_{P_j}. \quad (5.6)$$

⟨2⟩ 5 For $n \geq 2$,

$$\sum_{j=0}^{n-1} \frac{1}{r_j} \leq 1, \quad r_j \tau \leq \frac{12}{m}. \quad (5.7)$$

Because: From (5.3), $P_j - p_{j+1} = \frac{1}{3}e^{(j+1)\tau m}(e^{\tau m} - 1)$. Since $p_{j+1}, P_j \geq 4/3$,

$$\sum_j r_j^{-1} \leq \frac{3}{16}e^{\tau m}(e^{n\tau m} - 1) = \frac{3}{8}e^{\tau m} \leq \frac{3\sqrt{3}}{8} < 1.$$

Also $p_{j+1} \leq 2$, $P_j \leq 1 + \frac{1}{3}e^{(j+2)\tau m}$, and $\tau m \leq e^{\tau m} - 1$, whence

$$r_j \tau m = \frac{3\tau m, p_{j+1} P_j}{e^{(j+1)\tau m}(e^{\tau m} - 1)} \leq 6 + 2e^{\tau m} < 12.$$

⟨2⟩ 6 There is a constant D independent of j, n, K, R such that

$$\|e^{-\tau W_R}\|_{r_j} \leq D^{1/r_j}. \quad (5.8)$$

Because: Since $R > 0$, $e^{-r_j \tau W_R} \leq 1 + e^{-r_j \tau W}$. Equation (5.7) restricts $r_j \tau$ to $(0, 12/m]$, so the vacuum Nelson bound (2.13), with $p = 1$ and all σ in that compact interval, makes $\mathbb{E}e^{-r_j \tau W}$ uniformly bounded.

⟨2⟩ 7 Multiplication of (5.4), (5.6), and (5.8) yields

$$\|(e^{-\tau W_R} e^{-\tau H_0})^n\|_{4/3 \rightarrow 2} \leq D \sum_j 1/r_j \leq \max(1, D) =: M_0. \quad (5.9)$$

⟨2⟩ 8 Consequently, uniformly in K ,

$$\|T_K(s)\|_{2 \rightarrow 4} \leq M_0 e^{c_\lambda(s-s_4)} \quad (s \geq s_4), \quad (5.10)$$

and the same holds for $T(s)$ by strong convergence and Fatou.

⟨2⟩ 9 This proves ⟨1⟩ 11. □

⟨1⟩ 12 Upgrade strong convergence to operator-norm convergence at one time.

Set $s_2 = 2s_4$ and $\Delta V = V^{(K')} - V^{(K)}$.

⟨2⟩ 1 For bounded cylinder ϕ, ψ and every $0 < s < s_2$, both $T_K(s)\phi$ and $T_{K'}(s_2 - s)\psi$ lie in the form domains of both $K^{(K)}$ and $K^{(K')}$.

Because: The uniform form lower bound (2.22) puts both vectors in $Q(H_0)$. From (3.6), conditional Jensen, and the OU Nelson bound,

$$\|T_K(u)\phi\|_4 \leq C\|\phi\|_\infty.$$

For every $v \in L^4$ and either cutoff interaction, $\int |V^{(\cdot)}| |v|^2 d\mu \leq \|V^{(\cdot)}\|_2 \|v\|_4^2 < \infty$. Thus both vectors also lie in both multiplication-form domains.

⟨2⟩ 2 The form Duhamel identity is

$$\langle \phi, (T_K(s_2) - T_{K'}(s_2))\psi \rangle = \int_0^{s_2} \langle T_K(u)\phi, \Delta V, T_{K'}(s_2 - u)\psi \rangle du. \quad (5.11)$$

Because: Differentiate $u \mapsto \langle T_K(u)\phi, T_{K'}(s_2 - u)\psi \rangle$ in the common form domain. The two H_0 form terms cancel, leaving multiplication by ΔV . Integrability follows from the estimates in the next step, so the fundamental theorem for absolutely continuous form matrix elements applies.

⟨2⟩ 3 For $0 \leq u \leq s_4$, use exponents $(2, 4, 4)$ on $(T_K(u)\phi, \Delta V, T_{K'}(s_2 - u)\psi)$; for $s_4 \leq u \leq s_2$, reverse the two semigroup roles. Then

$$\|T_K(s_2) - T_{K'}(s_2)\|_{2 \rightarrow 2} \leq 18s_2 M_0 e^{2c_\lambda s_2} \|V^{(K)} - V^{(K')}\|_2. \quad (5.12)$$

Because: The long semigroup factor has elapsed time at least s_4 and is $L^2 \rightarrow L^4$ bounded by (5.10); the short factor is $L^2 \rightarrow L^2$ bounded. The interaction difference satisfies $\|\Delta V\|_4 \leq 9\|\Delta V\|_2$ by (1.13). Integrate the common bound over a length s_2 interval; the harmless factor 2 accommodates both halves.

⟨2⟩ 4 Equation (1.12) implies

$$\|T_K(s_2) - T(s_2)\|_{2 \rightarrow 2} \rightarrow 0. \quad (5.13)$$

⟨2⟩ 5 This proves ⟨1⟩ 12. □

⟨1⟩ 13 Prove discreteness, eigenvalue convergence, trace class, and trace-norm convergence for every positive time.

Let $\nu_0 \leq \nu_1 \leq \dots$ be the occupation-number eigenvalues of H_0 .

$\langle 2 \rangle 1$ $T(s_2)$ is compact, and if $\mu_j^{(K)}$ and μ_j are ordered eigenvalues of $K^{(K)}$ and K_L , then

$$\mu_j^{(K)} \rightarrow \mu_j, \quad \mu_j \geq \frac{1}{2}\nu_j - c_{1/2}. \quad (5.14)$$

Because: Each $T_K(s_2)$ is trace class by (3.7), hence compact. The operator-norm limit (5.13) is compact. The min–max/singular-value inequality gives convergence of every ordered heat eigenvalue $e^{-s_2\mu_j^{(K)}} \rightarrow e^{-s_2\mu_j}$; the logarithm is continuous on $(0, \infty)$. Apply min–max to (2.22) at $\delta = 1/2$ and pass to the limit.

$\langle 2 \rangle 2$ For every $s > 0$,

$$\mathrm{Tr} T_K(s) \rightarrow \mathrm{Tr} T(s) < \infty. \quad (5.15)$$

Because: By (5.14), each summand converges, while

$$e^{-s\mu_j^{(K)}} \leq e^{sc_{1/2}} e^{-s\nu_j/2}, \quad \sum_j e^{-s\nu_j/2} = Z_0(s/2) < \infty.$$

Dominated convergence on the counting measure gives (5.15), and shows that the spectrum of K_L is discrete and exhausts the μ_j .

$\langle 2 \rangle 3$ For $u = s/2$,

$$\|T_K(u) - T(u)\|_{\mathcal{J}_2}^2 = \mathrm{Tr} T_K(2u) - 2 \mathrm{Tr}(T_K(u)T(u)) + \mathrm{Tr} T(2u). \quad (5.16)$$

Because: Expand the square of the self-adjoint difference and use cyclicity for products of a trace-class and a bounded operator.

$\langle 2 \rangle 4$ The mixed trace in (5.16) converges to $\mathrm{Tr} T(2u)$.

Because: If (ϕ_n) diagonalizes $T(u)$ with eigenvalues σ_n , then

$$\mathrm{Tr}(T_K(u)T(u)) = \sum_n \sigma_n \langle \phi_n, T_K(u)\phi_n \rangle.$$

Each matrix element converges strongly, is bounded by e^{uc_λ} , and $\sum_n \sigma_n = \mathrm{Tr} T(u) < \infty$; dominated convergence applies.

$\langle 2 \rangle 5$ Thus $\|T_K(u) - T(u)\|_{\mathcal{J}_2} \rightarrow 0$, and

$$\|T_K(s) - T(s)\|_{\mathcal{J}_1} \leq (\|T_K(u)\|_{\mathcal{J}_2} + \|T(u)\|_{\mathcal{J}_2}) \|T_K(u) - T(u)\|_{\mathcal{J}_2} \rightarrow 0. \quad (5.17)$$

Because: Factor $T_K(s) - T(s) = T_K(u)[T_K(u) - T(u)] + [T_K(u) - T(u)]T(u)$ and apply the Schatten Hölder inequality. The Hilbert–Schmidt norms are bounded by (5.15).

$\langle 2 \rangle 6$ For every bounded B ,

$$\mathrm{Tr}(T_K(s)B) \rightarrow \mathrm{Tr}(T(s)B). \quad (5.18)$$

Because: The absolute difference is at most $\|T_K(s) - T(s)\|_{\mathcal{J}_1} \|B\|$.

$\langle 2 \rangle 7$ This proves $\langle 1 \rangle 13$. □

7 Periodic Feynman–Kac clause and Weyl insertions

Fix a coarse Weyl symbol $\xi = \varepsilon_M q + ip$ and its continuum refinement. Let H_p be the V2 continuum harmonic interpolant, $H_p^{(K)}$ its mode truncation, and let $J_q, J_q^{(K)}$ be the corresponding sharp-time sources.

$\langle 1 \rangle 14$ Pass every deterministic shift datum to the continuum cutoff limit.

$\langle 2 \rangle 1$ The action tail satisfies

$$0 \leq S_\infty(H_p) - S^{(K)}(H_p) \leq C_\xi (K/m)^{-\eta}. \quad (6.1)$$

Because: The tail is

$$\frac{1}{8L} \sum_{|k| > K} \gamma(k) \coth(t\gamma(k)/2) |b_\infty(k)|^2.$$

The D4 envelope is $|b_\infty(k)|^2 \leq C(1 + \varepsilon_M |k|)^{-2-\eta}$; $\coth(t\gamma/2) \leq \coth(t_0 m/2)$ and $\gamma(k) \leq m + |k|$. Integral comparison of the resulting $|k|^{-1-\eta}$ tail yields (6.1).

$\langle 2 \rangle 2$ The source-phase tail satisfies

$$\|\Phi(J_q) - \Phi(J_q^{(K)})\|_2^2 \leq C_\xi (K/m)^{-\eta}, \quad \|e^{i\Phi(J_q)} - e^{i\Phi(J_q^{(K)})}\|_2 \leq C_\xi (K/m)^{-\eta/2}. \quad (6.2)$$

Because: Its variance is the sharp-time thermal covariance paired with the source tail:

$$\sum_{|k| > K} \frac{\coth(t\gamma(k)/2)}{2\gamma(k)} |a_\infty(k)|^2.$$

Use the D4 envelope and integral comparison. The second assertion uses $|e^{ix} - e^{iy}| \leq |x - y|$.

$\langle 2 \rangle 3$ The shifted action difference obeys

$$\|\mathcal{U}_t^{(K)}(\eta + H_p^{(K)}) - \mathcal{U}_t(\eta + H_p)\|_2 \leq C_\xi [1 + \log(1 + K/m)]^2 (K/m)^{-1/2}. \quad (6.3)$$

$\langle 3 \rangle 1$ Expand every shifted Wick power using (3.12) and its quadratic analogue.

$\langle 3 \rangle 2$ Terms in which the random Wick chaos is truncated are bounded by (2.17)–(2.19).

$\langle 3 \rangle 3$ For deterministic coefficients,

$$\sup_s \frac{1}{2L} \sum_{|k| > K} |\hat{H}_p(s, k)|^2 \leq C_\xi (\varepsilon_M K)^{-1-\eta}, \quad \|H_p^{(K)}\|_\infty, \|H_p\|_\infty \leq C_\xi. \quad (6.4)$$

Because: The harmonic time factor has modulus at most one; square the D4 mode envelope and sum its tail. The uniform bound is the absolutely summable $|k|^{-1-\eta/2}$ envelope.

$\langle 3 \rangle 4$ The identity

$$A^{*n} - B^{*n} = \sum_{r=0}^{n-1} A^{*r} * (A - B) * B^{*(n-1-r)} \quad (6.5)$$

and Young/Cauchy–Schwarz bounds turn (6.4) into the required differences of all coefficient power symbols.

Because: Only $n \leq 4$ occurs and the untruncated coefficient sequences have uniform ℓ^1 bounds. Each term of (6.5) contains exactly one tail factor, so its ℓ^2 norm has the rate in (6.4). Wick

isometry pairs it with the bounded covariance convolution from V2. The slower common rate in (6.3) is therefore valid.

⟨2⟩ 4 For every finite p ,

$$\sup_K \|e^{-\mathcal{U}_t^{(K)}(\eta+H_p^{(K)})}\|_p < \infty, \quad e^{-\mathcal{U}_t(\eta+H_p)} \in L^p. \quad (6.6)$$

Because: Apply (1.4) pointwise with $u = \eta^{(K)}(s, x) + H_p^{(K)}(s, x)$ and integrate: this gives $\mathcal{U}_t^{(K)}(\eta + H_p^{(K)}) \geq -Ct\ell_K^2$, uniformly in the admissible coarse data. Equation (6.3) is the required L^2 increment bound; use $1 + \log(1 + K/m) \leq C_\varsigma(K/m)^\varsigma$ with $0 < \varsigma < 1/4$ to retain a positive power. Step ⟨1⟩5 then gives the uniform L^p bound, and an almost-everywhere convergent subsequence followed by Fatou gives the limiting assertion.

⟨2⟩ 5 This proves ⟨1⟩14. □

⟨1⟩ 15 Prove the continuum periodic partition and marked-Weyl formulas.

⟨2⟩ 1 For $t \geq t_0$,

$$\mathrm{Tr} e^{-tK_L} = Z_0(t) \mathbb{E}_{\mu_{t,L}^{\mathrm{per}}} [e^{-\mathcal{U}_t}] \in (0, \infty). \quad (6.7)$$

Because: Equation (3.8) holds at every K . Its left side converges by (5.15). For the right side, (4.2), (2.16), Hölder, and the uniform exponential bounds give $e^{-\mathcal{U}_t^{(K)}} \rightarrow e^{-\mathcal{U}_t}$ in L^1 . Finiteness follows. Positivity follows because the exponential is positive a.s.; alternatively Jensen gives $\mathbb{E}e^{-\mathcal{U}_t} \geq e^{-\mathbb{E}\mathcal{U}_t} > 0$, and the finite chaos expectation is finite.

⟨2⟩ 2 For the full refined coarse symbol,

$$\frac{\mathrm{Tr}(e^{-tK_L} W_{\mathrm{ct}}(R_M^\infty \xi))}{\mathrm{Tr} e^{-tK_L}} = e^{-S_\infty(H_p)} \frac{\mathbb{E}[e^{i\Phi(J_q)} e^{-\mathcal{U}_t(\eta+H_p)}]}{\mathbb{E}[e^{-\mathcal{U}_t}]}. \quad (6.8)$$

⟨3⟩ 1 Apply (3.10) to the symbol and interaction truncated at K .

⟨3⟩ 2 The Euclidean numerator converges.

Because: Split its difference into a phase difference and an exponential difference. Bound the phase part by (6.2) times the uniform L^2 norm in (6.6). Bound the exponential part using (4.2), (6.3), and (6.6). Equations (6.1) and (6.7) handle the prefactor and denominator.

⟨3⟩ 3 The operator numerator converges.

Because: Split

$$\mathrm{Tr}[(T_K(t) - T(t))W(\xi_K)] + \mathrm{Tr}[T(t)(W(\xi_K) - W(R_M^\infty \xi))].$$

The first term tends to zero by (5.17) and unitarity. The one-particle symbols ξ_K converge in norm by the D4 envelope. The Weyl relation and the Fock vacuum overlap

$$\langle W(\alpha)\Omega_0, W(\beta)\Omega_0 \rangle = e^{i\sigma(\alpha,\beta)/2} e^{-\|\alpha-\beta\|^2/4}$$

give strong continuity of $W(\xi_K)$. Expanding the trace-class $T(t)$ in an eigenbasis and using the bound $2e^{-t\mu_j}$ gives convergence of the second term by dominated convergence. Thus (6.8) follows.

⟨2⟩ 3 This proves ⟨1⟩15. □

8 Positivity improvement, simple ground state, and positive gap

⟨1⟩ 16 Prove the continuum ground-state assertions.

⟨2⟩ 1 $T(t)$ is positivity preserving.

Because: Each $T_K(t)$ has a nonnegative Feynman–Kac kernel; strong limits preserve the closed positive cone in L^2 .

⟨2⟩ 2 If $f, g \geq 0$ are nonzero and bounded, then

$$\langle f, T(t)g \rangle > 0. \quad (7.1)$$

Because: Equation (4.5) has a nonnegative integrand. The weight is strictly positive a.s. because $\mathcal{W}$ is finite a.s. ($\mathcal{W} \in L^2$). If $A = \{f > 0\}$ and $B = \{g > 0\}$, then

$$\mathbb{P}(X_0 \in A, X_t \in B) = \langle \mathbf{1}_A, \Gamma(e^{-t\gamma})\mathbf{1}_B \rangle > 0$$

by (2.3) and Simon’s positivity-improving theorem, since $\|e^{-t\gamma}\| = e^{-tm} < 1$. The integrand is strictly positive on an event of positive probability.

⟨2⟩ 3 Equation (7.1) extends to all nonzero $f, g \geq 0$ in L^2 .

Because: Replace each by its bounded truncations $f \wedge n, g \wedge n$ and use positivity preservation plus L^2 convergence.

⟨2⟩ 4 The largest eigenvalue of the compact positive operator $T(t)$ is simple and has an a.e. strictly positive eigenvector Ω_L .

Because: If u is a top eigenvector, positivity gives $T(t)|u| \geq |T(t)u| = \|T(t)\||u|$. The variational characterization forces equality, so $|u|$ is a nonnegative top eigenvector; (7.1) makes it strictly positive. If a second real top eigenvector h were orthogonal to it, then both h_+ and h_- would be nonzero. Positivity improvement gives $Th_+ > 0$ and $Th_- > 0$ a.e., hence $|Th| = |Th_+ - Th_-| < Th_+ + Th_- = T|h|$ a.e. Pairing with $\Omega_L > 0$ contradicts equality of their top Rayleigh quotients.

⟨2⟩ 5 If $\mu_0 < \mu_1 \leq \dots$ are the discrete eigenvalues of K_L ,

$$\gamma_L := \mu_1 - \mu_0 > 0. \quad (7.2)$$

Because: Simplicity says the eigenspace at μ_0 has dimension one; discreteness from Step ⟨1⟩13 prevents another spectral point from accumulating at μ_0 . Hence the next ordered eigenvalue is strictly larger.

⟨2⟩ 6 The vector functional

$$\omega_{L,\lambda}^{\text{ct}}(A) = \langle \Omega_L, A\Omega_L \rangle \quad (7.3)$$

is a state on the continuum Weyl algebra.

Because: Every Weyl operator whose symbol lies in $H^{-1/2}(\mathbb{T}_L) \oplus iH^{1/2}(\mathbb{T}_L)$ is unitary on the Fock/Gaussian representation. A normalized vector functional is positive and unital; it extends by continuity to the generated C^* -algebra. The D4 refined coarse symbols lie in this one-particle space by their decay envelope.

⟨2⟩ 7 This proves ⟨1⟩16. □

9 V3 conclusion and scope fence

Steps $\langle 1 \rangle 9$ – $\langle 1 \rangle 10$ construct the bounded-below self-adjoint K_L and identify its operator and quadratic form on $\mathcal{D}$. Steps $\langle 1 \rangle 11$ – $\langle 1 \rangle 13$ prove trace class for every positive time, trace-norm cutoff convergence, discrete spectrum, and eigenvalue convergence. Steps $\langle 1 \rangle 14$ – $\langle 1 \rangle 15$ prove the periodic partition and marked-Weyl formulas. Step $\langle 1 \rangle 16$ proves a simple strictly positive ground vector and $\gamma_L > 0$.

No assertion of essential self-adjointness of $(H_0 + V_L)|_{\mathcal{D}}$, and no assertion that $\mathcal{D}$ is a form core, is made or needed. The source V3’s compressed periodic N2 tail count is repaired explicitly in (2.17)–(2.19).

Part II V4 in Lamport notation

Fixed-time comparison, spectral transfer, ground-state limit, and assembly

Fresh reconstruction of `part_ii/v4_assembly.tex`

16 August 2026

This module proves the V4 assembly from the corrected V1–V3 modules. In particular, every use of V2’s shifted estimate refers to the repaired fundamental-band/alias proof, and every use of its fourth-chaos comparison includes the four-edge term omitted by the source V2. Thus the argument below validates the claimed endpoint after those repairs; it does not retroactively validate the original V2 proof.

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1 Objects and corrected input contract

Fix $L, m > 0$, $\lambda \geq 0$, a coarse scale M , and $\xi = \varepsilon_M q + ip \in h_{M,L}$. Let $N = M + r$, $\varepsilon_N = 2^{-r} \varepsilon_M$, and

$$\mathcal{L}_N = 1 + \log(1/\varepsilon_N), \quad \eta_D = 2 - \log_2 3 > 0. \quad (0.1)$$

The lattice Hamiltonian $K_{N,L}$ has ordered eigenvalues $E_0(N) < E_1(N) \leq \dots$, simple ground vector Ω_N , and ground state ω_N . Put

$$e_0(N) = \frac{1}{2} \sum_{k \in \Gamma_N} \gamma_N(k), \quad \tilde{E}_j(N) = E_j(N) - e_0(N), \quad (0.2)$$

$$D_N(t) = \sum_{j \geq 1} e^{-t(E_j(N) - E_0(N))}. \quad (0.3)$$

The continuum operator K_L constructed in the V3 module has eigenvalues $\mu_0 < \mu_1 \leq \dots$, normalized positive ground vector Ω_L , gap $\gamma_L = \mu_1 - \mu_0 > 0$, and

$$D_\infty(t) = \sum_{j \geq 1} e^{-t(\mu_j - \mu_0)} < \infty. \quad (0.4)$$

Define the thermal Weyl functionals

$$F_N(t; \xi) = \frac{\text{Tr}(e^{-tK_{N,L}} W_N(R_M^N \xi))}{\text{Tr} e^{-tK_{N,L}}}, \quad F_\infty(t; \xi) = \frac{\text{Tr}(e^{-tK_L} W_{\text{ct}}(R_M^\infty \xi))}{\text{Tr} e^{-tK_L}}. \quad (0.5)$$

Both have modulus at most one because the heat operator is positive trace class and the Weyl operator is unitary.

The input contract is as follows.

1. V1 supplies the exact lattice marked-slice mean-shift formula, with the free periodic Gaussian reference and with the interaction, not the full harmonic potential, in the path weight.
2. The repaired V2 module supplies the common-noise laws, unshifted comparison, the explicit four-edge correction, the exact fundamental-band/alias shifted comparison, and uniform plain/shifted Nelson bounds. Its standing time hypothesis is $t_0 L \geq 1$.
3. V3 supplies the continuum trace, marked-Weyl formula, trace-class semigroup, simple ground state, and positive gap.
4. The earlier free-rate and Weyl-reduction Lamport modules supply their stated elementary estimates and C^* -algebraic extension theorem.

No new printed theorem is imported in this module.

2 Exact coupled representations

Work on the repaired V2 common-noise probability space. Let η_N have the lattice periodic law and η the continuum periodic law. Let $\mathcal{U}_{N,t}, \mathcal{U}_t$ be the corresponding interactions, and $h^{(N)}, h^{(\infty)}$ the deterministic jump interpolants. Define

$$b_N(k) = \varepsilon_M^{1/2} P_r(\varepsilon_M k) \hat{p}_M(k), \quad a_N(k) = \varepsilon_M^{1/2} P_r(\varepsilon_M k) \hat{q}_M(k), \quad (1.1)$$

and define a_∞, b_∞ by replacing P_r with P_∞ .

$\langle 1 \rangle$ 1 Write both thermal functionals on this single coupling space.

$\langle 2 \rangle$ 1 Define

$$S_N = \frac{1}{8L} \sum_{k \in \Gamma_N} \gamma_N(k) \coth(t\gamma_N(k)/2) |b_N(k)|^2, \quad S_\infty = \frac{1}{8L} \sum_{k \in \Gamma_\infty} \gamma(k) \coth(t\gamma(k)/2) |b_\infty(k)|^2, \quad (1.2)$$

$$\Phi_N(J_q) = \varepsilon_N \sum_{x \in \Lambda_N} q_N(x) \eta_N(0, x), \quad \Phi(J_q) = \int_{\mathbb{T}_L} q_\infty(x) \eta(0, x) dx. \quad (1.3)$$

$\langle 2 \rangle$ 2 Put

$$\mathcal{N}_N = \mathbb{E}[e^{i\Phi_N(J_q)} e^{-\mathcal{U}_{N,t}(\eta_N + h^{(N)})}], \quad \mathcal{E}_N = \mathbb{E}[e^{-\mathcal{U}_{N,t}}], \quad (1.4)$$

$$\mathcal{N}_\infty = \mathbb{E}[e^{i\Phi(J_q)} e^{-\mathcal{U}_t(\eta + h^{(\infty)})}], \quad \mathcal{E}_\infty = \mathbb{E}[e^{-\mathcal{U}_t}]. \quad (1.5)$$

$\langle 2 \rangle$ 3 The exact representations are

$$F_N = e^{-S_N} \frac{\mathcal{N}_N}{\mathcal{E}_N}, \quad F_\infty = e^{-S_\infty} \frac{\mathcal{N}_\infty}{\mathcal{E}_\infty}. \quad (1.6)$$

Because: The first is V1's mode-by-mode Mehler mean-shift identity with the corrected interaction reference. The second is equation (6.8) of the V3 module. The repaired V2 covariance proposition identifies their Gaussian fields on one space. The Appell formula in V1/V3 shows that no extra Wick-subtraction term is present. All factors and normalizations appear explicitly in (1.1)–(1.5).

⟨2⟩ 4 This proves ⟨1⟩1. □

3 Action and marked-source comparisons

Throughout this section, $t \in [t_0, T]$, $t_0 L \geq 1$, and constants may depend on the fixed coarse data but not on N or t . Put $\bar{w} = \coth(t_0 m/2)$.

⟨1⟩ 2 Compare the action prefactors.

⟨2⟩ 1 $0 \leq S_N, S_\infty \leq C_\xi$.

Because: The D4 envelope gives

$$|b_\bullet(k)|^2 \leq C\varepsilon_M \|\widehat{p}_M\|_\infty^2 (1 + \varepsilon_M |k|)^{-2-\eta_D}. \quad (2.1)$$

Also $\gamma_N \leq \gamma$, $\coth(t\gamma_N/2) \leq \bar{w}$, and $\sum_k (m + |k|)(1 + \varepsilon_M |k|)^{-2-\eta_D} < \infty$.

⟨2⟩ 2 Let $\ell = \varepsilon_M |k|$ and split at $\ell_r = 2^{r/2}$. On $\ell \leq \ell_r$,

$$|\gamma_N \coth(t\gamma_N/2) - \gamma \coth(t\gamma/2)| \leq C(t_0, m) \varepsilon_N^2 k^4, \quad (2.2)$$

$$|b_N|^2 - |b_\infty|^2 \leq C_\xi 2^{-r} \ell (1 + \ell)^{-2-\eta_D}. \quad (2.3)$$

Because: For $w(y) = y \coth(ty/2)$, $|w'(y)| \leq \bar{w} + \sup_{x \geq t_0 m/2} x / \sinh^2 x$. The lattice dispersion estimate gives $0 \leq \gamma - \gamma_N \leq \varepsilon_N^2 k^4 / (24m)$, proving (2.2). The QMF telescope gives $|P_r - P_\infty| \leq C 2^{-r} \ell (1 + \ell)^{-1-\eta_D/2}$; multiply by $|b_N| + |b_\infty|$ and use (2.1) to obtain (2.3).

⟨2⟩ 3 Summing the low-mode dispersion term gives

$$C 4^{-r} \int_0^{\ell_r} \ell^4 (1 + \ell)^{-2-\eta_D} d\ell \leq C 2^{-r(1+\eta_D)/2}. \quad (2.4)$$

Because: The mode density in ℓ is constant. At large ℓ , the integrand is $\ell^{2-\eta_D}$, whose integral is $O(\ell_r^{3-\eta_D})$; multiply by 4^{-r} .

⟨2⟩ 4 Summing the low-mode filter term gives

$$C 2^{-r} \int_0^{\ell_r} \ell^2 (1 + \ell)^{-2-\eta_D} d\ell \leq C 2^{-r(1+\eta_D)/2}. \quad (2.5)$$

Because: The action weight grows at most linearly in ℓ ; (2.3) supplies the other factor. The integral is $O(\ell_r^{1-\eta_D})$.

⟨2⟩ 5 The high-mode tails of both actions obey

$$C \int_{\ell_r}^\infty \ell (1 + \ell)^{-2-\eta_D} d\ell \leq C 2^{-r\eta_D/2}. \quad (2.6)$$

⟨2⟩ 6 Therefore

$$|S_N - S_\infty| \leq C_\xi 2^{-r\eta_D/2}, \quad |e^{-S_N} - e^{-S_\infty}| \leq |S_N - S_\infty|. \quad (2.7)$$

Because: Add (2.4)–(2.6). The last inequality follows from the mean-value theorem because $S_N, S_\infty \geq 0$ and $|(-e^{-x})| \leq 1$ there.

$\langle 2 \rangle 7$ This proves $\langle 1 \rangle 2$. □

$\langle 1 \rangle 3$ Compare the marked-slice source phases, including the asymmetric edge.

$\langle 2 \rangle 1$ For $r \geq 1$, $a_N(\pm\pi/\varepsilon_N) = 0$.

Because: The refinement mask contains the factor $m_0(\pm\pi) = 0$ at the last scale, so $P_r(\pm\pi 2^r) = 0$. Thus the symmetrized V2 edge noise makes no contribution to the linear marked source. This fact applies only to this linear source; it does not erase edge terms from covariance powers.

$\langle 2 \rangle 2$ For $r \geq 1$, orthogonality of the common Gaussian Fourier family gives

$$\mathbb{E}[(\Phi_N - \Phi)^2] = \frac{1}{2Lt} \left[\sum_{\kappa \in D_N} |c_N(\kappa)^{1/2} a_N(-k) - c_\infty(\kappa)^{1/2} a_\infty(-k)|^2 + \sum_{\kappa \in D_\infty \setminus D_N} c_\infty(\kappa) |a_\infty(-k)|^2 \right]. \quad (2.8)$$

Because: After Step $\langle 2 \rangle 1$, the lattice source contains no edge coefficient, so its Fourier noise is literally the restriction of the common family. Subtract the two series and apply the Gaussian isometry. The edge mode $+\pi/\varepsilon_N$ is included in the continuum-tail sum.

$\langle 2 \rangle 3$ The filter-difference part of the first sum in (2.8) is at most

$$C_\xi 4^{-r} \int_0^{\pi 2^r} \ell^2 (1 + \ell)^{-2-\eta_D} d\ell \leq C_\xi 2^{-r(1+\eta_D)}. \quad (2.9)$$

Because: $t^{-1} \sum_{k_0} c_N(k_0, k) = \coth(t\gamma_N/2)/(2\gamma_N) \leq \bar{w}/(2m)$; square the QMF filter difference used in (2.3) and sum the spatial modes.

$\langle 2 \rangle 4$ The covariance-symbol part is $O(\varepsilon_N^2)$.

Because: $(\sqrt{x} - \sqrt{y})^2 \leq |x - y|$ and the repaired V2 symbol estimate is

$$|c_N - c_\infty|(k_0, k) \leq C\varepsilon_N^2 k^4 (m^2 + k_0^2 + k^2)^{-2}. \quad (2.10)$$

The exact k_0 count gives

$$t^{-1} \sum_{k_0} |c_N - c_\infty| \leq C\varepsilon_N^2 \frac{k^4}{(m^2 + k^2)^{3/2}} \leq C\varepsilon_N^2 (m + |k|).$$

Multiplying by the D4 envelope for $|a_\infty|^2$ leaves a summable $|k|^{-1-\eta_D}$ tail.

$\langle 2 \rangle 5$ The missing-mode term in (2.8) is $O(2^{-r(1+\eta_D)})$.

Because: Sum $t^{-1} \sum_{k_0} c_\infty \leq \bar{w}/(2\gamma(k))$ against $|a_\infty(k)|^2$ for $|k| \geq \pi/\varepsilon_N$ and use the D4 tail.

$\langle 2 \rangle 6$ After increasing the constant to cover the one fixed case $r = 0$,

$$\mathbb{E}[(\Phi_N - \Phi)^2] \leq C_\xi 2^{-r\eta_D/2}, \quad \|e^{i\Phi_N} - e^{i\Phi}\|_2 \leq C_\xi 2^{-r\eta_D/4}. \quad (2.11)$$

Because: For $r \geq 1$, combine (2.9)–(2.10) and the missing tail, then weaken the exponent to a common one. At $r = 0$ the replacement of the edge noise in (2.8) is not valid, but both variances are finite and bounded by a constant depending on the fixed coarse scale, which equals the right side after enlarging C_ξ . Finally use $|e^{ix} - e^{iy}| \leq |x - y|$.

$\langle 2 \rangle 7$ This proves $\langle 1 \rangle 3$. □

4 Numerators, denominators, and fixed-time convergence

⟨1⟩ 4 Bound the denominators away from zero.

⟨2⟩ 1 At the gate endpoint $(a, b) = (0, 0)$,

$$\mathbb{E}\mathcal{U}_{N,t} = 3\lambda d_N(t)^2(2Lt), \quad \mathbb{E}\mathcal{U}_t = 3\lambda d_\infty(t)^2(2Lt). \quad (3.1)$$

Because: The ensemble-Wick chaoses of orders two and four are centered. The Wick recombination constant term is exactly $3d^2$; integrate the constant over volume $2Lt$.

⟨2⟩ 2 Since $0 \leq d_N, d_\infty \leq \bar{d}$,

$$\mathcal{E}_N, \mathcal{E}_\infty \geq e^{-C_0}, \quad C_0 = 3\lambda \bar{d}^2(2LT). \quad (3.2)$$

Because: Jensen's inequality for the convex function e^{-x} gives $\mathbb{E}e^{-\mathcal{U}} \geq e^{-\mathbb{E}\mathcal{U}}$; insert (3.1).

⟨2⟩ 3 This proves ⟨1⟩4. □

⟨1⟩ 5 Compare the denominators.

⟨2⟩ 1 The repaired unshifted V2 comparison gives

$$\|\mathcal{U}_{N,t} - \mathcal{U}_t\|_2 \leq C(1+t)(1+L)\varepsilon_N \mathcal{L}_N^{3/2}. \quad (3.3)$$

Because: Its fourth-chaos computation contains the one-edge and two-edge classes and the nonzero four-edge class. The last contributes the explicit positive term

$$\Delta_{4e} = 4!(2Lt)^{-2} \frac{3}{2} \sum_{k_{0,1}+\dots+k_{0,4}=0} \prod_{i=1}^4 c_\times(k_{0,i}, \pi/\varepsilon_N), \quad (3.4)$$

bounded by $9t \coth(2t_0)^3 \varepsilon_N^5 / (256L^2)$. Hence it lies below the common rate in (3.3); it may not be deleted from the exact identity.

⟨2⟩ 2 The plain lattice and continuum Nelson bounds and $|e^{-x} - e^{-y}| \leq |x - y|(e^{-x} + e^{-y})$ imply

$$|\mathcal{E}_N - \mathcal{E}_\infty| \leq C(1+t)(1+L)\varepsilon_N \mathcal{L}_N^{3/2}. \quad (3.5)$$

Because: Apply Cauchy–Schwarz to the product of the exponent difference and $e^{-\mathcal{U}_N} + e^{-\mathcal{U}}$; (3.3) controls the first factor and the uniform L^2 Nelson bound the second.

⟨2⟩ 3 This proves ⟨1⟩5. □

⟨1⟩ 6 Compare the numerators.

⟨2⟩ 1 Split

$$\begin{aligned} |\mathcal{N}_N - \mathcal{N}_\infty| &\leq \mathbb{E}[|e^{i\Phi_N} - e^{i\Phi}| e^{-\mathcal{U}_{N,t}(\eta_N + h^{(N)})}] \\ &\quad + \mathbb{E}[|e^{-\mathcal{U}_{N,t}(\eta_N + h^{(N)})} - e^{-\mathcal{U}_t(\eta + h^{(\infty)})}|]. \end{aligned} \quad (3.6)$$

⟨2⟩ 2 The first term in (3.6) is at most $C_\xi 2^{-r\eta_D/4}$.

Because: Apply Cauchy–Schwarz, (2.11), and the uniform shifted lattice L^2 Nelson bound.

⟨2⟩ 3 The repaired shifted V2 theorem gives

$$\|\mathcal{U}_{N,t}(\eta_N + h^{(N)}) - \mathcal{U}_t(\eta + h^{(\infty)})\|_2 \leq C_\xi \left(2^{-r(1+\eta_D)/4} + \varepsilon_N \mathcal{L}_N^{3/2} \right). \quad (3.7)$$

Because: Decompose it into the unshifted difference (3.3) and the difference of the two shift increments. The latter is proved in the repaired V2 module by: (i) folding the lattice power symbols into the fundamental band; (ii) defining the exact-output cross and continuum symbols; (iii) isolating the mixed alias term $A_{\times,j}$; (iv) bounding filter, dispersion, high-output, and physical edge remainders separately; and (v) applying the convolution telescope to every deterministic coefficient power. Thus no undefined R_{tail} is used in (3.7).

⟨2⟩ 4 The second term in (3.6) is bounded by the right side of (3.7).

Because: Apply the exponential-difference inequality, Cauchy–Schwarz, (3.7), and the paired uniform shifted Nelson bounds: V2 for finite N , V3 for the continuum member.

⟨2⟩ 5 Therefore

$$|\mathcal{N}_N - \mathcal{N}_\infty| \leq C_\xi (2^{-r\eta_D/4} + \varepsilon_N \mathcal{L}_N^{3/2}), \quad (3.8)$$

and

$$|\mathcal{N}_N|, |\mathcal{N}_\infty| \leq B_1 < \infty. \quad (3.9)$$

Because: The slower of the two powers of 2^{-r} is $\eta_D/4$. Equation (3.9) follows by deleting the unit-modulus phase and applying the L^1 shifted Nelson bounds.

⟨2⟩ 6 This proves ⟨1⟩6. □

⟨1⟩ 7 Prove fixed-time convergence of the interacting thermal functionals.

⟨2⟩ 1 The exact algebraic difference is

$$\begin{aligned} F_N - F_\infty &= (e^{-S_N} - e^{-S_\infty}) \frac{\mathcal{N}_N}{\mathcal{E}_N} + e^{-S_\infty} \frac{\mathcal{N}_N - \mathcal{N}_\infty}{\mathcal{E}_N} \\ &\quad + e^{-S_\infty} \mathcal{N}_\infty \frac{\mathcal{E}_\infty - \mathcal{E}_N}{\mathcal{E}_N \mathcal{E}_\infty}. \end{aligned} \quad (3.10)$$

Because: Put (1.6) over the common denominators and add/subtract the two intermediate terms shown.

⟨2⟩ 2 Equations (2.7), (3.2), (3.5), (3.8), and (3.9) give

$$|F_N(t; \xi) - F_\infty(t; \xi)| \leq C_\xi (2^{-r\eta_D/4} + \varepsilon_N \mathcal{L}_N^{3/2}) \rightarrow 0. \quad (3.11)$$

⟨2⟩ 3 For every fixed $t \geq 1/L$, (3.11) is admissible by taking $t_0 = \max(t/2, 1/L)$ and $T = 2t$.

Because: Then $t_0 \leq t \leq T$ and $t_0 L \geq 1$. This is the genuine range of the V2 comparison; no conclusion at arbitrarily small t is inferred.

⟨2⟩ 4 This proves ⟨1⟩7 (A6). □

5 Partition functions and exact Laplace transfer

⟨1⟩ 8 Compare the normal-ordered free partition functions.

⟨2⟩ 1 For every $t > 0$,

$$\tilde{Z}_N^{(0)}(t) := \text{Tr} e^{-t(H_{N,L}^{(0)} - e_0(N))} = \prod_{k \in \Gamma_N} (1 - e^{-t\gamma_N(k)})^{-1}. \quad (4.1)$$

Because: Subtracting $e_0(N) = \frac{1}{2} \sum_k \gamma_N(k)$ removes the zero-point energy from the product of one-oscillator traces.

⟨2⟩ 2 For $k \in \Gamma_N$, the mean-value theorem gives

$$|\log(1 - e^{-t\gamma_N(k)}) - \log(1 - e^{-t\gamma(k)})| \leq \frac{te^{-t\gamma_N(k)}}{1 - e^{-tm}} |\gamma(k) - \gamma_N(k)|. \quad (4.2)$$

Because: The derivative of $y \mapsto \log(1 - e^{-ty})$ is $t/(e^{ty} - 1) \leq te^{-ty}/(1 - e^{-tm})$ for $y \geq m$, and $\gamma_N \leq \gamma$.

⟨2⟩ 3 Using

$$0 \leq \gamma(k) - \gamma_N(k) \leq \frac{\varepsilon_N^2 k^4}{24m}, \quad \gamma_N(k) \geq \max(m, 2|k|/\pi), \quad (4.3)$$

the sum of (4.2) is $O(\varepsilon_N^2)$.

Because: The remaining series $\sum_k e^{-t \max(m, 2|k|/\pi)} k^4$ converges with a bound independent of N .

⟨2⟩ 4 The omitted continuum modes contribute $O(e^{-2t/\varepsilon_N})$.

Because: The asymmetric complement begins at magnitude π/ε_N , including the positive edge. Use $|\log(1 - e^{-x})| \leq e^{-x}/(1 - e^{-tm})$ and $\gamma(k) \geq |k|$; sum the geometric spatial tail.

⟨2⟩ 5 Hence

$$|\log \tilde{Z}_N^{(0)}(t) - \log Z_0(t)| \leq C(m, L, t) \varepsilon_N^2, \quad \tilde{Z}_N^{(0)}(t) \rightarrow Z_0(t). \quad (4.4)$$

⟨2⟩ 6 This proves ⟨1⟩ 8. □

⟨1⟩ 9 Compare the interacting normal-ordered partition functions and obtain a uniform lower ground-energy bound.

Fix $t_* > 0$ with $t_* L \geq 1$.

⟨2⟩ 1 For every $t \geq t_*$,

$$\tilde{Z}_N(t) := \text{Tr} e^{-t(K_{N,L} - e_0(N))} = \tilde{Z}_N^{(0)}(t) \mathcal{E}_N(t). \quad (4.5)$$

Because: Apply the finite-dimensional cyclic Mehler trace construction from V1/V3 to the lattice dispersions and interaction. Normal ordering removes only the free zero-point factor. The repaired V2 proposition identifies the cyclic Gaussian law, including the symmetrized edge column, and Wick recombination identifies the path integrand with $\mathcal{U}_{N,t}$.

⟨2⟩ 2 Equations (3.5), (4.4), and the V3 continuum partition identity imply

$$\tilde{Z}_N(t) \rightarrow Z_0(t) \mathcal{E}_\infty(t) = \text{Tr} e^{-tK_L} =: Z_\infty(t) \quad (t \geq t_*). \quad (4.6)$$

Because: For this fixed t , use the admissible interval $t_0 = \max(t/2, 1/L)$, $T = 2t$ in (3.5). This avoids the invalid choice $t_0 = t/2$ when $t < 2/L$.

⟨2⟩ 3 There is $c_\lambda < \infty$ such that

$$\inf_N \tilde{E}_0(N) \geq -c_\lambda. \quad (4.7)$$

Because: Fix any $t_1 \geq t_*$. Then

$$e^{-t_1 \tilde{E}_0(N)} \leq \tilde{Z}_N(t_1) = \tilde{Z}_N^{(0)}(t_1) \mathcal{E}_N(t_1).$$

The first factor is uniformly bounded by (4.4), and the second by the unshifted L^1 Nelson bound. Take logarithms and divide by t_1 .

⟨2⟩ 4 This proves ⟨1⟩9. □

⟨1⟩ 10 Prove the Laplace-transfer lemma in the exact range needed here.

Let $\nu_N = \sum_{j \geq 0} \delta_{x_j(N)}$ be locally finite eigenvalue point measures, assume all $x_j(N) \geq c$, and suppose

$$\int e^{-tx} d\nu_N(x) \rightarrow \int e^{-tx} d\nu_\infty(x) \quad \text{for every } t \geq t_*. \quad (4.8)$$

⟨2⟩ 1 The span $\mathcal{A}_{t_*}$ of $x \mapsto e^{-tx}$, $t \geq t_*$, is uniformly dense in $C_0([c, \infty))$.

Because: It is a self-adjoint algebra: products add the two parameters and remain at least t_* . It separates points and vanishes nowhere. The locally compact Stone–Weierstrass theorem applies.

⟨2⟩ 2 For $f \in C_c([c, \infty))$, put $\tilde{f}(x) = f(x)e^{t_*x}$ and choose $g \in \mathcal{A}_{t_*}$ with $\|\tilde{f} - g\|_\infty < \delta$. Then

$$\left| \int f d\nu_N - \int g e^{-t_*x} d\nu_N \right| \leq \delta \int e^{-t_*x} d\nu_N. \quad (4.9)$$

Because: Multiply the pointwise approximation error by the finite measure $e^{-t_*x} d\nu_N$. This weighted step is necessary: ν_N itself generally has infinite total mass.

⟨2⟩ 3 The right side of (4.9) is uniformly $O(\delta)$, and the middle integrals converge.

Because: Equation (4.8) at t_* makes their masses uniformly bounded. Since g is a finite linear combination of e^{-tx} with $t \geq t_*$, ge^{-t_*x} is a finite linear combination with parameters at least $2t_*$, so (4.8) applies.

⟨2⟩ 4 Let $N \rightarrow \infty$ and then $\delta \downarrow 0$ in (4.9). Thus $\nu_N \rightarrow \nu_\infty$ vaguely.

⟨2⟩ 5 Vague convergence implies convergence of every ordered atom with multiplicity:

$$x_j(N) \rightarrow x_j(\infty). \quad (4.10)$$

Because: Around the first $j+1$ limit atoms choose finitely many pairwise disjoint intervals whose endpoints are not atoms and which contain no other limit atoms. Continuous upper and lower approximations to their indicators show that the integer masses in each interval eventually equal the limiting multiplicities. Shrink the intervals. Induction in j yields (4.10).

⟨2⟩ 6 For $t \geq t_*$,

$$D_N(t) = e^{tx_0(N)} \int e^{-tx} d\nu_N(x) - 1 \rightarrow D_\infty(t). \quad (4.11)$$

Because: Combine (4.8) and (4.10) for $j = 0$. No conclusion is asserted for $t < t_*$; the hypotheses supply no control there.

⟨2⟩ 7 This proves ⟨1⟩10. □

⟨1⟩ 11 Apply the transfer lemma to the interacting spectra.

⟨2⟩ 1 Take $x_j(N) = \tilde{E}_j(N)$ and $x_j(\infty) = \mu_j$. Equations (4.6) and (4.7) verify (4.8) and the common lower support bound.

⟨2⟩ 2 Therefore, for every fixed j ,

$$\tilde{E}_j(N) \rightarrow \mu_j. \quad (4.12)$$

⟨2⟩ 3 In particular,

$$E_1(N) - E_0(N) \circ \mu_1 - \mu_0 = \gamma_L > 0, \quad (4.13)$$

and for every $t \geq t_*$,

$$D_N(t) \rightarrow D_\infty(t) < \infty. \quad (4.14)$$

Because: The zero-point shift cancels from each eigenvalue difference and from D_N . Apply (4.10)–(4.11) and the V3 strict gap.

⟨2⟩ 4 This proves ⟨1⟩11 (A7). □

6 Thermal-to-ground-state interchange and the gate

⟨1⟩ 12 Prove the spectral tail estimate.

Let K have discrete spectrum $E_0 < E_1 \leq \dots$, normalized ground vector Ω , and let W be unitary.

⟨2⟩ 1 With $Z = e^{-tE_0}(1 + D(t))$,

$$\frac{\text{Tr}(e^{-tK}W)}{Z} - \langle \Omega, W\Omega \rangle = \frac{1}{Z} \sum_{j \geq 1} e^{-tE_j} (\langle \phi_j, W\phi_j \rangle - \langle \Omega, W\Omega \rangle). \quad (5.1)$$

Because: Expand the trace in an orthonormal eigenbasis and cancel the $j = 0$ term.

⟨2⟩ 2 Since every displayed matrix element has modulus at most one,

$$\left| \frac{\text{Tr}(e^{-tK}W)}{\text{Tr } e^{-tK}} - \langle \Omega, W\Omega \rangle \right| \leq \frac{2D(t)}{1 + D(t)} \leq 2D(t). \quad (5.2)$$

⟨2⟩ 3 This applies to every lattice $K_{N,L}$ and to K_L .

Because: Lattice compactness and ground simplicity were proved in the model sheet; V3 proves continuum trace class, discreteness, and ground simplicity.

⟨2⟩ 4 This proves ⟨1⟩12. □

⟨1⟩ 13 Prove the abstract interchange of fixed-temperature and ground-state limits.

Suppose $a_N(t)$, ℓ_N , and $D_N(t)$ obey

$$|a_N(t) - \ell_N| \leq c_0 D_N(t), \quad |a_N(t)| \leq c_0, \quad (5.3)$$

$a_N(t) \rightarrow a_\infty(t)$ for each $t \geq t_0$, $D_N(t_1) \rightarrow D_\infty(t_1) < \infty$ at one $t_1 \geq t_0$, and

$$\liminf_N (E_1(N) - E_0(N)) \geq \gamma > 0. \quad (5.4)$$

⟨2⟩ 1 For $t \geq t_1$,

$$D_N(t) \leq e^{-(t-t_1)(E_1(N)-E_0(N))} D_N(t_1). \quad (5.5)$$

Because: Factor $e^{-(t-t_1)(E_j-E_0)}$ from every summand and bound it by the $j = 1$ value.

⟨2⟩ 2 There are $N_0, D_* < \infty$ such that, for all $N \geq N_0$,

$$D_N(t) \leq D_* e^{-(t-t_1)\gamma/2}. \quad (5.6)$$

Because: Convergence at t_1 bounds $D_N(t_1)$; the liminf in (5.4) gives an eventual gap at least $\gamma/2$.

⟨2⟩ 3 The continuum tail independently satisfies

$$D_\infty(t) \leq D_\infty(t_1) e^{-(t-t_1)\gamma_L} \rightarrow 0. \quad (5.7)$$

⟨2⟩ 4 Given $\delta > 0$, choose t so that the right sides of (5.6)–(5.7) are below δ . Then

$$|\ell_N - \ell_\infty| \leq 2c_0\delta + |a_N(t) - a_\infty(t)|. \quad (5.8)$$

Because: Add and subtract $a_N(t), a_\infty(t)$ and apply (5.3) to the two tails.

⟨2⟩ 5 Let $N \rightarrow \infty$ in (5.8), then $\delta \downarrow 0$. Thus $\ell_N \rightarrow \ell_\infty$.

⟨2⟩ 6 This proves ⟨1⟩13. □

⟨1⟩ 14 Close the Weyl gate at every fixed coarse scale.

Set

$$t_* = \max(1, 1/L), \quad a_N(t) = F_N(t; \xi), \quad \ell_N = \omega_N(W_N(R_M^N \xi)), \quad (5.9)$$

with the corresponding continuum definitions.

⟨2⟩ 1 The bound (5.3) holds with $c_0 = 2$.

Because: $|F_N| \leq 1$ and the spectral estimate (5.2) applies on both sides.

⟨2⟩ 2 For each fixed $t \geq t_*$, $a_N(t) \rightarrow a_\infty(t)$.

Because: $t_* \geq 1/L$, so Step ⟨1⟩7 applies.

⟨2⟩ 3 $D_N(t_*) \rightarrow D_\infty(t_*) < \infty$.

Because: Step ⟨1⟩11 applies with the admissible lower time t_* .

⟨2⟩ 4 $\liminf_N (E_1(N) - E_0(N)) = \gamma_L > 0$.

Because: This is (4.13).

⟨2⟩ 5 Step ⟨1⟩13 therefore gives

$$\lim_{N \rightarrow \infty} \omega_N(\alpha_M^N(W_M(\xi))) = \omega_{L,\lambda}^{\text{ct}}(\beta_M(W_M(\xi))). \quad (5.10)$$

Because: The embeddings are defined by $\alpha_M^N(W_M(\xi)) = W_N(R_M^N \xi)$ and $\beta_M(W_M(\xi)) = W_{\text{ct}}(R_M^\infty \xi)$.

⟨2⟩ 6 This proves ⟨1⟩14 (the gate). □

7 Inductive-limit state, identification, and symmetry

⟨1⟩ 15 Construct the continuum embedding of the inductive-limit algebra.

Assume the already proved scaling identity $\beta_N \alpha_M^N = \beta_M$ and injectivity of all maps.

⟨2⟩ 1 On the algebraic union $\bigcup_M \alpha_M^\infty(\mathfrak{W}_M)$ define

$$\beta(\alpha_M^\infty(A)) = \beta_M(A). \quad (6.1)$$

⟨2⟩ 2 Definition (6.1) is independent of the representative.

Because: If $M \leq M'$ and $\alpha_M^\infty(A) = \alpha_{M'}^\infty(A')$, injectivity of $\alpha_{M'}^\infty$ gives $A' = \alpha_M^{M'}(A)$. Hence $\beta_{M'}(A') = \beta_{M'} \alpha_M^{M'}(A) = \beta_M(A)$.

⟨2⟩ 3 β is isometric on the union.

Because: An injective $*$ -homomorphism between C^* -algebras is isometric; thus $\|\beta(\alpha_M^\infty(A))\| = \|\beta_M(A)\| = \|A\| = \|\alpha_M^\infty(A)\|$.

⟨2⟩ 4 It extends uniquely to an injective $*$ -homomorphism

$$\beta : \mathfrak{W}_\infty \longrightarrow \mathfrak{W}_{\text{ct}}, \quad \beta \alpha_M^\infty = \beta_M. \quad (6.2)$$

Because: The algebraic union is dense in its inductive-limit completion. An isometry extends uniquely; multiplication and involution pass by continuity, and an isometry is injective.

⟨2⟩ 5 This proves ⟨1⟩15. □

⟨1⟩ 16 Apply the previously proved Weyl extension criterion.

Define

$$\chi_M(\xi) = \lim_{N \rightarrow \infty} \omega_N(\alpha_M^N(W_M(\xi))). \quad (6.3)$$

⟨2⟩ 1 The limit exists and equals

$$\chi_M(\xi) = \omega_{L,\lambda}^{\text{ct}}(\beta_M(W_M(\xi))). \quad (6.4)$$

Because: This is (5.10).

⟨2⟩ 2 For each M , there is a unique state ρ_M on $\mathfrak{W}_M$ satisfying $\rho_M(W_M(\xi)) = \chi_M(\xi)$, and

$$\omega_N(\alpha_M^N(A)) \rightarrow \rho_M(A) \quad (A \in \mathfrak{W}_M). \quad (6.5)$$

Because: The Weyl-reduction module proves: pointwise convergence on all Weyl generators, uniform state norm one, and density of finite Weyl linear combinations extend the limit uniquely to the whole C^* -algebra. Equation (6.4) also supplies positive definiteness and continuity directly as a restriction of a state.

⟨2⟩ 3 The family is consistent:

$$\rho_N \circ \alpha_M^N = \rho_M. \quad (6.6)$$

Because: Check it on $W_M(\xi)$ using $\alpha_M^N(W_M(\xi)) = W_N(R_M^N \xi)$ and the compatibility of the continuum refinement; extend by norm density.

⟨2⟩ 4 There is a unique state Ω_∞ on $\mathfrak{W}_\infty$ with

$$\Omega_\infty \circ \alpha_M^\infty = \rho_M, \quad \Omega_\infty = \omega_{L,\lambda}^{\text{ct}} \circ \beta. \quad (6.7)$$

Because: Consistency defines a norm-one positive functional on the dense algebraic union, hence a unique state on the completion. Equations (6.2), (6.4) show that it agrees there with the continuum pullback; density gives the second identity.

⟨2⟩ 5 All states in (6.5)–(6.7) are invariant under the field reflection $\vartheta(W(\zeta)) = W(-\zeta)$.

Because: Every refinement map is linear, hence intertwines ϑ . The lattice Hamiltonian is even and commutes with its reflection unitary. Ground-state simplicity forces that unitary to multiply Ω_N by a phase, so $\omega_N \circ \vartheta_N = \omega_N$. Pass to the limits in (6.5), then use consistency. Equivalently, the earlier Weyl-reduction symmetry proposition gives the same conclusion.

⟨2⟩ 6 This proves ⟨1⟩16 and the full fixed-coarse theorem. □

8 V4 conclusion and validity status

Step ⟨1⟩7 proves fixed- t thermal convergence with rate. Steps ⟨1⟩8–⟨1⟩11 prove normal-ordered partition convergence, eigenvalue convergence, the eventual lattice gap, and tail convergence. Steps ⟨1⟩12–⟨1⟩14 interchange the thermal and ground limits and prove the gate. Steps ⟨1⟩15–⟨1⟩16 identify the unique projective limit with the continuum vacuum and carry the $\mathbb{Z}_2$ symmetry.

This conclusion is conditional on replacing the original V2 Lemma 6.4/Proposition 6.5 proof by the exact repaired comparison in the preceding Lamport module. The original Part II files, without that replacement and the four-edge correction, do not prove their final theorem as written.
